DIAGNOSING GUIDANCE FAILURE IN DIFFUSION-BASED STRUCTURE-BASED DRUG DESIGN: A LEAKAGE-SAFE, FALSIFICATION-DRIVEN STUDY

THESIS

Submitted as one of the requirements to obtain
MAGISTER — academic degree as listed in the student handbook

By:

STUDENT NAME
STUDENT IDENTIFICATION NUMBER

FACULTY OF COMPUTER SCIENCE
MASTER OF INFORMATICS STUDY PROGRAM
PRESIDENT UNIVERSITY
CIKARANG
MONTH, YEAR OF THE DEFENSE

Copyright by
STUDENT NAME
YEAR

DIAGNOSING GUIDANCE FAILURE IN DIFFUSION-BASED STRUCTURE-BASED DRUG DESIGN: A LEAKAGE-SAFE, FALSIFICATION-DRIVEN STUDY

By

STUDENT IDENTIFICATION NUMBER

Approved:

Your Advisor Name Thesis Advisor	Ahmad Fadhil N, Ph.D. Program Head of MIT

Prof. Dr. Ir. Wiranto Herry Utomo, M.Kom.
Dean of Faculty of Computer Science

STATEMENT OF ORIGINALITY

In my capacity as an active student of President University and as the author of the thesis stated below:

Name	: STUDENT NAME
Student ID number	: STUDENT IDENTIFICATION NUMBER
Study Program	: Master of Informatics
Faculty	: Computer Science

I hereby declare that my thesis entitled “Diagnosing Guidance Failure in Diffusion-Based Structure-Based Drug Design: A Leakage-Safe, Falsification-Driven Study” is, to the best of my knowledge and belief, an original piece of work based on sound academic principles. If any plagiarism is detected in this thesis, I am willing to be personally responsible for the consequences of these acts of plagiarism, and will accept the sanctions against these acts in accordance with the rules and policies of President University.

I also declare that this work, either in whole or in part, has not been submitted to another university to obtain a degree.

Cikarang, DATE

(Full name & signature)

SCIENTIFIC PUBLICATION APPROVAL FOR ACADEMIC INTEREST

As an academic community member of President University, I, the undersigned:

Name : STUDENT NAME
Student ID number : STUDENT IDENTIFICATION NUMBER
Study program : Master of Informatics

for the purpose of development of science and technology, certify and approve to give President University a non-exclusive royalty-free right upon my final report with the title: “Diagnosing Guidance Failure in Diffusion-Based Structure-Based Drug Design: A Leakage-Safe, Falsification-Driven Study”.

With this non-exclusive royalty-free right, President University is entitled to converse, to convert, to manage in a database, to maintain, and to publish my final report, with the obligation of President University to mention my name as the copyright owner of my final report.

This statement I made in truth.

Cikarang, DATE

(Full name & signature)

ADVISOR APPROVAL FOR PUBLICATION

As an academic community member of President University, I, the undersigned:

Advisor Name	: Your Advisor Name
Employee ID number	: EMPLOYEE ID
Study program	: Master of Informatics
Faculty	: Computer Science

declare that the following thesis:

Title of thesis	: Diagnosing Guidance Failure in Diffusion-Based Structure-Based Drug Design: A Leakage-Safe, Falsification-Driven Study
Thesis author	: STUDENT NAME
Student ID number	: STUDENT IDENTIFICATION NUMBER

will be published in: journal / institution's repository / proceeding / unpublished / other (choose one, underline that applies).

Cikarang, DATE

(Advisor full name & signature)

SIMILARITY INDEX REPORT

Insert the similarity-check (Turnitin) report for the final version of this thesis here.

ABSTRACT

Structure-based drug design increasingly relies on three-dimensional diffusion models that generate ligands inside a protein pocket. A proposed way of improving such models is guidance with learned predictors of binding affinity and synthesizability. It was tested whether this idea survives when judged by independently computed evaluation outcomes rather than by the predictor's own score, and, where it does not, which of three mechanisms explains the failure: an uninformative guidance signal, a correctly informative signal misapplied through the guidance mechanism, or an objective whose apparent gain can be obtained without a genuine improvement. A frozen TargetDiff model was guided by affinity and synthesizability predictors, and results were evaluated with AutoDock Vina docking, PoseBusters validity, ligand efficiency and the independently computed RA-score, using Kolmogorov–Smirnov tests with Benjamini–Hochberg correction, bootstrap intervals and negative controls.

Five guidance routes were investigated: gradient guidance with an EGNN affinity predictor; a physics-anchored GIGN+PIGNet2 predictor and three mechanism variants; non-gradient rejection sampling; guidance toward synthesizability; and a pre-registered residual (delta-learning) training target. A leakage-safe, target-disjoint data split showed that an earlier generalisation gap was largely a split artefact, yet none of the routes produced a defensible improvement. Affinity guidance had no detectable effect across three orders of magnitude of guidance strength, and the predictor's gradient magnitude was positively correlated with the distance of an atom from the pocket ($r = +0.24$), while a signed measurement of the gradient's direction over 23,202 atom-states found it indistinguishable from random (mean cosine toward the nearest contact -0.019, 95% CI $[-0.046, +0.008]$): the signal is uninformative

about the interface rather than pointed away from it. Rejection sampling with the trained predictor improved the raw docking score in 12 of 15 pockets but worsened ligand efficiency in 14 of 15 (on average 12 more heavy atoms, and 29 percentage points lower PoseBusters validity). A pre-registered positive control then replaced the trained predictor with AutoDock Vina's own zero-shot score in the identical selection mechanism and pockets: this succeeded, with a significant improvement in predicted docking score (Wilcoxon $p = 1.9 \times 10^{-6}$) that survived size normalisation (ligand efficiency unchanged, $p = 0.756$) but came with significantly larger and significantly less synthesizable molecules (both $p < 10^{-4}$), so it is a harness validation and a multi-objective trade-off, not a demonstration of successful design. Because the selection score and the evaluation score are two modes of the same docking program rather than independent measurements, the control establishes that the *selection* harness can detect a genuine signal; it localises the failure of surrogate-based *selection* directly to the predictor's information content, and supports the same reading for the gradient routes only indirectly, since a non-differentiable signal cannot be put through a gradient mechanism. An eight-block information-content audit supports this localisation: no representation of the pocket tested, including a chemically typed interaction representation, reliably outperformed a ligand-only model after correction, and a within-target label-permutation control, run for both a descriptor pipeline and the deployed architecture itself, showed that 70 to 90% of apparent predictive performance is a class-level prior rather than within-target structure–activity learning. Synthesizability guidance gave no supported gain; in an exploratory design it raised the predictor's own score while lowering the independently computed RA-score, a divergence that was not statistically confirmed after retraining on a leakage-safe split. The delta-target predictor (test $R^2 = 0.117$; target-clustered 95% CI −0.20 to 0.32) was

significantly worse than a two-feature linear baseline ($R^2 = 0.362$) and than an EGNN (0.34–0.41), because the residual was not learnt better than by a constant, and augmenting training data with a larger external data set (BindingNet), after removing a compound-level leak found before the production run, did not close the gap to a ligand-only baseline at matched training-set size.

The thesis contributes a reproducible, leakage-aware evaluation framework with a validated positive control for its selection harness, a systematic information-content audit of what predicts measured affinity on this split, and a documented negative result that locates where learned guidance signals fail under the tested configurations, with direct evidence that signal quality, rather than the selection mechanism, limits the rejection-sampling route and converging indirect evidence that it also limits the gradient routes.

Keywords: structure-based drug design; diffusion models; classifier guidance; binding affinity prediction; positive control; data leakage; negative results

DEDICATION

To be written by the author (optional page).

ACKNOWLEDGMENTS

The author wishes to express… (to be written by the author).

TABLE OF CONTENTS

Page

DEDICATION.....ii

ACKNOWLEDGMENTS.....iii

LIST OF TABLES.....x

LIST OF FIGURES.....xii

LIST OF ABBREVIATIONS.....xvi

CHAPTER

I. INTRODUCTION.....1

I.1 Background.....1

I.2 Problem Statement and Importance.....2

I.3 Related Work.....3

I.4 Solution Employed and Novelty.....4

I.5 Summary of Results.....5

I.6 Research Objectives and Experimental Map.....6

I.7 Organization of the Thesis.....9

II. LITERATURE REVIEW.....10

II.1 Deep Generative Models for Structure-Based Drug Design.....10

II.2 Guidance of Diffusion Models.....12

II.3 Predicting Binding Affinity.....13

II.3.1 Docking scores and their biases.....13

II.3.2 Learned scoring functions.....14

II.3.3 Data leakage and shortcut learning.....14

II.4 Evaluating Generated Molecules.....16

II.5 Estimating Synthesizability.....17

II.6 Residual (Delta) Learning for Scoring.....17

II.7 Multi-Task Learning and Interaction Supervision.....18

II.8 QSAR Controls: y-Randomisation, Activity Cliffs and Matched Molecular Pairs.....19

II.9 Positioning and Research Gap.....20

III. RESEARCH METHODOLOGY AND EXPERIMENTAL INFRASTRUCTURE.....23

III.1 Research Design.....23

III.1.1 Three candidate mechanisms for a guidance failure.....25

III.1.2 Pre-registration status of each route.....25

III.2 Data.....29

III.2.1 CrossDocked2020 and the pocket representation.....29

III.2.2 Affinity labels.....30

III.2.3 Splits used for predictor training.....30

III.2.4 Synthesizability labels....32
III.3 Generative Model and Guidance Mechanism....32
III.3.1 The frozen diffusion model....32
III.3.2 Guidance on the clean-data estimate....32
III.4 Predictors....33
III.5 Evaluation Metrics....35
III.5.1 What each metric does not certify....35
III.5.2 What "independent" means in this thesis, and what it does not36
III.6 Statistical Protocol....38
III.6.1 Tests, corrections and intervals....38
III.6.2 What is shared and what is independent across a comparison38
III.6.3 Decision bands and negative controls....39
III.6.4 A note on the wording of null results....40
III.6.5 Units of analysis....41
III.6.6 Target-clustered statistics for predictor comparisons....42
III.7 Computing Environment and Engineering....43
III.8 Integrity Checks....43
IV. AFFINITY PREDICTOR QUALITY, THE SPLIT ARTEFACT AND BASELINE GUIDANCE ABLATIONS....45
IV.1 The Original Affinity Predictor....45
IV.1.1 Training set-up....45
IV.1.2 Held-out quality....46
IV.2 Leakage and Coverage Audit....47
IV.3 Hyper-Parameter Sweep and Diagnostics on the Old Split....48
IV.3.1 Learning rate and depth....48
IV.3.2 Five diagnostics....49
IV.4 The Split Artefact: Leakage-Safe Re-Evaluation....50
IV.4.1 Construction of the split....50
IV.4.2 Result....51
IV.5 Baseline Guidance Ablations....53
IV.5.1 Task F: the original ablation....53
IV.5.2 Strength re-sweep with the original predictor....55
IV.5.3 What a range of λ does and does not cover....55
IV.5.4 Strength re-sweep with the repaired predictor....56
IV.6 A Fuller Information-Content Audit: What Representation Predicts Affinity?....57
IV.6.1 The representation ladder....58
IV.6.2 Ligand size, not interaction identity, is the dominant signal. 60
IV.6.3 Structure rarely beats ligand-only, after correction....61
IV.7 Does More Training Data Close the Gap? Augmenting with BindingNet62
IV.7.1 A compound-level leak, found and removed before it could inflate a result....62

IV.7.2 Arm A versus Arm C: pure CrossDocked outperforms BindingNet augmentation at matched size....63
IV.8 How Much of the Predictor's Performance Is a Class-Level Prior?....65
IV.8.1 Two baselines under within-target permutation....65
IV.8.2 Consequence: a variance decomposition and a narrower window than it first appears....67
IV.9 Where Does the Predictor Fail? Activity Cliffs, Matched Pairs and Protein Families....69
IV.9.1 Activity cliffs....69
IV.9.2 Matched molecular pairs....70
IV.9.3 Protein-family stratification....71
IV.10 Summary of the Chapter....71
V. ARCHITECTURE AND MECHANISM ROUTES: TRACKS A AND B, AND THE GRADIENT DIAGNOSTIC....74
V.1 Decision Rule for the Two Routes....74
V.2 Track A: A Physics-Anchored Predictor....75
V.2.1 Architecture and training....75
V.2.2 Predictive quality....76
V.2.3 Exploratory tier....77
V.2.4 Full tier....79
V.3 Track B: Three Mechanism Variants....81
V.3.1 B3: timestep-windowed guidance....81
V.3.2 B1: gradient-norm normalisation....82
V.3.3 B2: classifier-guidance reformulation....83
V.4 Diagnostic of the Gradient Direction....85
V.4.1 Method....85
V.4.2 Results....85
V.4.3 Interpretation and its limits....86
V.5 Conclusions of Tracks A and B....90
VI. REJECTION SAMPLING: THE AFFINITY PREDICTOR AS A POST-HOC RANKER (TRACK C)....91
VI.1 Design....91
VI.1.1 Pools, ranker and evaluation....91
VI.1.2 Statistics....92
VI.2 Results....93
VI.2.1 Raw docking score....93
VI.2.2 Ligand efficiency, size and validity....94
VI.2.3 Two pockets with the opposite behaviour....97
VI.2.4 Sensitivity to the selection fraction....97
VI.3 Post-Hoc Analysis of the Validity Loss....98
VI.4 Interpretation....99

VI.5 A Positive Control: Reranking by a Genuine, Non-Surrogate Affinity Signal....100
VI.5.1 Design....101
VI.5.2 Result....102
VI.5.3 The confound, disclosed directly....103
VI.5.4 Interpretation: what this isolates....103
VII. SYNTHESIZABILITY GUIDANCE (TRACK D)....106
VII.1 Target, Predictor and Guidance Design....106
VII.2 The Original Design: Divergence between the Own Score and the Independently Computed Score....107
VII.3 Re-Examination: Convergence and Leakage of the Synthesizability Predictor....108
VII.4 Guidance with the Leakage-Safe Predictor....109
VII.5 Diagnostic of the Synthesizability Gradient....113
VII.6 Conclusion of Track D....114
VIII. REDESIGN OF THE TRAINING TARGET: DELTA-LEARNING AGAINST THE DOCKING SCORE (TRACK E)....115
VIII.1 Rationale and Premise Checks....115
VIII.1.1 The two modifications....115
VIII.1.2 Premise checks that changed the plan....116
VIII.2 Design....118
VIII.2.1 Architecture and target....118
VIII.2.2 Anchor and its semantics....119
VIII.2.3 The auxiliary head and label reliability (E2, designed)....120
VIII.3 Pre-Registration....122
VIII.3.1 Arms, metrics and decision rules....122
VIII.3.2 Power, disclosed in advance and recomputed....124
VIII.4 Infrastructure and Verification....124
VIII.5 Results of the First Stage....126
VIII.5.1 Predictive quality....126
VIII.5.2 Pre-registered criteria....128
VIII.6 Why the Delta Target Failed....129
VIII.7 Limits of the Conclusion....132
VIII.8 Position in the Literature....132
VIII.9 Summary of the Chapter....133
IX. SYNTHESIS AND DISCUSSION....134
IX.1 Summary of the Evidence....134
IX.2 What the Evidence Supports....138
IX.2.1 Main finding....138
IX.2.2 Predictor quality is not guidance usefulness....139
IX.2.3 Grading the explanations....140

IX.2.4 The role of statistical power....143
IX.3 Methodological Findings....144
IX.4 Threats to Validity....147
IX.4.1 Internal validity....147
IX.4.2 External validity....148
IX.4.3 Construct validity....149
IX.5 Relation to the Literature....149
IX.6 Implications and Recommendations....151
X. CONCLUSIONS AND RECOMMENDATIONS....153
X.1 Conclusions....153
X.2 Contributions....158
X.3 Limitations....160
X.4 Recommendations for Future Work....162
X.4.1 A proposed extension: uncertainty-aware acquisition (not tested)....163
REFERENCES....165
APPENDIX A. REPRODUCIBILITY AND REPOSITORY....172
A.1 Hardware and Software....172
A.2 Exact Identifiers for Reproduction....174
A.3 Data Provenance and What a Reproducer Needs....175
A.4 Test-Set Isolation, Per Training Run....176
A.5 Reproducibility Checklist....178
A.6 Repository Layout....179
A.7 Reproducing the Results and This Document....180
APPENDIX B. LITERATURE SEARCH AND VERIFICATION....181
B.1 Search Strategy....181
B.2 Verification and Its Limits....181
APPENDIX C. TRACK A FULL-TIER RESULTS: ALL COMPARISONS....184
C.1 Per-Pocket, Per-Strength Results....184
APPENDIX D. TRACK E: PRE-REGISTRATION AMENDMENTS....186
D.1 Summary of the Amendments....186
APPENDIX E. POST-SYNTHESIS FOLLOW-UP: TESTING THE DIAGNOSED MECHANISM DIRECTLY....188
E.1 Scope and Relation to the Main Synthesis....188
E.2 Phase 1: Deepening the Pattern on the Predictor Itself....190
E.2.1 Noise-matched and Vina-derived training recipes....191
E.2.2 Gradient-alignment: optimising the diagnosed correlation directly....192
E.2.3 A global protein-language-model feature (ESM2)....194
E.2.4 Kitchen-sink: all four interventions combined....198

E.3 Phase 2: Independent Validation and the Hardest Generalisation Test 199
E.3.1 Does guidance change chemically meaningful interactions, not just the raw score? (PLIP).....199
E.3.2 Generalisation to a protein family never seen in training….201
E.3.3 A side-finding: AlphaFold-predicted pockets as a robustness check.....203
E.4 Phase 3: Sharpening the Representation and an Independent Label Check.....206
E.4.1 A pocket-restricted protein-language-model feature.....206
E.4.2 Independent cross-validation against ChEMBL.....208
E.5 Direct Mechanistic Trace of the Guidance Nudge.....210
E.5.1 Method.....211
E.5.2 Result: the gap grows smoothly and ends two orders of magnitude below the raw nudge sum.....211
E.5.3 A puzzle checked, not glossed over.....213
E.5.4 Scope and what would have rejected this hypothesis.....215
E.6 Pooled Statistical Correction Across the Entire Follow-up Phase.....217
E.7 What This Appendix Adds to the Thesis's Central Claim.....219

LIST OF TABLES

TABLE		Page
1.1	Research objective, experiment, metric, result and conclusion	7
2.1	Relation of the closest prior work to the experiments of this thesis	21
3.1	Where a decision gate was applied, by route	27
3.2	Master summary of every experimental route, including the Appendix E follow-up	28
3.3	Data splits used for affinity predictors	31
3.4	Predictors used as guidance signals or ranking functions	34
3.5	Independent evaluation metrics	37
3.6	Unit of analysis and seed count, by route	42
4.1	Old split: validation and test quality of the affinity EGNN at selected epochs	46
4.2	Old split: learning-rate and depth sweep (validation n = 4,061, test n = 27 pockets)	49
4.3	Old split: diagnostics of the generalisation gap	49
4.4	Stage 0 checkpoint on the leakage-safe split (test n = 11,855)	51
4.5	Task F: unguided versus affinity-guided sampling (λ = 1.0), 20 pockets	54
4.6	Repaired predictor: single-pocket strength sweep (n = 8 per value)	57
4.7	Representation ladder: held-out R^2 of each information source (test, 127 target clusters, B = 2,000)	59
4.8	Surrogate predictor, training-data composition (3 seeds per arm, validation R^2, B = 2,000)	63
4.9	Variance decomposition of apparent predictive performance	67
5.1	Track A: predictive quality on the leakage-safe test set (n = 11,855)	77
5.2	Track A exploratory tier: 9 comparisons with the unguided baseline	78
5.3	Track A full tier: summary over eight pockets by guidance strength	79
5.4	Track B: summary of the three mechanism variants (Stage 0 EGNN, 2 to 3 pockets, n = 20 per condition)	84
6.1	Track C: top 10% selection by the Stage 0 ranker, per pocket	94
6.2	Reranking by a genuine Vina signal versus the unguided baseline, 20 pockets × 3 seeds (n = 597 baseline, 600 reranked molecules)	102
7.1	Original (placeholder) design, one pocket, n = 8 per value	107
7.2	SynthPredNet: original and leakage-safe evaluation	109
7.3	Leakage-safe predictor, bond-aware design, pocket BSD_ASPTE, n = 30 per value	110
7.4	Track D diagnostic: gradient magnitude against three atom-level proxies (n = 144 states)	113

8.1 Track E: reference predictors on the leakage-safe test set (127 targets; target-clustered 95% CI)....117
8.2 Track E, first stage: test-set results (127 targets; seed-averaged; target-clustered 95% CI)....126
8.3 Pre-registered criteria of the first stage, applied to E1 (seed-averaged; difference = E1 minus reference)....128
9.1 Summary of the experimental routes....134
9.2 Claim, evidence and scope, Tracks A to E....136
A.1 Measured training budget of the six production surrogate arms (Chapter IV.7)....173
A.2 Exact package versions of the `drugdisc` environment, as actually installed....174
A.3 Checksums and identifiers of the principal frozen artefacts....174
A.4 Provenance of the derived data artefacts, with the script that generates each....175
A.5 Test-set isolation by training run, with the evidence for each claim….177
A.6 Reproducibility checklist, marked only where actually verified....178
A.7 Principal elements of the repository....179
C.1 Track A full tier: all 24 comparisons with the unguided baseline....184
D.1 Track E amendments....186
E.1 Test-set predictive quality (leakage-safe split, Section III.3) of every checkpoint trained in the follow-up investigation....191
E.2 DIAG1 diagnostic (Section V.4 protocol) across every checkpoint of the follow-up investigation....193
E.3 ESM2 checkpoint, exploratory (n=8) versus confirmatory (n=24) independently computed docking test, paired by seed....195
E.4 AlphaFold-predicted versus crystal pocket geometry, full-scale result (n=835)....204
E.5 Independent cross-validation of this project's affinity labels against ChEMBL....208
E.6 Mean all-atom position gap between the guided and unguided trajectory, by diffusion step....211
E.7 Sources pooled into the single Benjamini–Hochberg correction across the follow-up phase....217

LIST OF FIGURES

FIGURE Page

3.1 Overall pipeline. A frozen TargetDiff denoiser proposes a clean-data estimate at each reverse step; a guidance predictor's gradient with respect to atom positions modifies that estimate; and the finished molecules are evaluated by tools that are independent of the guidance predictor and analysed with corrected statistics.....24

3.2 Decision logic applied to each comparison. The null-consistency check guards against implementation errors; the ligand-efficiency, heavy-atom and PoseBusters checks are applied to every apparent positive before it may be called a clear signal.....40

4.1 Test R^2 of the affinity predictor on the old split (n = 27 pockets) and on the leakage-safe split (n = 11,855 entries). Ligand-only random forests are shown in grey. Error bars are bootstrap 95% confidence intervals where available.....52

4.2 Test Pearson correlation of every guidance predictor trained in this thesis on the leakage-safe splits. The synthesizability predictor evaluated on its original random split (orange), which had no target-level separation and no test set, was superseded by the leakage-safe evaluation (green; Chapter VII).....53

5.1 Track A, full tier (8 pockets, 29–30 molecules per condition, 24 comparisons corrected together with Benjamini–Hochberg). (A) Mean shift of the independent Vina Dock score for each pocket and guidance strength (negative is better; no shift was significant after correction). (B) PoseBusters pass rate of the unguided samples ($\lambda = 0$) and of the guided samples, for each pocket (grey) and on average (red).....80

5.2 Track B: median shift of the independent Vina Dock score for every comparison (positive is worse). Open symbols mark comparisons in which the PoseBusters pass rate decreased. None of the 23 comparisons was significant after correction. Medians are shown because a single +286.58 kcal/mol outlier dominates the mean of one B1 and one B2 comparison.....84

5.3 Correlation of the per-atom gradient magnitude of the Stage 0 predictor with the heaviness of the atom type and with its distance to the pocket, over 144 captured states (3 pockets, 3 timesteps, 16 samples). Points are means of per-sample Pearson correlations; bars are bootstrap 95% confidence intervals.....86

6.1 Track C, top 10% selection by the Stage 0 ranker in 15 pockets. (A) Shift of the raw Vina Dock score with 95% bootstrap intervals; (B) shift of ligand efficiency (red: worse, green: better); (C) drop of the PoseBusters pass rate in percentage points.....96

6.2 Three top-ranked molecules from Track C pools (SQHC_ALIAD, XANLY_BACGL, CD38_HUMAN), re-docked and rendered with

hydrogen-bond partner residues labelled. The large, high-affinity-scoring poses illustrate the kind of molecule the ranker favours before the size normalisation of Figure 6.1 is applied.....97

7.1 Track D. (A) Original design (placeholder features, n = 8, one pocket): the predictor's own score and the independently computed RA-score as a function of guidance strength. (B) Leakage-safe predictor with bond-aware features (n = 30, same pocket): shift of the independently computed RA-score with 95% bootstrap intervals; grey points rest on 6 and 2 scoreable molecules.....111

8.1 Design of Track E. A shared GIGN encoder feeds the E1 delta head (physics-sum readout with learned bias and free-sign scale, primary) and a scalar-MLP readout (ablation); the prediction is reconstructed as the Vina anchor plus the predicted correction. The E2 auxiliary head with PLIP labels was designed but not run.....119

8.2 The `vina` column of the annotation file against Vina 1.2.6 scores recomputed with this project's preparation for 100 reference poses. (A) Score-only evaluation; (B) local minimisation. The dashed line is the identity.....120

8.3 Stability of PLIP interaction labels under a 0.2 Å Gaussian perturbation of the ligand coordinates (100 complexes): (A) agreement at the level of atoms (Jaccard index) and (B) agreement at the level of complexes (presence of at least one interaction of the class).....121

8.4 Test R^2 of the Track E arms and the reference predictors on 127 held-out targets. Bars: target-clustered 95% confidence intervals; ticks: individual seeds; dotted and dashed vertical lines: the Stage 0 EGNN and the linear Vina + heavy-atom baseline.....127

8.5 Mean within-target Spearman correlation over the 73 test targets with at least 10 complexes (95% bootstrap intervals). All arms overlap with raw Vina.....127

8.6 Best validation error (standardised units; 1.0 corresponds to the variance of the target) of the three arms for each seed. Dashed lines: error of a constant predictor. The absolute target is learnt; the delta target is not learnt better than by a constant.....131

9.1 Fraction of comparisons significant after Benjamini–Hochberg correction, for every checkpoint that used the KS-test protocol of Section III.6.1. Counts in parentheses are the number of comparisons in that checkpoint. This fraction is a measure of statistical significance, not of successful molecular design: the large surviving fraction corresponds to the Track C ranker, whose raw docking improvement did not survive the size-normalised evaluation of Chapter VI and was accompanied by reduced physical validity.....138

9.2 Minimum Vina Dock shift detectable at 80% power as a function of the number of molecules per condition, computed from the two-sample normal-approximation formula at the range of within-pocket standard deviations measured in this project (1.1 to 1.6 kcal/mol), at the nominal

5% level and at a Bonferroni-equivalent level for 24 simultaneous tests. Vertical lines mark the sample sizes actually used in Track B (20) and in Track A's full tier (30); the Task F ablation used 10 molecules per pocket per seed, which pools to about 30 per pocket across its three seeds and is shown at the same mark.....144

9.3 Number of pockets or targets against the number of molecules or complexes evaluated per pocket or condition, for every checkpoint of Chapters IV to VIII (logarithmic scales). The pocket/molecule counts are those already stated in the corresponding chapter.....148

E.1 Direction consistency (fraction of comparisons favouring guidance, by median shift) for all five original thesis tracks and all seven distinct post-thesis follow-up branches, with sample size and, where greater than zero, the percentage significant after correction annotated per route. The one open marker (Track C ranker) flags the route independently shown to be confounded by molecular size (Chapter VI), not a route this appendix's checkpoints fall short of.....190

E.2 Test-set predictive quality (R^2, Pearson, Spearman) of all eight checkpoints trained in the follow-up investigation (Table E.1 lists them), read from each checkpoint's own training log. ESM2 (whole-protein) is the best of any variant; ESM2 (pocket-only, Section E.4.1) is the worst, including below the original checkpoint.....195

E.3 Vina Dock shift under guidance for the ESM2 checkpoint, exploratory tier (n=8, dashed) against the confirmatory tier (n=24, solid), by guidance strength λ. The effect size shrinks and a validity cost (loss of single-fragment rate) appears at the larger sample that was invisible at n=8, the same pattern documented for the soft-v relaxation result of Section V.2.....197

E.4 The diagnostic (distance-to-pocket gradient correlation, Table E.2) against the independently computed docking outcome, for every checkpoint of this appendix. The checkpoint with the strongest, most directly optimised diagnostic fix (gradient-alignment) has the flattest docking result; the checkpoint with the best incidental diagnostic (ESM2) has the closest approach to a corrected effect, which still does not survive correction. Diagnostic quality and the independently computed docking outcome dissociate across the full range tested.....197

E.5 PLIP-detected interaction-count shift under guidance on the ESM2 checkpoint, exploratory tier (n=8) against confirmatory tier (n=16, new seed). The sign of the effect reverses, not merely weakens, between the two tiers.....201

E.6 Test-set predictive quality of the Vina-target recipe evaluated on its ordinary, same-distribution test set against the same recipe trained with the entire kinase (PF00069) family held out and evaluated only on that family. Quality does not collapse on the unseen family.....203

E.7 Pocket RMSD between AlphaFold-predicted and crystal structures against pocket-local pLDDT confidence, for 835 leakage-safe-split

targets. The strong negative correlation (Spearman −0.686) supports pLDDT as an average-case accuracy guide; the points in the lower right are confident predictions that are nonetheless substantially wrong at the binding site.....205

E.8 Pocket-local pLDDT against whole-protein pLDDT for the same 835 targets. Pocket confidence exceeds whole-protein confidence by 7.4 points on average, consistent with binding pockets lying in well-ordered regions, but the gap does not eliminate the overconfident minority of Figure E.7.....206

E.9 This project's affinity-label range against the independently curated ChEMBL Ki range, for the four targets used repeatedly throughout this appendix. Three of four targets overlap; CD38 does not.....210

E.10 Mean all-atom position gap between the guided (λ=1.0) and unguided (λ=0) trajectory, by reverse-diffusion step, for the paired deterministic trace of Section E.5. The gap is exactly zero until guidance is first applied, grows smoothly, and only accelerates in the final third of sampling.....212

E.11 2D structures of the two molecules of the paired trace: identical starting noise and identical stochastic draws, diverging only through the guidance nudge. The geometric divergence is at most 0.13 Å per atom, yet the discrete atom-type channel produces two chemically distinct, valid molecules.....214

E.12 The unguided (λ=0) molecule of the paired trace, docked into its pocket (PDB 1h36) and rendered with its hydrogen-bond partners. Renders the same molecule referenced in Figure E.11 and Table E.6 in its 3D binding context.....214

E.13 The guided (λ=1.0) molecule of the same paired trace, from bit-identical starting noise, docked into the same pocket. The two renders answer directly what happens to one generated molecule when the affinity gradient is applied from one diffusion step to the next: a smoothly growing, late-accelerating positional nudge that ends sub-Ångström, alongside a discrete identity change the positional trace alone does not explain.....215

E.14 Effect size and 95% confidence interval of every one of the 48 pooled comparisons of Table E.7, ordered by checkpoint. No interval is both significant after correction and in the beneficial direction.....218

LIST OF ABBREVIATIONS

AMP	Automatic mixed precision
BH	Benjamini–Hochberg (false-discovery-rate correction)
CI	Confidence interval
DDPM	Denoising diffusion probabilistic model
EGNN	E(n)-equivariant graph neural network
GIGN	Geometric interaction graph neural network
HA	Heavy-atom count
KS	Kolmogorov–Smirnov (test)
LE	Ligand efficiency (Vina Dock score per heavy atom)
LMDB	Lightning memory-mapped database
LP split	Leakage-proof (target-disjoint) data split
MSE	Mean squared error
PB	PoseBusters physical-validity check
PLIP	Protein–ligand interaction profiler
pK	Negative decadic logarithm of an affinity constant (Ki, Kd or IC50)
RA-score	Retrosynthetic accessibility score
SA score	Synthetic accessibility score
SBDD	Structure-based drug design
Vina	AutoDock Vina docking program and scoring function
$\hat{x}_0$	Clean-data estimate at a diffusion step

CHAPTER I

INTRODUCTION

I.1 Background

Structure-based drug design (SBDD) seeks small molecules that bind a protein of known three-dimensional structure. Classically, candidates are proposed by medicinal chemists or drawn from large compound libraries, placed in the binding pocket by docking programs, and ranked by scoring functions such as the empirical function of AutoDock Vina [10], [50]. In recent years, deep generative models have been added to this workflow. Pocket-conditioned three-dimensional diffusion models, of which TargetDiff is a representative example [20], learn the joint distribution of atom types and coordinates of ligands inside a pocket and can produce new molecules directly in the binding site, following the general principle of denoising diffusion probabilistic models [21].

A generative model that reproduces the training distribution is not, by itself, a drug-design tool: most sampled molecules bind poorly, cannot be synthesized, or are physically implausible. A natural remedy, borrowed from image generation, is guidance. In classifier guidance, the gradient of an auxiliary predictor is used to steer each denoising step toward inputs that the predictor scores highly [9]. For drug design, the auxiliary predictors are learned models of binding affinity and of synthesizability, and the appeal is obvious: the generator remains fixed, and only a small, cheaply trained predictor must be supplied. This idea motivated the original

formulation of this thesis, the *coupled* use of an affinity signal and a synthesizability signal to steer a target-conditional diffusion model.

I.2 Problem Statement and Importance

The central question of this thesis is whether learned guidance signals improve generated molecules on independently computed evaluation outcomes, that is, on quantities that a medicinal chemist would recognise as meaningful and that are computed independently of the guidance model itself. The question is important for three reasons. First, published evaluations of generative SBDD methods rely heavily on docking scores, and it has been reported that docking scores can be inflated simply by making molecules larger or by adding particular functional groups, without a corresponding gain in binding [18], [57]. A guidance signal that raises such a score is therefore not automatically a success. Second, learned affinity predictors are known to be sensitive to data leakage between training and test sets, and to exploit ligand-only shortcuts [19], [34], [53]; a predictor that looks accurate on a leaky benchmark may be useless for a new target. Third, negative and null results about widely used techniques are rarely reported, although they determine where a field should invest its effort.

The problem can therefore be stated precisely: given a frozen, pretrained pocket-conditioned diffusion model and learned predictors of affinity and synthesizability that are accurate on leakage-safe held-out targets, does gradient-based guidance, or post-hoc selection with the same predictors, produce statistically detectable and non-degenerate improvements in independently computed docking score, physical validity, ligand efficiency and synthesizability?

I.3 Related Work

Four bodies of work are most closely related. The first concerns diffusion models for SBDD and their guidance or conditioning by affinity or interaction information: interaction-conditioned generators [32], [56], [65], diffusion models that inject priors derived from affinity networks [23], a general binding-affinity guidance framework whose classifier-guidance variant applies a gradient-based affinity signal to a diffusion SBDD generator in a plug-and-play fashion and reports up to a 60% improvement in binding affinity [24], and a target-conditioned diffusion model that integrates bond diffusion with affinity and drug-likeness guidance trained jointly with the generator [22]. The second concerns learned scoring functions and the reliability of their evaluation, including physics-informed and interaction-explicit architectures [38], [59], leakage-controlled benchmarks [19], [34] and critiques of shortcut learning [45], [53]. The third concerns the evaluation of generative SBDD methods themselves, where independent work has questioned the reliability of docking scores as a success criterion and the practice of selecting checkpoints on test data [18]. The fourth, directly relevant to the positive control of Chapter VI, is best-of-N and best-of-K selection applied to diffusion SBDD generation itself, which BoKDiff reports as a novel generation technique in its own right, ranking an oversampled pool by a combined QED, SA and Vina score [29]; this thesis reuses the same mechanism for a different purpose, as a signal-swapped positive control rather than as a proposed generation method (Section VI.5). Chapter II reviews these areas in detail and identifies the specific gap addressed here: to the best of the author's knowledge, no published evaluation of learned guidance for a 3D diffusion SBDD generator, including the gradient-guidance study above, reports a leakage-safe, target-level split

together with statistically corrected significance testing, an explicit positive control built from a genuine, non-surrogate signal, and the quality-degradation checks used in this thesis.

I.4 Solution Employed and Novelty

Rather than proposing a single new guidance method, this thesis employs a *falsification design*. Guidance was implemented on the clean-data estimate of a frozen TargetDiff model, and four mechanistically distinct routes were then tested against the same evaluation harness: in-loop gradient guidance with two predictor architectures and three mechanism variants (Chapters IV and V), non-gradient rejection sampling with the same affinity predictor as a ranker (Chapter VI), and gradient guidance toward synthesizability (Chapter VII). Each route was assigned pre-declared decision bands, corrected for multiple comparisons, and accompanied by quality-degradation checks (ligand efficiency, molecular size and PoseBusters validity) and negative controls. Two diagnostics were used to ask *why* a route failed, not only whether it did.

A fifth route, described in Chapter VIII, is the redesign of the predictor's training target as a residual against the docking score (delta-learning), which was pre-registered, implemented on a resumable training infrastructure with unit tests and a test-set-selection audit, and evaluated with target-clustered statistics. A sixth and decisive addition, reported in Chapter VI.5, is a *positive control*: the same rejection-sampling mechanism of Chapter VI, applied with AutoDock Vina's own zero-shot score in place of the trained surrogate, on the identical generator, pockets and independent evaluation protocol. The novelty of the work lies in (i) the leakage-safe, target-level evaluation of predictors used for guidance; (ii) the systematic, statistically disciplined mapping of where gradient guidance and rejection sampling fail; (iii) the

mechanistic diagnostics, extended in Chapter IV to a full information-content audit (an eight-block representation ladder, a within-target class-prior decomposition run for the deployed architecture itself, and failure diagnostics by activity cliff, matched molecular pair and protein family) and in Chapter VI.5 to a genuine-signal positive control that localises the guidance failure to the signal rather than the mechanism; and (iv) the methodological audits that accompany the results, including the recognition that an earlier generalisation gap was largely a split artefact, and a compound-level data-leakage audit that preceded, rather than followed, a production training-data-composition experiment.

I.5 Summary of Results

None of the five guidance routes produced a defensible improvement, and the information-content audit of Chapter IV shows why the two most carefully tested routes could not have been expected to. Affinity guidance produced no detectable effect on the independently computed docking score at any tested strength, and the magnitude of the predictor's gradient was positively correlated with the distance of an atom from the pocket, not with proximity to binding contacts. Rejection sampling with the trained surrogate improved raw docking scores in 12 of 15 pockets, but ligand efficiency worsened significantly in 14 of 15, the selected molecules were on average 12 heavy atoms larger, and PoseBusters validity fell by 29 percentage points. A positive control that replaced the trained surrogate with Vina's own zero-shot score, otherwise identical in mechanism, pockets and evaluation protocol, *did* produce a significant, largely size-independent improvement in independently computed docking score (Wilcoxon $p = 1.9 \times 10^{-6}$; ligand efficiency flat, $p = 0.756$), directly validating the evaluation harness and localising the preceding two failures to the

quality of the surrogate signal rather than to the selection or gradient mechanism. The information-content audit behind this localisation found that no representation of the binding pocket tested in this thesis, including a chemically typed interaction representation, reliably outperformed a ligand-only baseline after correction for multiple comparisons, and that 70 to 90% of the predictor's apparent performance, depending on architecture, is reproducible from a class-level prior with no within-target structure–activity signal at all. Synthesizability guidance gave no supported gain: in the exploratory design the guidance model's own score rose while the independently computed RA-score fell, and after two fixes this divergence was still present in sign but was not statistically confirmed. Finally, the delta-target predictor was significantly worse than simple baselines, because the residual was not learnt better than by a constant, and a dedicated attempt to close the gap by augmenting training data with BindingNet, after removing a compound-level leak found in the combined pool, did not outperform training on CrossDocked alone at matched size. These findings are reported as legitimate scientific outcomes, together with the limits within which they hold.

I.6 Research Objectives and Experimental Map

The problem statement of Section I.2 is one question; Table 1.1 unpacks it into the specific sub-objective that each route was designed to answer, the metric that decided it, and the conclusion actually drawn. Every row maps to exactly one chapter, so this table also serves as a map of the thesis.

Table 1.1 Research objective, experiment, metric, result and conclusion

Research objective	Experiment / Track	Primary metric	Result	Conclusion
Does gradient guidance with the original predictor affect the independently computed docking score?	Baseline ablation (IV)	Vina Dock (Mann–Whitney, Wilcoxon)	No effect at any of 3 orders of magnitude of λ	Predictor repair alone does not produce an effect
What information does the predictor actually use, and does a chemically typed interaction representation help?	Representation ladder (IV.6)	Target-clustered R^2, BH-corrected pairwise tests	12/40 wins for structure, 0 for ligand-only; typed PLIP interactions collapse to a size proxy (R^2 0.08 → 0.32 after adding one size covariate)	No representation of the pocket tested reliably beats a ligand-only baseline
Does augmenting training data with a larger external affinity data set (BindingNet) close the gap to ligand-only?	Arm A vs. Arm C (IV.7)	Validation R^2 (3 seeds/arm)	Arm A (pure CrossDocked, matched size) 0.362 vs. Arm C (BindingNet-augmented, leakage-filtered) 0.272	More or different training data did not close the gap at matched size
How much of the apparent performance is a class-level prior rather than within-target structure–activity learning?	Within-target y-randomisation (IV.8)	R^2 under label permutation, descriptor pipeline and the deployed EGNN	70% (descriptor) and 90% (EGNN) of real-label performance reproduced with no within-target signal	The resolvable window beyond the class prior is narrower than one confidence interval
Does the predictor's architecture limit gradient guidance?	Track A (V)	Vina Dock (KS, BH)	0/24 significant; direction consistency weakened with more data	Architecture is not the limiting factor at the scale tested
Does the guidance mechanism (scale, objective, timing) limit it?	Track B (V)	Vina Dock (KS, BH)	0/23 significant; one variant worse by median in 6/6	Mechanism variants do not rescue guidance on a fixed representation

Research objective	Experiment / Track	Primary metric	Result	Conclusion
What does the affinity gradient actually encode?	Diagnostic (V.4)	Gradient-magnitude/ distance correlation; signed projection toward the nearest pocket contact	Magnitude correlates +0.240 with distance and is not explained by atom heaviness; signed direction gives mean cosine −0.019, 95% CI [−0.046, +0.008]	Gradient magnitude increases with distance from the pocket, but its direction is indistinguishable from random rather than inverted
Can non-gradient, post-hoc selection succeed where gradient guidance failed?	Track C (VI)	Vina Dock, ligand efficiency, PoseBusters	Raw gain in 12/15 pockets, confounded by size in 14/15	The apparent docking gain does not survive size-normalised evaluation and is accompanied by reduced physical validity
Does a genuine, non-surrogate signal succeed with the identical selection mechanism, generator and pockets?	Positive control (VI.5)	Vina Dock (Wilcoxon, target-clustered bootstrap), ligand efficiency, SA/RA-score	Significant improvement ($p = 1.9 \times 10^{-6}$); ligand efficiency flat ($p = 0.756$); size and synthesizability both worsen significantly	Harness validated; the guidance failure localises to the signal, not the selection mechanism
Does the same guidance mechanism work for synthesizability?	Track D (VII)	Independently computed RAscore (KS, BH)	0/5 significant at the powered scale; own-score/independent-score divergence not confirmed	No supported gain; the mechanism of the divergence is undiagnosed
Does a residual (delta) training target improve the predictor?	Track E (VIII)	Target-clustered R^2	$R^2 = 0.117$, worse than a linear baseline (0.362) and an EGNN (0.34–0.41)	Falsified at this scale; the auxiliary arm (E2) was not run
Does directly repairing the diagnosed gradient mechanism produce an effect?	Appendix E, Phase 1	Vina Dock (Wilcoxon); DIAG1 correlation	Strongest diagnostic fix of any checkpoint (−0.78); still 0/35 pooled tests significant	Repairing the diagnosed mechanism does not, by itself, produce an effect

Research objective	Experiment / Track	Primary metric	Result	Conclusion
Does the null survive independent, non-docking validation and the hardest generalisation split available?	Appendix E, Phase 2	PLIP interaction counts; R^2 on an unseen protein family	Interaction-count signal reverses sign on confirmation; predictive quality holds (R^2 = 0.507) on an unseen kinase family, docking effect still null	The null is not an artefact of a docking-only metric or of family-level leakage
What happens, mechanistically, to a single molecule when the gradient is applied?	Appendix E, Section E.5	Per-step, per-atom position gap	Gap ≈ 0.05 Å mean after 1,000 steps, ≈1% of the raw cumulative nudge	Consistent with the nudge being overwhelmed by the sampler's own step noise for most of the trajectory, rather than actively cancelled

I.7 Organization of the Thesis

Chapter II reviews the literature. Chapter III describes the data, models, guidance mechanism, evaluation metrics, statistical protocol and integrity checks. Chapter IV reports predictor quality, the discovery of the split artefact, the baseline guidance ablations, a fuller information-content audit of what the predictor reads, a training-data-composition experiment with BindingNet, and a class-prior decomposition and failure analysis by activity cliff, matched molecular pair and protein family. Chapter V reports the architecture and mechanism routes (Tracks A and B) and the gradient diagnostic. Chapter VI reports rejection sampling (Track C) and the positive control that replaces the trained surrogate with a genuine, non-surrogate affinity signal; Chapter VII reports synthesizability guidance (Track D) and Chapter VIII the delta-learning redesign (Track E). Chapter IX synthesises and discusses the findings across tracks, and Chapter X concludes and outlines future work.

CHAPTER II

LITERATURE REVIEW

This chapter reviews the literature on which the experimental chapters build. It proceeds from the generative models that are guided, through the predictors that supply the guidance signal and the pitfalls in evaluating both, to the two methodological ideas (residual learning and interaction supervision) that motivate the final experiment. Throughout, statements are limited to what the cited sources report. Where a body of work could not be examined in full, this is stated in the text.

II.1 Deep Generative Models for Structure-Based Drug Design

The generation of ligands conditioned on a protein pocket has been approached with several model families. Early voxel-grid and autoregressive atom-placement models were followed by non-autoregressive models based on denoising diffusion [21], which learn to reverse a gradual noising process. TargetDiff [20] applies an equivariant diffusion model to the joint distribution of ligand atom coordinates and atom types conditioned on a protein pocket, and additionally uses the learned representation for affinity prediction. It is trained on the CrossDocked2020 data set [15], which contains 22.5 million poses of ligands docked into multiple similar binding pockets of the Protein Data Bank. Because CrossDocked2020 was produced by cross-docking, the poses are docking outputs rather than experimental complexes, a property whose consequences for evaluation are discussed in Section II.4.

Equivariant graph networks [44] provide the architectural basis for such models: node features are updated from invariant quantities such as distances, and coordinates are updated along relative position vectors, so that rotating or translating the complex rotates or translates the output. The same family of networks is used in this thesis for the affinity predictor that supplies guidance (Chapter III).

Recent work has attempted to inject interaction knowledge into 3D generators. DeepICL conditions an autoregressive generator on atom-level interaction labels of four types (hydrogen bond, salt bridge, hydrophobic and π-stacking) extracted from reference structures with the protein–ligand interaction profiler PLIP during training, and reports improved docking scores against a baseline without interaction information [65]. InterDiff conditions a diffusion model on interaction "prompts" detected with BINANA, uses cross-attention to inject them, and adds a cross-entropy loss that classifies protein-atom features into interaction types inside the denoiser [56]. BInD treats non-covalent interactions extracted with PLIP as an additional categorical channel that is generated jointly with atoms and bonds under its own loss, and applies knowledge-based guidance terms after the network prediction [32]. An earlier recurrent generator was constrained by an interaction fingerprint computed on docking poses [63]. DPDiff, a diffusion model on CrossDocked2020, extracts "interaction priors" from two pretrained affinity-prediction networks during sampling, using intermediate diffusion predictions as input [23]. Its ablation reports that a geometry-based prior that was pretrained on accurate poses degraded generation when queried on noisy intermediate structures, because of the mismatch between training and sampling inputs [23]. This observation is directly relevant to guidance and is returned to in Chapters V and IX.

II.2 Guidance of Diffusion Models

Classifier guidance modifies the reverse process of a diffusion model with the gradient of the log-probability of a target class under an auxiliary classifier evaluated on noisy inputs, thereby trading diversity for fidelity to the class [9]. In molecular generation, the same principle has been applied with regressors of properties such as affinity. Two design decisions distinguish implementations. First, the guidance predictor may be trained on noisy inputs at every diffusion time, or on clean structures and then evaluated on the model's clean-data estimate. The latter avoids retraining and is cheaper, but the predictor then receives inputs that differ from its training distribution, particularly at early, noisy steps. Second, the guidance may act on continuous coordinates only, or also on discrete atom types; discrete types require a differentiable relaxation and, for predictors that use bond-derived features, an estimate of connectivity that does not exist during sampling. Both decisions are made explicitly in Chapter III, and the second is examined empirically in Chapter VII.

The approach closest to the gradient-guidance routes of this thesis is BADGER, whose classifier-guidance variant applies the gradient of a separately trained affinity predictor to a frozen diffusion SBDD generator at sampling time, in essentially the same plug-and-play arrangement as Chapters IV to VI, and reports up to a 60% improvement in the binding affinity of sampled molecules [24]; its companion classifier-free variant instead folds affinity conditioning into the generator's own training, which this thesis's frozen-generator design does not attempt. DiffGui integrates bond diffusion with affinity and drug-likeness guidance trained jointly with the generator, rather than through a separately frozen predictor, and reports improved binding affinity together with validation on a small set of wet-

laboratory assays [22]. Neither paper's abstract or summary materials available to this search state whether the reported affinity improvement was evaluated on a target-disjoint split, tested for statistical significance with correction for multiple comparisons, or checked against a negative control or a quality-degradation criterion (ligand efficiency, molecular size, physical validity) of the kind used throughout this thesis; this search did not extend to their full methods sections, and no claim is made here about what those sections do or do not report (Appendix B). Section IX.5 returns to the apparent tension between their reported positive result and this thesis's null result once the full evidence is in.

II.3 Predicting Binding Affinity

II.3.1 Docking scores and their biases

AutoDock Vina combines an empirical scoring function with efficient conformational search and has been extended in later releases with additional features and Python bindings [10], [50]. Its score is reported in kcal/mol and, being fast, is used ubiquitously both to evaluate generated molecules and as a label or baseline for machine-learned scoring. Its limitations are well documented. In a systematic analysis of docking successes and failures on the DUD-E benchmark, the Vina scoring function was reported to show a bias toward compounds of higher molecular weight [57]. Re-parameterisations of the function, such as Vinardo, were introduced because the conventional approach of fitting empirical terms by linear regression to experimental affinities does not yield a function with optimal docking capabilities [39]. The choice of docking-based baseline therefore matters, a point that becomes central to the delta-learning experiment of Chapter VIII.

II.3.2 Learned scoring functions

Machine-learned scoring functions have been built on convolutional and graph architectures. GIGN unifies covalent and non-covalent interactions in one heterogeneous message-passing layer that is invariant to rotations and translations of the complex, and reports state-of-the-art performance on three external test sets [59]. PIGNet predicts atom–atom pairwise interaction terms with physics-informed equations whose parameters are produced by a neural network and sums them to obtain the affinity, and augments the training data with additional poses and ligands to improve generalisation [38]. Interformer combines a graph-transformer with an interaction-aware mixture density network for docking and affinity prediction [31], and PLANET is trained with three objectives (affinity, protein–ligand contact map and ligand distance matrix) [64]. A framework that adds an adjustable affinity term to the training of a mixture density network was proposed to balance scoring, docking, ranking and screening powers, which the authors note are rarely achieved simultaneously by a single machine-learned function [47]. The GIGN layers and the PIGNet physics decomposition are the two architectural ingredients of the physics-anchored predictor examined in Chapter V.

II.3.3 Data leakage and shortcut learning

A large part of the recent literature concerns the reliability of the reported performance of these models. A study with modular graph networks showed that an explicit description of protein–ligand non-covalent interactions provided no advantage over ligand or protein descriptors alone, and that simple models that infer the affinity of a test sample from the closest ligands or proteins in the training set already performed well, which the authors interpreted as memorisation dominating true

learning [53]. Retraining top models on PDBbind CleanSplit, which removes structure-based train–test leakage and redundancy within the training set, caused their benchmark performance to drop substantially, indicating that reported performance was largely driven by leakage [19]. The Leak Proof PDBBind data set re-organises PDBbind so that proteins and ligands with high sequence and structural similarity do not cross the split, and reports that models retrained on it perform better on a new independent set [34]. A further line of work argues that the problem is not only leakage in the conventional sense of a complex crossing the split, but *redundancy*: MIRAGE reports that apparent generalisation in affinity prediction tracks how much historical protein-family support a test target has in public databases, that controls unable to exploit the test family stay flat, and that model rankings reverse on genuinely novel families, and it proposes reporting performance as the excess over a support-insensitive baseline [60]. That proposal is closely related to the class-prior decomposition this thesis arrives at independently, and contemporaneously, by a different construction in Chapter IV.8. Complementary evidence comes from work in which the injection of additional geometric descriptors into a graph network gave no statistically significant improvement on leak-controlled splits [3], and from a scoring function whose attributions were compared with a distance-based interaction profiler in order to test whether it had learned interactions rather than data-set biases [45]. The leakage-safe, target-level split used in this thesis (Chapter III) follows the spirit of these studies; it is an adaptation of the Leak Proof PDBBind procedure to CrossDocked2020 and not the published data set itself.

II.4 Evaluating Generated Molecules

The reliability of docking scores as the success criterion for generative models has been questioned directly. A study that proposed practical alternative metrics reported that Vina scores can be inflated simply by increasing the number of atoms in a molecule, and that current SBDD models with high Vina scores fall short on measures of practical usability [16], [18]. In its published (ICLR 2025) version, the same work states that "some models use trained Vina predictors to guide sampling, prioritizing the hacking of Vina scores rather than generating reliable and actual effective molecules", and that "many existing models use the test set as a validation set to select checkpoints during training, which risks data leakage" [18]. It also observes that the structures in CrossDocked2020 are generated by docking software rather than being real complexes, and that the selection of the data relies on docking software, so that the ligands chosen for training tend to be favoured by the docking software [18]. That study defines a "delta score" as the mean difference between the docking score of a molecule against a randomly sampled other target and against its own target, i.e. a measure of specificity [17], [18], [41]. This quantity is unrelated to the residual-against-docking-score target ("delta-learning") used in Chapter VIII, and the shared name is a terminological collision that is disambiguated wherever it arises in this thesis.

Physical plausibility is an independent axis of evaluation. PoseBusters is a suite of chemical and geometric checks (stereochemistry, planarity of aromatic rings, bond lengths, protein–ligand clashes) which showed that several deep-learning docking methods frequently produce physically implausible poses even when their root-mean-square deviation from the native pose is low [6]. A related study of

interaction recovery reported that ignoring interaction fingerprints can lead to an overestimation of model performance, notably for co-folding models that often fail to recapitulate key interactions [12]. Together, these works motivate the multi-criterion evaluation harness of Chapter III, in which a docking score is never interpreted without a size-normalised measure (ligand efficiency) and a physical-validity check.

II.5 Estimating Synthesizability

Two families of synthesizability estimates are in common use. The synthetic accessibility score (SA score) combines fragment contributions derived from one million PubChem molecules with a complexity penalty and maps a molecule to a value between 1 (easy) and 10 (very difficult) [13]. The retrosynthetic accessibility score (RAscore) is a machine-learned classifier that predicts whether a synthetic route can be found for a compound by the computer-aided synthesis planning tool AiZynthFinder, and is at least 4500 times faster than the underlying planning [48]. In this thesis, the RAscore is used as the target of the synthesizability guidance predictor and, independently, as an evaluation metric computed with the reference implementation (Chapters III and VII). A pre-screening tool for enumerated and generated compounds, it does not by itself certify that a route exists.

II.6 Residual (Delta) Learning for Scoring

Delta-learning trains a model to predict the difference between a high-quality target and a cheap baseline, so that the baseline supplies the bulk of the prediction. It was introduced for quantum chemistry, where machine-learning corrections added to approximate legacy methods reproduce high-level enthalpies of large molecular sets after training on 1–10% of the data [40], and it remains in use, for example for correcting a semiempirical method in the prediction of reaction barrier heights [27].

For protein–ligand scoring, ΔVinaRF20 adds a random forest with 20 descriptors to the Vina score; its authors report that direct random-forest scoring functions gain in scoring power but lose docking and screening power relative to classical functions, whereas the delta formulation was superior in all power tests of two Comparative Assessment of Scoring Functions (CASF) benchmarks [54]. Related work re-scored Vina poses with random forests [33], and a delta model built on a linear empirical baseline and gradient boosting reported Pearson correlations of 0.853, 0.839 and 0.813 on locally optimised, redocked and ensemble-docked poses of the CASF-2016 core set [58]. All of these use tree ensembles on hand-crafted features and are evaluated on the CASF core set, which the leakage literature of Section II.3.3 shows to overlap with common training data. To the best of the author's knowledge, a matched comparison of a residual target against an absolute target for the same graph neural network on a leakage-controlled split was not found in the literature searched; the search strategy and its limits are documented in Appendix B.

II.7 Multi-Task Learning and Interaction Supervision

Multi-task training with auxiliary objectives is a common way of regularising a main task. Its central practical difficulty is balancing the losses. Uncertainty weighting learns per-task weights from homoscedastic uncertainty [28]; gradient normalisation equalises gradient magnitudes at a shared layer [7]; gradient surgery projects a task gradient off the gradient of a conflicting task [61]; and dual balancing acts on both loss scales and gradient magnitudes [36]. Auxiliary tasks can also cause negative transfer, for which dedicated remedies have been proposed [25], and in molecular graph networks auxiliary learning combined with gradient surgery has been

used to counter it [8]. Multi-task learning has also been surveyed specifically for drug design [2].

Interaction supervision requires a source of interaction labels. PLIP detects seven types of non-covalent interaction at single-atom level from a three-dimensional complex, accepts custom complexes, for example from docking, and is designed for high-throughput use [1], [43], [46]. Alternative encodings include the interaction fingerprints of ProLIF [5] and of related tools [14], [55]. Whether such labels are reliable on poses that are not experimental is not settled: predicted poses can differ from experimental ones in their interaction profiles [12], and the stability of PLIP labels under coordinate perturbation was not characterised in the literature found, which is why it was measured in this thesis (Chapter VIII).

II.8 QSAR Controls: y-Randomisation, Activity Cliffs and Matched Molecular Pairs

Three standard controls from the QSAR literature, none applied to this project's predictors before Chapter IV.6–IV.9, were adopted there to characterise what the affinity predictors actually learn. y-Randomisation, refitting a model after permuting the response variable and confirming performance collapses to chance, is a long-standing control against chance correlation in QSPR/QSAR modelling [42]; Chapter IV.8 adapts it further, by permuting labels *within* each protein target rather than globally, into a decomposition of apparent predictive performance into a class-level prior and a within-target structure–activity component. Activity cliffs, pairs of structurally similar ligands with disproportionately different measured affinity, are a standard stress test for whether a model has learned local structure–activity relationship rather than coarse, class-level trends; the MoleculeACE benchmark

reports that, across 24 machine-learning approaches evaluated on activity cliffs in 30 targets, descriptor-based models outperform more complex deep-learning architectures specifically on cliff compounds [51]. Matched molecular pairs, which isolate the effect of a single substituent change, are a related and stricter test of local structure–activity capture, reported in the literature to be the easiest case for any model, machine-learned or otherwise, to predict correctly [30]. Chapter IV.9 applies all three to this project's own predictors.

II.9 Positioning and Research Gap

Table 2.1 summarises how the closest bodies of work relate to the experiments of this thesis. Interaction information has been used to *condition* or *co-generate* with 3D generators [23], [32], [65], and, in one case, as an auxiliary classification loss inside the generator's own denoiser [56]. Residual learning has been applied to scoring with tree ensembles [54], [58]. Multi-objective affinity networks with structural auxiliary targets exist [64]. Gradient-based affinity guidance of a diffusion SBDD generator has been reported, with a positive result, by BADGER [24]; a jointly trained, non-frozen variant with the same guidance aim is DiffGui [22]. Neither was found, within the limits of the search described in Appendix B, to report whether its result was evaluated on a target-disjoint split, tested with correction for multiple comparisons, or checked against a negative control, although this search did not extend to their full methods and no claim is made that these checks are absent from them. Separately, best-of-N and best-of-K selection applied to diffusion SBDD generation itself, ranked by a combined QED, SA and Vina score, has been proposed as a novel generation technique by BoKDiff [29]; this thesis's own positive control (Chapter VI.5) reuses the same mechanism, but for a different purpose, as a diagnostic

instrument that isolates signal quality from guidance mechanism, and was not found to be anticipated by that study. What was not found, within the limits of the searches described in Appendix B, is a study that (i) evaluates learned affinity or synthesizability guidance for a 3D diffusion generator against independently computed outcomes under a leakage-safe, statistically corrected protocol with an explicit positive control built from a genuine, non-surrogate signal, or (ii) combines a residual target against a docking score with typed interaction supervision on a separate guidance predictor whose gradient steers the generator. The first gap motivates Chapters IV to VII; the second motivates the design of Chapter VIII, in which only the residual-target component was ultimately tested.

Table 2.1 Relation of the closest prior work to the experiments of this thesis

Work	**3D diffusion SBDD**	**Learned guidance predictor (separate from the generator)**	**Interaction information**	**Residual (delta) target**	**Target-disjoint, leakage-safe evaluation**	**Evaluation independent of the model, with negative/positive controls**
DeepICL [65]	No (autoregressive)	No (conditioning, not guidance)	PLIP labels as conditioning	No	Not the focus of that study, as reviewed	Not the focus of that study, as reviewed
InterDiff [56]	Yes	No (in-denoiser conditioning)	BINANA types; classification loss in denoiser	No	Not the focus of that study, as reviewed	Not the focus of that study, as reviewed
BInD [32]	Yes	No (co-generated, not a separate predictor)	PLIP interactions co-generated	No	Not the focus of that study, as reviewed	Not the focus of that study, as reviewed
DPDiff [23]	Yes	Yes (affinity-network priors)	Latent priors from affinity networks	No	Not the focus of that study, as reviewed	Not the focus of that study, as reviewed

Work	3D diffusion SBDD	Learned guidance predictor (separate from the generator)	Interaction information	Residual (delta) target	Target-disjoint, leakage-safe evaluation	Evaluation independent of the model, with negative/positive controls
BADGER [24]	Yes	Yes (classifier-guidance variant; a classifier-free variant also folds affinity into training)	No (affinity, QED and SA guidance, not typed interactions)	No	Not stated in the abstract/summary reviewed; full methods not read (Appendix B)	Not stated in the abstract/summary reviewed; full methods not read (Appendix B)
DiffGui [22]	Yes	No (affinity and drug-likeness guidance trained jointly with the generator, not a separate frozen predictor)	No (affinity, QED, SA, LogP and TPSA guidance, not typed interactions)	No	Not stated in the abstract/summary reviewed; full methods not read (Appendix B)	Not stated in the abstract/summary reviewed; includes wet-laboratory validation, which this thesis does not
BoKDiff [29]	Yes	No (best-of-N/K selection by an external QED+SA+Vina score, no separate trained predictor)	No	No	Not the focus of that study, as reviewed	Proposed as a generation technique, not framed as a positive control against a prior guidance failure
PLANET [64]	No (scoring model only)	n/a	Contact map (not typed)	No	Not the focus of that study, as reviewed	Not the focus of that study, as reviewed
ΔVinaRF20 [54]; Δ-XGBoost [58]	No (scoring functions)	n/a	No	Yes (tree ensembles)	No (CASF core set; Section II.3.3)	Not the focus of that study, as reviewed
This thesis	Yes (frozen TargetDiff)	Yes (EGNN / GIGN+PIGNet2 / SynthPredNet); also a genuine, non-surrogate Vina control	PLIP labels (tested as a guidance representation, Chapter IV.6; collapses to a size proxy)	Yes (graph network; falsified at this scale)	Yes (LP-split adaptation; Section III.2)	Yes (KS/Wilcoxon + BH, bootstrap CIs, a signal-swapped positive control; Section III.6 and Chapter VI.5)

The table lists what each source reports, as read during the search of Appendix B; "not the focus of that study, as reviewed" means the question this thesis asks was not what that work set out to answer, not that the work is deficient. It does not claim that no other work anywhere addresses these questions (see Appendix B for the limits of the search).

CHAPTER III

RESEARCH METHODOLOGY AND EXPERIMENTAL INFRASTRUCTURE

This chapter describes the research design, data, generative model, guidance mechanism, predictors, evaluation metrics, statistical protocol and integrity checks that are shared by all experiments. Track-specific settings are given in the chapter of the corresponding track.

III.1 Research Design

The research was organised as a falsification study. A hypothesis, that a learned predictor can steer a frozen diffusion model toward molecules that are better on independently computed evaluation outcomes, was tested by mechanistically distinct routes, each of which was assigned a decision rule before its results were inspected. The routes are named Tracks A to E in the experiment logs of the project and in this thesis: Track A (physics-anchored predictor architecture), Track B (three variants of the guidance mechanism on a fixed predictor), Track C (non-gradient rejection sampling), Track D (synthesizability guidance) and Track E (a redesigned training target). A standing rule required two independent routes, one varying the predictor architecture and one varying the mechanism, to both fail before the approach was considered not viable, because a failure of the mechanism alone does not exclude a better predictor, and vice versa.

Every route followed a two-tier protocol. An exploratory tier, with few pockets, few molecules and partially trained models, was used to obtain a cheap first reading and to derive stable operating ranges. A full tier with converged models, more

pockets and more molecules was run only when its cost was justified, and its verdict was taken from the pre-declared decision bands. Each checkpoint was dual-criterion: the predictor had to be accurate on held-out targets, and its guidance had to change an independently computed outcome. These two criteria are deliberately kept separate and are never substituted for one another: a predictor's held-out accuracy is a property of the predictor alone, measured once, on a fixed test set, independently of any generator; its usefulness as a guidance signal is a property of the coupled predictor-plus-sampler system, measured separately, on generated molecules, by tools that do not see the predictor's output. Chapter IX (Section IX.2.2) returns to this distinction once the full evidence is in; it is stated here because every route of this thesis measures the two quantities on this footing from the outset, not only in hindsight. Figure 3.1 shows the overall pipeline.

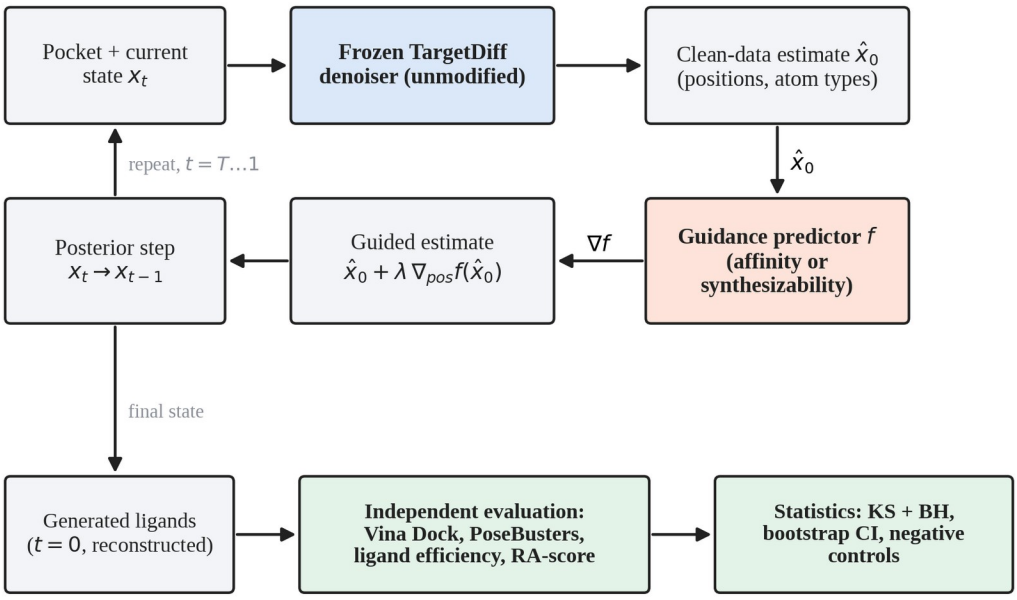


Figure 3.1 Overall pipeline. A frozen TargetDiff denoiser proposes a clean-data estimate at each reverse step; a guidance predictor's gradient with respect to atom positions modifies that estimate; and the finished molecules are evaluated by tools that are independent of the guidance predictor and analysed with corrected statistics.

III.1.1 Three candidate mechanisms for a guidance failure

A route failing its pre-declared criteria does not, by itself, say *why* it failed, and several of this thesis's methodological choices, the positive control of Chapter VI.5 and the information-content audit of Chapter IV chief among them, were made so that a null result could be read against three candidate mechanisms rather than left unexplained. The three are not mutually exclusive, and a single checkpoint rarely adjudicates all of them at once. **F1**, an uninformative signal: the guidance predictor carries little usable information about the quantity it is meant to steer toward, specifically the true binding interface, so that no mechanism built on it could succeed regardless of how the signal is used. **F2**, a misaligned mechanism: the signal is informative in principle, but the specific way it is turned into guidance, a gradient computed on a clean-data estimate, a particular scale or timing, a top-k selection rule, misapplies it. **F3**, a gameable objective: the guidance objective can be satisfied in a way that does not correspond to a genuine improvement in the quantity a practitioner cares about, for instance by exploiting a correlate of the evaluation metric, such as molecular size, that a differently shaped objective would not reward. This framework is introduced here because it is used to interpret results throughout the thesis, most directly in Chapter VI.5's positive control, which varies the signal while holding the mechanism fixed, and in Chapter IX.2's synthesis of the evidence for each of the three once every route has reported.

III.1.2 Pre-registration status of each route

Not every route in this thesis carries the same evidentiary status, and the distinction matters for how its result should be read. Four categories are used throughout, and every analysis in this thesis belongs to exactly one of them. *Pre-*

registered means a frozen, version-controlled document fixed the hypothesis, the primary metric, the decision thresholds and the stopping rule before any model of that route was trained; only Track E (Chapter VIII) meets this bar, and its amendments are logged, dated and never edited in place (Appendix D). *Planned, not pre-registered* means the decision bands, correction procedure and quality-degradation checks were declared in the project's working log before the results of that specific checkpoint were inspected, following the shared protocol of this chapter, but without a frozen external document; Tracks A, B, C and D, the baseline ablation of Chapter IV, and the gradient diagnostic of Section V.4 are of this kind. *Post hoc* means a question was formed after seeing a result and is reported as exploratory and uncorrected for multiple comparisons unless stated otherwise; this applies to specific, individually flagged analyses, such as the pocket-size correlation of Section VI.3 and the pooled Mann–Whitney check of Chapter IV. *Appendix E follow-up* means the analysis was conducted after Chapter IX's synthesis had been written and frozen, outside the decision sequence of Tracks A to E described below; it follows this project's general exploratory-then-confirmatory discipline (the two-tier protocol of Section III.1) but was not pre-registered, and it is reported in full as Appendix E with its own, explicitly dated chronology (Section E.1).

Table 3.1 makes explicit where a pass/fail gate was actually applied, since not every route had one. Tracks A to D were stopped or extended by the shared decision bands of Section III.6.3 (no comparison may be called a clear signal without surviving correction and the degradation checks), applied after each tier; no route was permitted to continue to a more expensive tier without first clearing this check. Track E additionally carried its own, pre-registered numerical gates, frozen before training

(Appendix D). Appendix E's follow-up phase used the same confirm-before-reporting discipline as a working practice (no encouraging small-sample result was reported before a larger, confirmatory run) but applied no frozen stopping rule, because it was not pre-registered.

Table 3.1 Where a decision gate was applied, by route

Route	Gate type	Condition to continue or to declare a positive result	Consequence if not met
Baseline ablation (IV)	Shared decision bands (planned)	No significant Mann–Whitney/Wilcoxon shift; no PoseBusters/LE degradation	Verdict recorded as no signal; motivated Tracks A–D
Track A (V)	Shared decision bands (planned)	Exploratory tier's cost of the full tier was justified by its yellow-flag result; full tier itself gated by the same bands	Full tier was run; its own null stopped the route
Track B (V)	Shared decision bands (planned)	Same as above, applied per variant (B1–B3)	Each variant's null closed that variant; no further variants of the same kind were run
Track C (VI)	Shared decision bands (planned) + negative control	Scrambled-filter control must agree with the random-subset control	Disagreement would have been read as an implementation bug, not a result; it did not occur
Track D (VII)	Shared decision bands (planned)	Divergence at the exploratory scale must be resolved before the multi-pocket confirmation (step D.4) was run	Divergence persisted only partially; D.4 was explicitly skipped by this pre-declared rule
Track E (VIII)	Pre-registered numerical gate (C1a, then C4)	Point estimate and interval of the primary comparison must clear a stated margin; the gradient-distance gate C4 applies only if C1a passes	C1a's upper bounds excluded the required margin; E1 declared falsified at this scale; C4 and E2 were not run

Route	Gate type	Condition to continue or to declare a positive result	Consequence if not met
Appendix E (post-synthesis)	Working discipline, not a frozen gate	Exploratory tier (n≈8) must be confirmed at larger n before being reported as a finding	Applied uniformly (e.g. the ESM2 and PLIP exploratory signals); no frozen pass/fail threshold existed to stop the phase itself

Table 3.2 summarises every route at a glance, including the follow-up investigation of Appendix E, with the predictor used, what was intervened on, the scale of evidence, the primary outcome metric, the correction applied, and the verdict. It is a map for the chapters that follow, not a substitute for them.

Table 3.2 Master summary of every experimental route, including the Appendix E follow-up

Route	Predictor	Intervention	Pockets / targets	Units per condition	Primary metric	Correction	Verdict
Baseline (IV)	Stage 0 EGNN	Guidance strength λ	20 pockets; 8 pockets × 3 λ	8–10 molecules	Vina Dock (Mann–Whitney, Wilcoxon)	Not explicitly corrected (pre-dates the KS/BH protocol below)	No effect at any tested λ
Track A (V)	GIGN + PIGNet2	Predictor architecture	8 pockets (full tier)	29–30 molecules	Vina Dock (KS)	BH, 24 tests	0/24 significant
Track B (V)	Stage 0 EGNN	Gradient scale (B1), objective (B2), timing (B3)	2–3 pockets	20 molecules	Vina Dock (KS)	BH, 23 tests	0/23 significant
Diagnostic (V.4)	Stage 0 EGNN	None (read-only probe)	3 pockets, 144 states (magnitude); 20 pockets, 23,202 atom-states (direction)	n/a	Gradient–distance correlation; signed projection toward nearest contact	Bootstrap CI only; no gate	Magnitude +0.240 (largest far from the pocket); direction indistinguishable from random (mean cos −0.019, CI [−0.046, +0.008])
Track C (VI)	Stage 0 EGNN (as ranker)	Post-hoc top-10% selection	15 pockets	~590 molecules/pocket pool	Vina Dock, LE, PoseBusters (KS)	BH, 15 tests	Raw gain confounded by size

Route	Predictor	Intervention	Pockets / targets	Units per condition	Primary metric	Correction	Verdict
Track D (VII)	SynthPredNet	Synthesizability guidance	1 pocket (powered tier)	30 molecules	Independently computed RAscore (KS)	BH, 5 tests	0/5 significant
Track E (VIII)	GIGN delta head	Training target (residual vs. Vina)	127 test targets	n/a (predictor comparison)	Target-clustered R^2	Pre-registered gate (bootstrap CI)	Falsified at this scale
Follow-up (App. E)	8 further checkpoints (EGNN / GIGN variants)	4 mechanism-level fixes, alone and combined, plus validation and a mechanistic trace	1 fixed pocket + CDK6 (unseen family) + 835 (AlphaFold side-check)	8–24 molecules per λ	Vina Dock, PLIP interactions, R^2	Pooled BH, 48 tests	0/48 significant

"n/a" marks routes that compare trained predictors on a held-out test set rather than generated molecules under guidance. The diagnostic of row 4 is not a guidance checkpoint and carried no pass/fail gate (Section V.4).

III.2 Data

III.2.1 CrossDocked2020 and the pocket representation

All generative sampling used the pretrained TargetDiff model, which was trained on the CrossDocked2020 data set [15], [20]. The processed data set used here is the pocket-cropped version of the original TargetDiff release (residues within 10 Å of the ligand; a stored ligand pose within 1 Å RMSD of the docked pose), held in a lightning memory-mapped database (LMDB) with 166,500 protein–ligand entries, verified directly against the file. The official split of that release assigns only 99,990 of these pocket instances to training (1,905 unique targets) and 100 to testing (93 unique targets), with no target overlap between the two; the remaining 66,410 LMDB entries are outside both official partitions and were not used directly, except where the affinity-labelled subset of Section III.2.2 and the leakage-safe split of Section III.2.3 draw on them by target identity. Each entry stores protein atom positions and element/amino-acid identifiers, ligand atom positions, elements and bonds, and file

names; it does not store residue indices or chains. Since the poses are cross-docking outputs [18], they were treated throughout as docked poses, not as experimental complexes.

III.2.2 Affinity labels

Experimental binding affinities were taken from the annotation file distributed with the same release, which contains 184,087 entries keyed by the ligand file name, of which 76,803 carry a non-zero pK. These are experimental pKd, pKi or pIC50 values derived from PDBbind, not docking scores, which excludes a possible form of label leakage in which training labels are computed with the same docking configuration that is later used for evaluation. The file also stores, for each pose, an RMSD value and a Vina-type score; the latter is used in Chapter VIII as the anchor of the residual target. Its semantics were examined explicitly (Section VIII.3).

III.2.3 Splits used for predictor training

Three target-level structures were used. The *old split* (the first version of the affinity predictor) took a random 90/10 division of pocket instances of the official training partition as training and validation sets, and the labelled officially-held-out pockets as the test set; this gave 6,000 training instances (507 targets, after a subsampling made necessary by the training time), 4,061 validation instances (456 targets) and only 27 test pockets (26 targets). Because validation and training shared 74% of their targets, validation measured interpolation within seen targets rather than transfer to a new target.

The *leakage-safe (LP) split* was constructed for this thesis by adapting the Leak Proof PDBBind procedure [34] to CrossDocked2020 at *target* granularity, so that all entries of a target move together. Targets were assigned to families using

UniProt information; a small random test set was seeded per family and expanded with any target whose protein sequence identity exceeded 0.9, whose ligand similarity exceeded 0.99, or whose pocket-histogram cosine similarity exceeded 0.97 to a current test member; a validation set was formed analogously with a protein-identity threshold of 0.5; and any training target with ligand similarity above 0.99 to a validation or test target was discarded. The pocket-similarity threshold was recalibrated from the originally sketched 0.8 and 0.95, because an empirical check on 3,000 random target pairs showed that a 20-dimensional amino-acid-composition histogram is too coarse for two-tier thresholding (mean 0.69, 99.9th percentile 0.94 among unrelated pairs). The pool entering this construction, restricted to entries with a non-zero pK and an RMSD below 2 Å, held 1,041 targets and 76,803 entries. Of these, 1,002 targets had a UniProt accession that could be resolved and assigned to a family; the remaining 39 targets were dropped at this step for an unresolvable identifier, before the family-based assignment and contamination checks described above were applied. The resulting split, after the 31 further targets discarded for contamination (Section IV.4.1), is summarised in Table 3.3.

Table 3.3 Data splits used for affinity predictors

Split	Train (targets / entries)	Validation	Test	Comment
Old split	507 / 6,000	456 / 4,061	26 / 27	Validation shares 74% of targets with training
LP split	779 / 46,964	65 / 6,069	127 / 11,855	Target-disjoint; 31 targets discarded
Synthesizability (LP-derived)	8,835 entries	1,015	1,940	3,210 of 15,000 discarded (target not assigned)

Entries are complexes (pocket–ligand pairs). Test-set statistics in Chapters VIII use the 127 test targets as the resampling unit.

III.2.4 Synthesizability labels

Synthesizability labels were obtained by computing the RAscore [48] of 15,000 ligands of the training partition, giving a target in [0, 1]. The original random 90/10 split of these labels had no held-out test set and no leakage check; it was replaced (Chapter VII) by the target-level assignment of the LP split, applied by target identity without recomputing any label.

III.3 Generative Model and Guidance Mechanism

III.3.1 The frozen diffusion model

The generator was the pretrained TargetDiff checkpoint [20], used without modification. It models ligand atom coordinates and 13 atom-type classes (atomic number and aromaticity pairs) over 1,000 diffusion steps, conditioned on the pocket. Bonds are not modelled: they are assigned after sampling by a reconstruction procedure that turns coordinates and types into a molecule. The number of ligand atoms is drawn from a prior before sampling begins and is therefore fixed for the whole trajectory.

III.3.2 Guidance on the clean-data estimate

The guidance loop re-implements the reverse sampling step so that the pretrained model and its posterior functions are used unchanged, and a guidance term is added to the clean-data estimate before the posterior step. Denoting the estimate by $\hat{x}_0$, the guided estimate is

$$\hat{x}_0' = \hat{x}_0 + \lambda \nabla_{pos} f(\hat{x}_0) \quad (3.1)$$

where f is the guidance predictor and λ the guidance strength. With λ = 0 the loop performs exactly the same computation as the unguided sampler (verified against

the original implementation, up to the non-determinism of GPU scatter operations, which already makes two calls of the original sampler with the same seed differ at the level of 10^{-6}). The equation above was written with the gradient taken only through the guidance predictor and not through the diffusion network, which is a deliberate compute trade-off on a 4 GB GPU: back-propagation through the denoiser at each of 1,000 steps would multiply memory and runtime by roughly the depth of the network. Consequently, the guidance predictors are trained on clean ligand structures and evaluated on $\hat{x}_0$, which are the conditions under which the literature reports the mismatch problem for noisy intermediate structures [23]. Two further scope decisions were made and are disclosed. First, guidance acts on positions only; the gradient with respect to atom types is set to zero, because the predictors use bond-derived features (degree, hydrogen count and hybridisation) that do not exist during sampling. These fields were filled with a fixed placeholder for the affinity predictors, and an alternative, differentiable bond-aware estimate was tested for the synthesizability predictor (Chapter VII). Second, guidance can be restricted to a window of diffusion steps, and the gradient can be rescaled to unit norm per atom before λ is applied; both are used only in the Track B variants.

III.4 Predictors

Table 3.4 summarises the predictors. All were trained on clean structures with mean-squared-error (regression) or cross-entropy (classification) objectives and selected by validation loss.

Table 3.4 Predictors used as guidance signals or ranking functions

Predictor	Architecture	Parameters	Training set	Held-out test result
Stage 0 EGNN (affinity)	6-layer EGNN, hidden 256, 48 nearest neighbours, 10 Å cut-off, MSE	2.55 M	LP split, 6,000-entry subsample	R^2 0.342, Pearson 0.609 (n = 11,855)
GIGN + PIGNet2 (affinity)	3 GIGN interaction layers, PIGNet2 physics readout, hidden 256	0.49 M	LP split, all 46,964	R^2 −0.189, Pearson 0.581 (n = 11,855)
Classifier head (affinity)	Linear head on frozen Stage 0 embedding	Head only	Median-binarised pK (7.03)	AUROC 0.677 (test)
SynthPredNet (RA-score)	Ligand-only EGNN, sigmoid output	0.43 M	LP-derived, 8,835 entries	R^2 0.414, Pearson 0.647 (n = 1,940)

Reference-model details are reproduced in the chapters of the corresponding tracks. Parameter counts are those printed in the training logs.

The *Stage 0 EGNN* is the affinity predictor of the project's original guidance implementation, an E(n)-equivariant network [44] applied to the pocket–ligand point cloud, with a scatter-sum readout and an MLP head. It was trained with Adam (learning rate 10^{-4}), batch size 1, a Gaussian position noise of 0.1 Å, and a plateau learning-rate scheduler with early stopping; the same configuration was retrained, unchanged, on the LP split, which defines the Stage 0 checkpoint. The *GIGN + PIGNet2* predictor combines the heterogeneous interaction layers of GIGN [59], in which covalent (or covalent-proxy) and non-covalent edges share one message pass and are differentiated by separate MLPs, with the physics-decomposed readout of PIGNet [38]: pairwise Morse-type van der Waals, hydrogen-bond, metal and hydrophobic terms between ligand and protein atoms are summed to the prediction. Protein covalent edges, which are not available in the LMDB, were approximated by a distance cut-off. The *classifier head* is a linear head fitted on the pooled embedding

of the frozen Stage 0 network for the task of predicting whether the pK is above the median of the training set; its gradient log p(y = 1 | x) is the classical classifier-guidance target [9]. *SynthPredNet* is a ligand-only EGNN that regresses the RAscore; it has no protein input.

III.5 Evaluation Metrics

Guidance effects were never evaluated with the guidance predictor's own output. Every generated molecule was reconstructed and, when valid, evaluated as summarised in Table 3.5.

III.5.1 What each metric does not certify

Each metric in Table 3.5 answers one narrow question, and the combination is still far short of a usability assessment. PoseBusters checks that a *pose* is geometrically and chemically well formed: correct stereochemistry, planar aromatic rings, plausible bond lengths, no steric clash with the receptor. A molecule passing it is not thereby a reasonable compound; it has not been checked for internal strain beyond those geometric tests, for promiscuous or assay-interfering substructures, or for anything pharmacological. Ligand efficiency normalises a docking score by heavy-atom count and so detects one specific artefact, the preference for larger molecules; it does not detect others. The RA-score estimates whether a retrosynthetic planner can find a route, not whether that route is practical, cheap or safe. Throughout this thesis, "valid" means "passes the stated check", never "suitable for synthesis or testing", and the gap between those two is the subject of Section X.3.

III.5.2 What "independent" means in this thesis, and what it does not

The phrase *independently computed* recurs throughout this thesis and carries a deliberately narrow, operational meaning that is fixed here once: the quantity is produced by a program that does not read, and was not fitted to, the guidance predictor's output or parameters. It is a statement about the provenance of a number, not about its physical validity. Three consequences follow and are relied on in later chapters. First, independence in this sense rules out the circularity of reporting a model's own score as its own success criterion; it is the minimum any evaluation of guidance must satisfy, and the main thesis's gradient routes satisfy it by construction. Second, it does not make two numbers *orthogonal*: two quantities can both be independent of the predictor while sharing an upstream bias with each other, which is exactly the situation of the positive control in Section VI.5, where selection and evaluation use two modes of one scoring function and therefore share that function's documented preference for larger ligands. Third, it does not make a computed quantity an *experimental* measurement. Vina Dock is a predicted docking score, not a measured dissociation constant, and no claim in this thesis about a molecule being a better binder rests on anything stronger than that proxy; where this distinction affects the reading of a result, it is restated locally rather than assumed. A genuinely orthogonal evaluation would require a scoring function sharing neither parameters nor functional form with the selector, or an assay; neither was available within this project's scope (Section X.3).

Table 3.5 Independent evaluation metrics

Metric	Definition	Direction
Vina Dock	AutoDock Vina 1.2.6 [10] re-docking against the real receptor of the pocket, exhaustiveness 8 or 16 (kcal/mol)	Lower is better
Ligand efficiency (LE)	Vina Dock divided by the number of heavy atoms	Lower is better
PoseBusters validity	Fraction of poses passing the PoseBusters checks [6] in docking configuration	Higher is better
RA-score	Independently computed RAscore model [48] applied to the reconstructed molecule	Higher is better
QED, SA score	Drug-likeness; synthetic accessibility [13] rescaled to [0, 1] following the TargetDiff evaluation code	Higher is better
Validity, uniqueness, diversity	Fraction reconstructable; fraction unique; mean 1 − Tanimoto (Morgan radius 2, 1,024 bits)	Higher is better
Standard success	Vina Dock < −8.18, QED > 0.25 and SA > 0.59	Higher is better
Honest success	Standard success, additionally PoseBusters-valid with LE within a stated tolerance of the reference ligand	Higher is better

The receptor files used for docking were coordinate-matched to the processed data (Chapter IV describes an incident in which files from a different mirror produced meaningless scores). Ligand efficiency and the heavy-atom count were computed in every experiment because raw Vina scores can be improved mechanically by enlarging molecules [18], [57]; PoseBusters validity was computed because a favourable score can be obtained from physically implausible poses [6]; and the RAscore was computed with the reference implementation, independently of the guidance model's own approximation.

III.6 Statistical Protocol

III.6.1 Tests, corrections and intervals

The primary test of a guidance effect was the two-sample Kolmogorov–Smirnov (KS) test of the Vina Dock distribution of guided versus unguided molecules within a pocket. For every checkpoint, all tests of that checkpoint were corrected together with the Benjamini–Hochberg (BH) procedure for the false discovery rate [4]. Effect sizes were reported with 95% confidence intervals obtained by case-resampling bootstrap [11] with 2,000 resamples, and the median shift was reported alongside the mean, because a single pathological pose can dominate a mean (an example, a Vina Dock of +286.58 kcal/mol from a malformed pose, is documented in Chapter V). Mann–Whitney U tests (unpaired) and Wilcoxon signed-rank tests (paired by pocket) were used in the earlier ablations. Correlations were Pearson or Spearman with bootstrap intervals; partial correlations were computed by linear residualisation.

III.6.2 What is shared and what is independent across a comparison

Two different designs are used in this thesis, and they are not interchangeable. In the *independent-sample* design, used for the KS tests above, each molecule of the guided condition and each molecule of the unguided condition is sampled from its own, separately drawn initial noise; nothing is shared between the two groups beyond the pocket and the model, and the two samples are treated as independent draws from their respective distributions. In the *paired* design, used for the Wilcoxon signed-rank tests of Track B, the gradient-alignment trace of Section E.5, and several of the n = 8 exploratory tiers of Appendix E, the same initial noise (the same random seed, consumed in the same order) is reused across every λ value of a given pocket, so that

a guided and an unguided molecule at that position in the sequence begin from bit-identical noise and encounter the same per-step stochastic draws up to the first point at which guidance is applied; only the guidance intervention itself then differs between the paired molecules. This is why the two trajectories of the mechanistic trace (Appendix E.5) are described as diverging "only through the guidance nudge": nothing else differs between them by construction. Pairing is appropriate here because it removes the between-sample noise contributed by independently drawn initial positions and step noise, which is the dominant source of variance at the small n of the exploratory tiers, leaving the guidance effect as the main remaining source of a paired difference; it is reported as a Wilcoxon signed-rank test, not a paired t-test, because the shift is not assumed to be normally distributed. Different pockets, and different seeds within the same pocket in a multi-seed design (Track C, Track E), are independent of one another; pairing is never used across pockets or across seeds, only across λ within one seed of one pocket.

III.6.3 Decision bands and negative controls

Each comparison was assigned one of five verdicts by pre-declared rules (Figure 3.2): *no signal* (fails BH correction); *signal, wrong direction*; *signal but confounded by size* (ligand-efficiency degradation surviving BH correction, or a significant heavy-atom shift); *signal but quality degraded* (PoseBusters validity dropping by more than 10 percentage points); and *clear signal*. A verdict of *suspect implementation bug* was assigned if a scrambled-filter negative control disagreed with a random-subset control (Track C). Because a scrambled top-k selection is mathematically equivalent to a random subset, this control is an implementation sanity check and not an independent hypothesis test.

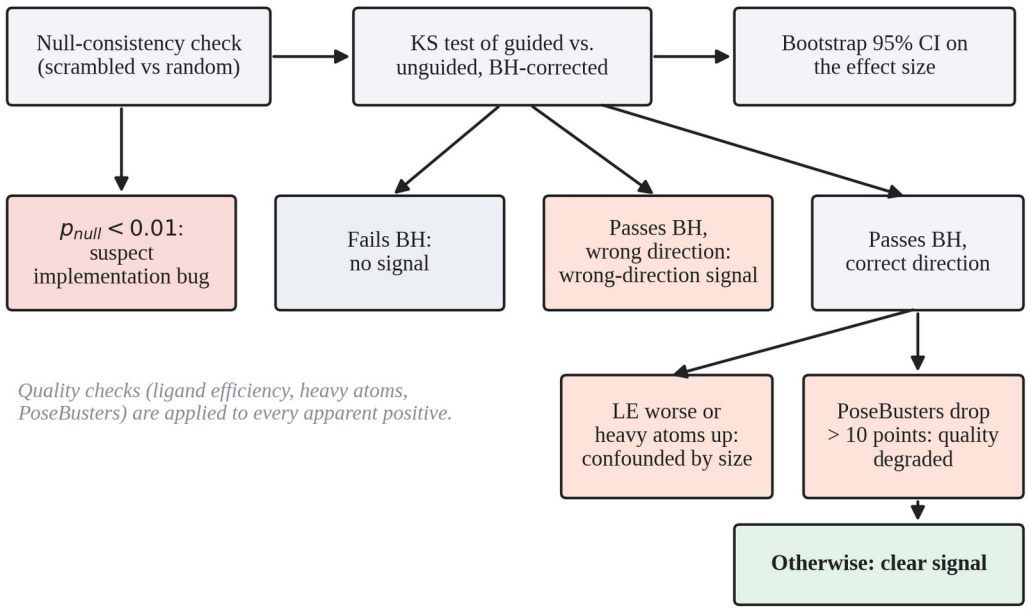


Figure 3.2 Decision logic applied to each comparison. The null-consistency check guards against implementation errors; the ligand-efficiency, heavy-atom and PoseBusters checks are applied to every apparent positive before it may be called a clear signal.

III.6.4 A note on the wording of null results

This thesis uses five distinct phrases for a comparison that did not support a positive claim, and they are not interchangeable. *No significant effect* is the narrowest and most frequent: a specific test did not survive the stated correction at the stated sample size, and nothing is implied about effects of a different size or at a different scale. *No evidence of an effect* is used when several such tests are taken together (for example, across an entire checkpoint or phase) and none of them individually or in combination supports a positive claim. *No effect of practical size* is reserved for cases in which a power calculation (Section V.1, Figure 9.2) bounds the effect that could have been missed, so the claim is about magnitude and not only about significance. *Failed criterion* or *failed its pre-declared criterion* is used for a route governed by explicit decision bands (Section III.1.1) that did not clear them, independently of

whether a numerical test was involved. *Falsified at this scale* is reserved for Track E, the one route with a pre-registered numerical margin (Appendix D): it means the pre-declared rule was applied exactly as written and returned a negative verdict, not merely that a result failed to reach significance. None of these phrases is used to claim that no effect exists under conditions that were not tested; that distinction is kept explicit throughout and restated in Chapter X.

III.6.5 Units of analysis

This thesis reports counts at several different levels, and conflating them would misstate the amount of independent evidence behind a number. From smallest to largest: an *atom* is one position in one molecule, the unit of the gradient diagnostics (Section V.4, Section VII.4) and of the mechanistic trace (Appendix E.5); a *molecule* is one sampled, reconstructed ligand, the unit that a per-condition count such as "n = 30 molecules" refers to, and molecules from the same condition and pocket are not independent of one another in the strict sense when they share initial noise under the paired design of Section III.6.2, only across conditions; a *complex* is one protein-ligand pair in a training or test set, the unit of Table 3.3 and of the predictor-quality tables of Chapters IV, V and VIII; a *pocket* or *target* is one protein binding site, the unit that a KS test or a target-clustered bootstrap resamples, and is the appropriate independent unit whenever several complexes or molecules of the same pocket are compared, because within-pocket observations share the receptor and are not independent; a *seed* indexes one full, independent repetition of a stochastic procedure (sampling, or training) with different random initialisation, and is the unit that bounds the variance not captured by resampling a fixed set of results; and a *statistical comparison* (a row of Table C.1, for example) is one test, the unit that a multiple-

comparison correction is applied across. Table 3.6 states, for every route, which of these levels was treated as the independent replication unit for its primary test, and how many seeds were used.

Table 3.6 Unit of analysis and seed count, by route

Route	Statistical / replication unit	Molecules or complexes per unit	Seeds
Baseline ablation (IV)	Pocket	8–10	1 (strength sweeps); 3 (Task F)
Track A (V)	Pocket	20 (exploratory); 29–30 (full tier)	1
Track B (V)	Pocket	20	1 (paired across λ within a pocket)
Diagnostic (V.4, VII.4)	Captured state (pocket × timestep × sample)	n/a (144 states total)	n/a (read-only probe, no training)
Track C (VI)	Pocket	~590 pool members	2 (combined per pocket)
Track D (VII)	Pocket	8 (original); 30 (leakage-safe)	1 (paired across λ)
Track E (VIII)	Target (127 test targets; complexes within a target are not independent)	n/a (predictor comparison, not generated molecules)	3 per arm (plus 2 extra EGNN seeds)
Follow-up (App. E)	Pocket (1 fixed); target (CDK6 unseen family; 835 for the AlphaFold side-check)	8 (exploratory); 16–24 (confirmatory)	1 (paired across λ within a tier)

"n/a" marks a row for which the concept does not apply (the diagnostic captures states, not molecules; Track E compares predictors on an existing test set rather than molecules it generated). A single seed per λ-sweep does not mean a single molecule: it means the molecules within that sweep share one initial-noise draw per the paired design of Section III.6.2, while the sweep itself is not independently repeated. The baseline ablation's 3-seed Task F run used 10 molecules per pocket per seed; pooled across its three seeds this is about 30 molecules per pocket, the figure Section IX.2.4 compares against Track A's full tier.

III.6.6 Target-clustered statistics for predictor comparisons

For comparisons of predictors on the LP test set (Chapter VIII), the independent unit was the *target* and not the complex, since complexes of one target are strongly dependent. Confidence intervals were obtained by resampling the 127 test targets with replacement (2,000 resamples, fixed seed), and paired differences between two predictors were computed inside each resample so that shared sampling

noise cancels. For predictors trained with several seeds, the value of an arm inside a resample was the mean over its seeds. One-sided p-values were BH-adjusted over the tests of a stage, and intervals were reported at 90, 95 and 99%. This resampling unit produces intervals that are several times wider than complex-level bootstrap intervals; the difference is quantified in Chapter VIII.

III.7 Computing Environment and Engineering

All experiments ran on a single laptop with an RTX 3050 GPU (4 GB), a 20-thread CPU and 31 GB of RAM. Long runs were made resumable: sampling and docking jobs wrote one result directory per (variant, pocket, seed) unit and could be restarted after a shutdown, which occurred and was recovered from without loss during the Task F run; the Track E trainers additionally wrote atomic, mid-epoch checkpoints and handled termination signals (Chapter VIII). Two optimisations were adopted for the largest run (Track C) after their effect on quality had been checked: bf16 automatic mixed precision for the diffusion core's forward pass only (about 14% faster sampling, with a shift of the Vina Dock mean of 0.1–0.13 kcal/mol, within the run-to-run variation between two fp32 baselines), and docking of two molecules in parallel processes (about twice as fast). Increasing the sampling batch size gave only about 3% higher throughput, and `torch.compile` was not attempted because of the risk of graph breaks in the conditional sampling loop. Together the two adopted changes reduced the per-pool time from about 5.75 h to about 4.6 h.

III.8 Integrity Checks

Several checks were made part of the method because they had changed conclusions during the project. (i) *Leakage.* Before any retraining, the target identifiers of the 20 pockets of the guidance ablation were compared with the training

and validation index sets of the affinity model, and the intersection was empty; the labels were verified to be experimental affinities and not docking scores. A coarse gene-symbol check found one coincidental mnemonic match between unrelated proteins; because no sequence-level clustering was available at that time, a bill of health at the fold-family level was not asserted, and the LP split was later built precisely to close that gap. (ii) *Coordinate frames.* An early attempt that used receptor files from a different data mirror produced meaningless zero-interaction docking scores because of a coordinate-frame mismatch; it was caught in a dry run and the coordinate-matched files were used. (iii) *Outliers.* A single pathological docking score was investigated and shown to be a genuine, if degenerate, outcome and not a sentinel value (Chapter V). (iv) *Analysis errors.* An exploratory analysis that truncated a pooled sample to its first 300 molecules gave a wrong conclusion and was corrected by re-deriving the numbers from a saved script (Chapter VI). (v) *Checkpoint selection.* All trainers select checkpoints by validation criteria. The Stage 0, Track A and synthesizability trainers also compute and log test metrics at every validation improvement, so that test numbers were visible during development, although they were not used for selection; the Track E trainers evaluate the test set once, after training, and this is enforced by an automated test (Chapter VIII). One older hyper-parameter sweep labelled a configuration as best on the basis of its test metrics (27 pockets), and this configuration was reused in a learning-curve diagnostic; it was not used for any deployed or guidance model, and the diagnostic is read with that caveat (Chapter IV).

CHAPTER IV

AFFINITY PREDICTOR QUALITY, THE SPLIT ARTEFACT AND BASELINE GUIDANCE ABLATIONS

This chapter reports the first phase of the experimental work. It starts with the affinity predictor of the original guidance implementation, shows that its poor held-out quality was investigated through a leakage audit, a hyper-parameter sweep and five diagnostics, and then shows that most of the apparent generalisation gap was caused by the way the data had been split. After the predictor had been repaired in this sense, the first guidance ablations were run, on the original and on the repaired predictor. All of them returned null results, which motivated the routes reported in Chapters V to VIII.

IV.1 The Original Affinity Predictor

IV.1.1 Training set-up

The first guidance predictor was the Stage 0 EGNN architecture of Table 3.4, trained to regress the experimental pK of a pocket–ligand pair from the docked pose. Training used the official CrossDocked2020 training partition with a random 90/10 division of pocket instances into training and validation sets, a random subsample of 6,000 training instances (507 targets) because of the training time on a 4 GB GPU, and the 27 labelled pockets of the official test partition as the held-out test set (26 targets, disjoint from training). The label distributions were similar (training 6.76 ± 1.84, validation 6.85 ± 1.89, test 5.71 ± 1.81 pK units), although the test mean was

lower. Training was run for at most 20 epochs with early stopping, and the checkpoint of the last epoch, which had the lowest validation loss (1.375), was deployed.

IV.1.2 Held-out quality

The validation and test results of the deployed predictor differed strongly (Table 4.1). Validation Pearson correlation rose steadily to 0.79, whereas test Pearson correlation fluctuated between 0.12 and 0.39 with no trend, and test R^2 was negative at every logged epoch, meaning that the predictions were worse than always predicting the training mean. The deployed checkpoint was not even the best on the test set: the highest test correlation (0.386) was recorded at epoch 3, which was not a running-best validation epoch.

Table 4.1 Old split: validation and test quality of the affinity EGNN at selected epochs

Epoch	Val Pearson	Val R^2	Test Pearson	Test Spearman	Test R^2	Test RMSE (pK)
3	n/a	n/a	0.386	0.321	−0.454	2.188
5	0.682	0.452	0.315	0.237	−0.401	2.148
10	0.745	0.548	0.381	0.313	−0.161	1.955
15	0.768	0.590	0.121	0.150	−0.797	2.433
19	0.795	0.616	0.327	0.304	−0.275	2.049
20 (deployed)	0.788	0.616	0.233	0.141	−0.537	2.250

Validation n = 4,061 pocket instances; test n = 27 pockets. Only epochs at which the validation loss improved were logged. Test metrics were logged during training but were not used for checkpoint selection.

The disagreement between validation and test had a structural explanation that was recognised at this point. Because the validation set was a random division of *pocket instances*, and CrossDocked2020 contains many poses of each target, the validation set shared 74% of its targets with the training set (507 training targets and 456 validation targets drawn from one pool of about 1,900 targets). This 507-target figure is itself a consequence of two compounded subsampling steps and not of the

instance-level 90/10 division alone: the 90/10 split was drawn across the full training partition (99,990 instances, 1,905 targets), after which only a 6,000-instance subsample of the resulting 90% portion was used for training because of the 4-GB GPU's training time (Section IV.1.1); it is this second, smaller subsample that yields 507 covered targets and, with the independently subsampled validation portion, the 74% figure, rather than the near-complete target overlap that a single clean instance-level split of a 52-poses-per-target pool would otherwise be expected to produce. The validation loss therefore measured interpolation among poses of partly seen targets, whereas guidance requires transfer to an unseen pocket, which only the test set measured. With 27 test pockets, however, the test numbers themselves carried large sampling uncertainty.

IV.2 Leakage and Coverage Audit

Before any retraining, it was checked whether the near-zero test quality, or the later null guidance results, could be explained by leakage. The audit found the following.

(i) Pocket level: the 20 target identifiers of the guidance ablation of Section IV.5 were compared with the training and validation index sets rebuilt with the true training seed, and no target was shared. Seven of these 20 targets were, however, members of the 27-pocket test set of the predictor, for which metrics (not gradients) had been computed during training; this is disclosed because it is relevant for a strict reading of "independent test".

(ii) Label provenance: the labels were experimental pKd, pKi and pIC50 values, not docking scores. This rules out the subtler leakage in which a predictor is

trained on labels produced by the same docking configuration that is later used for evaluation.

(iii) Family level: only a coarse gene-symbol comparison was possible at that time. One coincidental match was found between two proteins with the same four-letter mnemonic but different organisms and folds. Because no sequence-identity clustering was available, the audit did not claim the absence of leakage at the fold-family level. The leakage-safe split of Section IV.4 was built to close this gap.

(iv) Coverage: 76,803 of the 184,087 annotated entries had a non-zero pK. The training subsample was wide but shallow, with about 11.8 examples per target.

The audit therefore found no disqualifying leakage in the sense of the guidance evaluation, and it directed the investigation toward generalisation.

IV.3 Hyper-Parameter Sweep and Diagnostics on the Old Split

IV.3.1 Learning rate and depth

A light sweep varied the learning rate and the depth of the EGNN, with all other settings, the data and the seed held fixed, and with a full 20-epoch budget per configuration (Table 4.2). Lowering the learning rate to 3×10^{-5} gave the only configuration with a positive test R^2 (0.046) and a clearly better test correlation (0.538) than the deployed configuration (0.233). A shallower, four-layer network had the best validation quality of all four configurations (Pearson 0.806) and the worst test quality (0.119), an illustration of how validation loss on shared-target data can select for models that transfer worse.

Table 4.2 Old split: learning-rate and depth sweep (validation n = 4,061, test n = 27 pockets)

Configuration	Val Pearson	Val R^2	Test Pearson	Test R^2	Test RMSE (pK)
Baseline (lr 1e-4, 6 layers; deployed)	0.788	0.616	0.233	−0.537	2.250
lr 3e-5, 6 layers	0.803	0.628	0.538	0.046	1.772
lr 3e-4, 6 layers	0.727	0.523	0.272	−0.442	2.179
lr 1e-4, 4 layers	0.806	0.641	0.119	−0.671	2.346

The best configuration was labelled as such on the basis of its test metrics, so its test performance is optimistic; it was used only in the learning-curve diagnostic below and was not deployed as a guidance model.

The improvement must, however, be read with the bootstrap intervals for n = 27. The 95% intervals of the test R^2 of the best configuration (−0.601 to 0.445) and of the deployed configuration (−1.769 to 0.183) overlapped, so a positive R^2 could not be claimed. The correlation intervals were more informative: 0.230 to 0.775 for the best configuration, which excludes zero, and −0.145 to 0.595 for the deployed one.

IV.3.2 Five diagnostics

Five diagnostics were applied to characterise the gap (Table 4.3).

Table 4.3 Old split: diagnostics of the generalisation gap

Diagnostic	Result	Reading
Ligand-only baseline (ECFP4, no protein)	Random forest: test Pearson 0.493, R^2 0.047; ridge: 0.427, −0.769	A model without the pocket matches the best EGNN (0.538, 0.046)
Heavy-atom count vs. prediction	Pearson 0.799 ($p < 0.001$, n = 27)	Much of the prediction is explained by ligand size
Pocket scrambling (20 pairs)	Mean change of 1.49 pK units (SD of predictions 1.77)	The model reacts to the pocket, without evidence of benefit from it
Per-target error	26 of 26 groups are singletons	No per-target correlation computable; inconclusive
Learning curve (25/50/75/100% of targets)	Pearson 0.298, 0.305, 0.296, 0.538 (CIs overlap)	Too imprecise to resolve whether more targets help

The learning curve used 127, 254, 380 and 507 targets with the configuration lr 3e-5. All intervals overlapped with their neighbours.

Two readings are supported by these diagnostics. First, on this split, a representation without any protein information was indistinguishable from the pocket-aware network, so the gap was not specific to the EGNN. Second, the predictions correlated strongly with the ligand's heavy-atom count, the same kind of shortcut that is known to inflate docking scores of generated molecules [18], [57]. A third possible reading, that more training data would close the gap, was not confirmed: the learning curve was flat between 25 and 75% of the targets and rose only at 100%, but the four intervals overlapped, so the test set with 27 pockets was too small to resolve this question. At this point the investigation concluded that the gap was more consistent with data scarcity and superficial features than with an architecture flaw. This conclusion was revised in the next section.

IV.4 The Split Artefact: Leakage-Safe Re-Evaluation

IV.4.1 Construction of the split

The diagnostics above had been obtained on a split in which the validation set overlapped with the training set by target and the test set contained only 27 pockets. To repeat the evaluation under a defensible protocol, the target-level leakage-safe (LP) split described in Section III.2.3 was constructed. Two calibration problems were found and repaired during its construction: a rule that seeded the test set had to be bounded to avoid its runaway growth, and a ligand-identity overlap between validation and test that had passed the original algorithm was removed by an explicit check. The final split had 779 training targets (46,964 entries), 65 validation targets (6,069 entries) and 127 test targets (11,855 entries). Of the 1,002 targets carried forward into this assignment (Section III.2.3), 31 were discarded for ligand contamination: 10 on the training side and 21 where a validation target remained

contaminated against the already-finalised test set, leaving 971 kept targets (779 + 65 + 127), with verified zero target overlap between any two subsets.

IV.4.2 Result

The Stage 0 EGNN configuration was retrained on this split without changing any architecture or hyper-parameter (6,000-entry training subsample, learning rate 10^{-4}; best validation loss at epoch 4, early stopping after epoch 9). On the 11,855 test entries it reached a Pearson correlation of 0.609 (95% CI 0.597 to 0.621) and R^2 of 0.342 (0.322 to 0.361), as shown in Table 4.4 and Figure 4.1. A ligand-only random forest reached R^2 0.195, so on the repaired split, the pocket-aware model had a nominal advantage over the ligand-only baseline, opposite to the pattern on the old split (this difference was not tested pairwise, and the complex-level interval is optimistic; see Chapter VIII). The correlation of the predictions with the heavy-atom count decreased modestly, from 0.799 to 0.719, and remained strong.

Table 4.4 Stage 0 checkpoint on the leakage-safe split (test n = 11,855)

Model	**Val Pearson**	**Val R^2**	**Test Pearson (95% CI)**	**Test R^2 (95% CI)**
Ligand-only ridge	0.274	−0.689	0.375	−0.602
Ligand-only random forest	0.405	0.123	0.462	0.195
EGNN (lr 1e-4, 6 layers, unchanged)	0.637	0.394	0.609 (0.597, 0.621)	0.342 (0.322, 0.361)

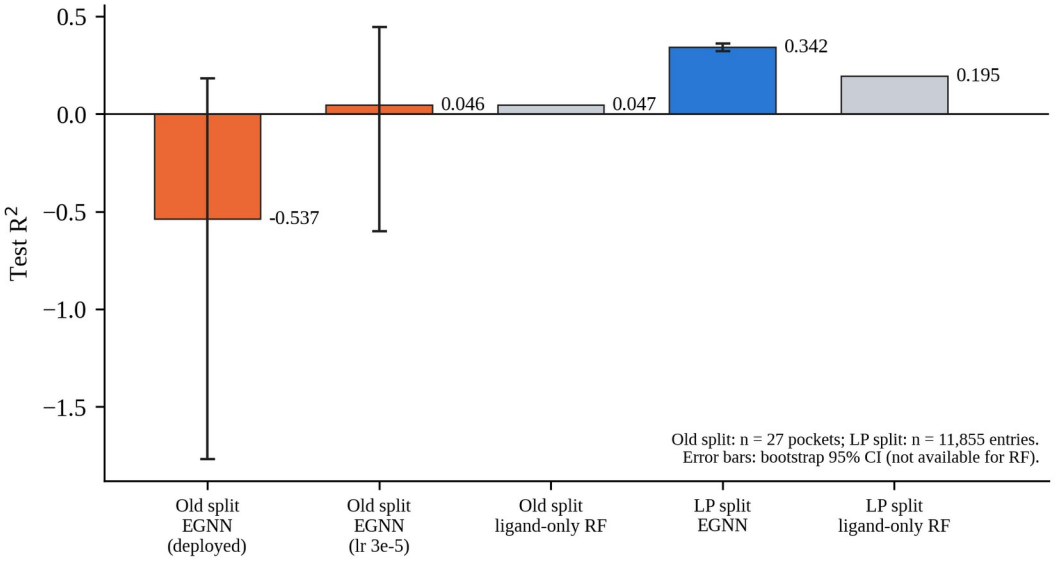


Figure 4.1 Test R^2 of the affinity predictor on the old split (n = 27 pockets) and on the leakage-safe split (n = 11,855 entries). Ligand-only random forests are shown in grey. Error bars are bootstrap 95% confidence intervals where available.

This result contradicted the expectation stated when the split was built, namely that the split alone would not repair the model, and it contradicted the reading of the previous section. It is reported as such. A substantial part of the earlier gap was a consequence of the split construction (target-overlapping validation data and a 27-pocket test set), and not of the EGNN architecture or an unavoidable data limit. The earlier diagnostics were correct as measurements on that split, but the causal story built upon them was answering the wrong question. Three limits of this conclusion should be noted. The first is that "the split" changed in more than one way at once, so the attribution is to the change as a whole and not to any single component of it: the new split simultaneously removed the target overlap between training and validation, enlarged the test set from 27 pockets to 127 targets, and altered which complexes fall on each side. Each of these alone could move the measured gap, the second for reasons of precision rather than of generalisation, and this design does not separate them; an intermediate split that fixed only the validation overlap while keeping a 27-

pocket test set would be needed to apportion the effect, and was not built. What the comparison supports is that the earlier gap was largely an artefact of how the data were divided, not which component of the division was responsible. The second is that the comparison is between models evaluated on different test sets, so the two R^2 values are not paired estimates. The third is that the size shortcut (correlation 0.719) is still present, so a better-than-baseline predictor is not the same as a predictor that has learned binding chemistry.

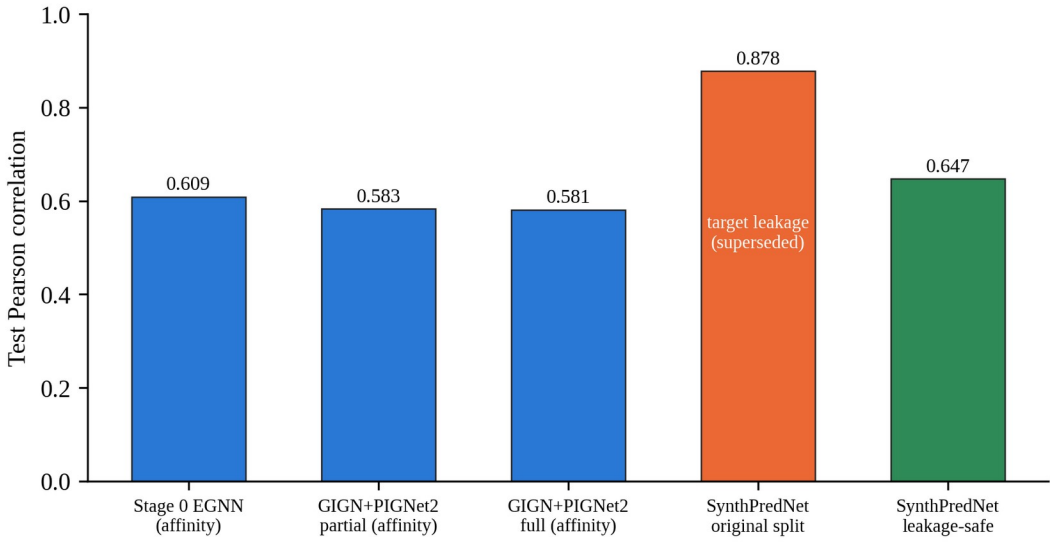


Figure 4.2 Test Pearson correlation of every guidance predictor trained in this thesis on the leakage-safe splits. The synthesizability predictor evaluated on its original random split (orange), which had no target-level separation and no test set, was superseded by the leakage-safe evaluation (green; Chapter VII).

IV.5 Baseline Guidance Ablations

IV.5.1 Task F: the original ablation

The first guidance ablation compared unguided sampling with affinity-guided sampling, with the original predictor at $\lambda = 1.0$. It used 20 of the 100 official test pockets (every fifth pocket in the official order, with one substitution because a receptor file was missing from the data mirror), each a distinct protein by gene

symbol (the coarse family proxy available at this stage of the project; Section III.8), 3 seeds and 10 molecules per (variant, pocket, seed), giving 1,193 scored molecules (597 unguided and 596 guided; 0 failed runs). Docking used AutoDock Vina with exhaustiveness 16 against coordinate-matched receptors; an earlier attempt with receptor files from a different data mirror had produced meaningless zero-interaction scores because of a coordinate-frame mismatch, and was caught in a dry run. The value $\lambda = 1.0$ had been chosen in an earlier single-pocket sweep because it was the strongest tested value that left validity unaffected.

No metric differed significantly between the variants (Table 4.5); all p-values were above 0.27 and most above 0.6, for both the pooled Mann–Whitney test and the Wilcoxon test paired by pocket ($n = 20$). The standard deviation of the unguided Vina Dock (10.08 kcal/mol) was four times that of the guided condition (2.41), which indicated a heavy-tailed distribution produced by a few extreme poses; the paired test, which is more robust to such outliers, agreed with the unpaired test. The numerically lower guided mean Vina Dock (−7.25 against −6.81 kcal/mol) was not distinguishable from noise ($p = 0.96$ unpaired, 0.55 paired), and the ligand efficiency did not worsen (−0.372 against −0.340), so no signature of Vina hacking [18] was found. Because there was no significant gain, there was also no gain to be tested for authenticity.

Table 4.5 Task F: unguided versus affinity-guided sampling ($\lambda = 1.0$), 20 pockets

Metric	Unguided	Guided	Mann–Whitney p	Wilcoxon (paired) p
Validity rate	0.995	0.993	n/a	n/a
QED	0.449 ± 0.185	0.452 ± 0.185	0.889	0.648
SA score	0.601 ± 0.122	0.600 ± 0.121	0.798	0.277
RA-score	0.435 ± 0.386	0.429 ± 0.387	0.816	0.312
Vina Dock (kcal/mol)	−6.813 ± 10.079	−7.251 ± 2.409	0.961	0.546
Ligand efficiency	−0.340 ± 0.749	−0.372 ± 0.109	0.956	0.330

Metric	Unguided	Guided	Mann–Whitney p	Wilcoxon (paired) p
PoseBusters pass rate	0.551	0.545	n/a	n/a
Standard success rate	0.0704	0.0705	n/a	n/a
Honest success rate	0.0302	0.0268	n/a	n/a

Values are means ± standard deviations over molecules. The dual-guidance configuration was identical to the affinity-only variant because the synthesizability weight had been set to zero after the finding reported in Chapter VII.

IV.5.2 Strength re-sweep with the original predictor

Two explanations were considered for the null result: that λ = 1.0 was too weak, or that the predictor's gradient carried no useful signal. The first was tested by docking molecules generated at nine values of λ between 1 and 30 on the bundled example pocket (n = 8 per value). Fragmentation of the molecules started between λ = 20 and λ = 30 (50% single-fragment rate at λ = 30), which defined 20 as the practical safety ceiling. Within the safe range, no comparison with λ = 1.0 was significant (Mann–Whitney p between 0.79 and 1.00 for all eight comparisons), and the mean Vina Dock moved non-monotonically within a band of about 2 kcal/mol. The predictor's own score rose by only about 1.2% across the range, and this drift did not correspond to any trend in independent Vina Dock. The first explanation was therefore refuted, and the null result was not an artefact of an overly conservative operating point.

IV.5.3 What a range of λ does and does not cover

A sweep spanning three orders of magnitude sounds exhaustive, but λ is a dimensionless multiplier on a gradient whose own scale is a property of the trained predictor, so the nominal range does not by itself establish that the *effective* regime was covered. Two measurements made elsewhere in this thesis bound what the range

meant physically, and they are quoted here so the sweep is not read as broader than it was. The gradient magnitudes of the deployed predictor average about 1×10^{-3} in the captured states (Section V.4), and the direct trajectory trace of Appendix E.5 shows that at $\lambda = 1.0$ the summed magnitude of the per-step nudges reaches roughly 9.5 Å over 1,000 steps while the net displacement between the guided and unguided molecule ends below 0.1 Å. The operative quantity is therefore not λ itself but the per-step displacement relative to the sampler's own posterior noise, and across the tested range that ratio stayed small until the point where molecules began to fragment, which is what set the upper bound (Section IV.5.2) rather than any judgement about sufficiency. A regime in which guidance displaces atoms comparably to the posterior noise without destroying the molecule may exist under a different gradient normalisation, a different predictor scale, or a schedule that concentrates the nudge in late steps; variant B1 of Chapter V tests the first of these and does not rescue the result, but the space is not exhausted by a λ sweep alone.

IV.5.4 Strength re-sweep with the repaired predictor

The same question was asked again after the repair of Section IV.4, because the improved predictor could have been expected to change the result. The sweep covered eight values of λ from 0 to 100, with $n = 8$ molecules per value. Robustness improved markedly: fragmentation appeared only at $\lambda = 100$ (one of eight molecules), against $\lambda = 30$ for the original predictor. The independent Vina Dock, however, showed no dose–response (Table 4.6); the comparison of each value against unguided sampling had p between 0.69 and 1.00.

Table 4.6 Repaired predictor: single-pocket strength sweep (n = 8 per value)

λ	Single-fragment rate	Mean own score	Mean Vina Dock (kcal/mol)	SD
0 (unguided)	1.00	7.891	−10.500	1.411
0.1	1.00	7.904	−10.370	1.562
0.3	1.00	7.904	−10.427	1.490
1.0	1.00	7.904	−10.503	1.353
3.0	1.00	7.901	−10.689	1.438
10.0	1.00	7.893	−10.495	1.115
30.0	1.00	7.914	−10.503	1.393
100.0	0.875	7.993	−10.715	0.834

Mann–Whitney U tests against λ = 0: p ≥ 0.69 for every value.

A confirmation over eight diverse pockets (one seed, n = 8 per condition, λ = 0.3, 10 and 100, independent Vina docking with exhaustiveness 16, 32 runs and no failure) gave all 24 pocket-level comparisons as non-significant (p from 0.105 to 1.000). A paired-by-pocket Wilcoxon test at λ = 0.3 reached nominal significance (p = 0.023), but the effect was a consistent, tiny, wrong-directed shift (+0.030 kcal/mol, that is, slightly worse binding under guidance; seven of eight pockets shifting by +0.01 to +0.055), an order of magnitude below the typical per-molecule standard deviation of 1.1–1.6 kcal/mol. With six primary tests, one nominal $p < 0.05$ is expected by chance, and the other two λ values showed no pattern (p = 0.547 and 0.742). It was reported for completeness, and was not treated as evidence that guidance works.

IV.6 A Fuller Information-Content Audit: What Representation Predicts Affinity?

The comparison of Table 4.4 established that a ligand-only random forest on ECFP4 fingerprints (R^2 0.195) did not match the EGNN (0.342). This section asks the same question with a wider and, as it turns out, more damaging set of ligand-only representations, together with every other feature block that could plausibly be

swapped in as a guidance signal: pocket-only descriptors, raw and size-normalised contact-count fingerprints (ECIF), and the typed non-covalent interaction counts (PLIP) considered as a guidance representation in Chapter III. All comparisons below use the leakage-safe split of Section IV.4, target-clustered bootstrap confidence intervals (B = 2,000) and Benjamini–Hochberg correction across every pairwise comparison reported together, following the statistical protocol of Section III.6.

IV.6.1 The representation ladder

Table 4.7 reports the held-out R^2 of eight representation blocks, each paired with the model (ridge regression or a gradient-boosted tree) that performed best on it in a light hyper-parameter search.

Because this section's conclusion turns on a comparison between models, the budget each one received has to be stated, or the comparison is not interpretable. Two asymmetries exist and they point in opposite directions. Within the ladder the budget is deliberately *symmetric*: every block, ligand-only and pocket-aware alike, is fitted with the same five ridge penalties and the same single gradient-boosting configuration, the better of the two families is chosen on the validation set only, and the test set is scored once. Both families are tried for every block precisely so that a block is not penalised for happening to suit one regressor, since the question asked is what the *information* is worth rather than which regressor fits it best. Between the ladder and the EGNN the budget is sharply *asymmetric*, and the asymmetry favours the structure-aware model: the EGNN received hours of GPU training with early stopping on a validation set, while the ligand-only ridge that matches it received six cheap fits. A reader inclined to object that the baselines were under-tuned should note that the objection runs the wrong way here. The finding is that a model given far less

compute, and no protein information at all, is statistically indistinguishable from one given far more and the entire pocket; a *better*-tuned ligand-only baseline could only widen that result, not narrow it. A 14-descriptor ligand-only block (molecular weight, logP, topological polar surface area, rotatable-bond count and ten related RDKit descriptors, no protein information at all) reaches $R^2 = 0.3531$, matching the EGNN's 0.3420 (single seed) and 0.3437 (three-seed mean, SD 0.0377). A single feature, the ligand's heavy-atom count alone, reaches $R^2 = 0.3069$, 90% of the structure-aware model's variance explained. A 38-descriptor pocket-only block, with no ligand information and evaluated on unseen targets, reaches $R^2 = 0.2889$.

Table 4.7 Representation ladder: held-out R^2 of each information source (test, 127 target clusters, B = 2,000)

Representation block	Dim.	Best model	Test R^2	95% CI
Ligand-only (14 RDKit descriptors)	14	ridge	0.3534	—
Heavy-atom count alone	1	—	0.3069	—
Pocket-only (38 descriptors, no ligand)	38	ridge	0.2889	—
ECIF, raw contact counts	72	ridge	0.3701	—
ECIF, size-normalised	72	gradient-boosted tree	0.2077	—
PLIP typed interaction counts, raw	4	gradient-boosted tree	0.0743	[−0.093, 0.173]
PLIP typed interaction counts, size-normalised	4	gradient-boosted tree	0.0822	[−0.043, 0.173]
PLIP counts + one size covariate (heavy-atom count)	5	ridge	0.3241	[0.167, 0.430]
Ligand + pocket (both sources combined)	52	ridge	≈0.35 (tie with ligand alone, see below)	—

Representation block	Dim.	Best model	Test R^2	95% CI
EGNN Stage 0 (three-seed mean)	—	—	0.3437	SD 0.0377, n = 3

Blocks without a listed CI are point estimates from the representation-ladder run (`guidance/cheminformatics/representation_ladder_results.json`) and are compared pairwise, not individually bootstrapped, in the tests below; the model column records which family won on the validation set for that block, not a family chosen in advance. The ligand-only block appears twice in this thesis with slightly different values, and the difference is a fitting detail rather than a disagreement: the ladder refits it inside its own alpha grid and reaches $R^2 = 0.3534$, while the dedicated ligand-only baseline script reaches $R^2 = 0.3531$ on the same features and split. The latter is the figure used wherever this thesis refers to the ligand-only ceiling, including the power analysis of Section IV.8, because that script is also the source of the heavy-atom and ECFP baselines quoted alongside it.

IV.6.2 Ligand size, not interaction identity, is the dominant signal

Three findings converge. First, combining ligand and pocket information produced no gain over either source alone (ligand+pocket vs. ligand: tie; vs. pocket: tie, after correction), so the two sources are largely redundant rather than complementary on this split. Second, the raw ECIF contact-count fingerprint (0.3701) significantly outperformed its own size-normalised variant (0.2077; $\Delta R^2 = +0.1625$, $q = 0.013$), indicating that most of its apparent signal is a re-encoding of ligand size through contact *count* rather than contact *identity*; a specific prior claim that a size-normalised interaction fingerprint of this kind should retain the relevant signal (the CORDIAL finding of Brown et al., reviewed in Chapter II) was not supported on this split, since the normalised fingerprint tied with both ligand-only and pocket-only marginals rather than exceeding them.

Third, and sharper still, the same pattern holds for a representation that is chemically explicit by design rather than only implicitly size-correlated. Four typed non-covalent interaction counts (hydrogen bond, hydrophobic contact, π-stacking and salt bridge, extracted with PLIP and summed per complex, the coarsest possible typed-interaction representation and the one considered for guidance in Chapter III)

explain almost no variance on their own (R^2 0.07–0.08) and lose significantly to both the ligand-only block ($\Delta R^2 = -0.2712$, $q = 0.003$) and the pocket-only block ($\Delta R^2 = -0.2066$, $q = 0.003$). Unlike the ECIF pair, the raw and size-normalised PLIP counts did not differ from each other ($\Delta R^2 = -0.0080$, $q = 0.870$), so this particular block carries no detectable confound between its raw and normalised forms considered alone; what moves the result is adding a single explicit size covariate (heavy-atom count) to the four interaction counts, which raises R^2 from 0.0822 to 0.3241 ($\Delta R^2 = +0.2419$, $q < 0.001$). A representation built specifically to describe *what kind* of contact a ligand makes is, on this split, dominated by *how big* the ligand is to a degree comparable to, and in this instance sharper than, the simpler ECIF count fingerprint.

IV.6.3 Structure rarely beats ligand-only, after correction

A systematic pairwise comparison of every representation block against every other, 40 comparisons in total with Benjamini–Hochberg correction, found 12 cases where a structure-aware block significantly outperformed a ligand-only block, 28 ties, and zero cases where a ligand-only block significantly outperformed a structure-aware one. Three nominally significant wins for structure ($p = 0.039$, 0.044 and 0.049 before correction) did not survive the correction ($q = 0.12$–0.13) and are reported as ties. This is consistent with, and sharpens, the single comparison of Table 4.4: on a target-disjoint, leakage-safe split, no representation of the binding pocket tested in this thesis, including a chemically typed interaction representation, reliably outperforms a 14-descriptor model of the ligand alone.

IV.7 Does More Training Data Close the Gap? Augmenting with BindingNet

The predictor examined so far was trained only on CrossDocked2020. A separate question is whether its ceiling is a property of the architecture and representation, which the previous section addresses, or simply of training-set size and diversity. BindingNet is a larger, structurally paired affinity data set; this section reports a dedicated training-data-composition study (Arms 0, B, A and C) that augments the CrossDocked training pool with BindingNet rows and asks whether the resulting predictor closes the distance to the ligand-only ceiling (0.3531) or exceeds the class-level prior established in Section IV.8.

IV.7.1 A compound-level leak, found and removed before it could inflate a result

Before the production arms were run, a compound-level leakage audit of the combined CrossDocked–BindingNet pool found that 423 of 124,973 candidate training rows (0.34%) shared a ligand, by exact structure match, with a row in the validation or test partition. Measured the other way, the affected ligands accounted for 10.12% of evaluation records (1,814 of 17,924) and 8.6% of distinct validation/test ligands, a thirty-fold difference between the two ways of expressing the same leak that illustrates why a leakage audit must be read from the evaluation side, not only from the training side, to judge its practical severity. A seed-by-seed replay of the exact sampling used for an earlier exploratory run (Arm B, seed 2021) found actual per-seed exposure of 3.06–4.65% of evaluation records across three seeds. A filter was built and applied before any production arm was trained, removing the 423 rows (123 distinct compounds) and is enforced in code, not only upstream in a cached file,

by `train_surrogate_arm.py`'s `compound_filter=True` default, which raises explicitly if its reference summary file is missing.

IV.7.2 Arm A versus Arm C: pure CrossDocked outperforms BindingNet augmentation at matched size

Two arms, each with three seeds, were trained under the batch-size-1, no-mixed-precision regime validated in Section III.7 (an earlier batch-size-4, mixed-precision configuration was rejected at an explicit quality gate, Section III.7.2, because of oscillating validation R^2 and markedly worse final quality). Arm A trained on 20,000 CrossDocked rows only. Arm C trained on the production composition (6,000 CrossDocked plus 14,000 BindingNet rows, matched to the total row count of Arm A) drawn from the leakage-filtered pool. Table 4.8 reports the result, alongside the earlier Arm B-match run (the same composition as Arm C, but from the pool *before* the compound filter existed, included as a check of the filter's practical effect).

Table 4.8 Surrogate predictor, training-data composition (3 seeds per arm, validation R^2, B = 2,000)

Arm	Composition	Seed R^2 values	Mean	95% CI
Arm A	20,000 rows, CrossDocked only	0.351, 0.344, 0.391	0.362	[0.299, 0.425]
Arm C	6,000 CrossDocked + 14,000 BindingNet, leakage-filtered	0.271, 0.292, 0.253	0.272	[0.224, 0.320]
Arm B-match (pre-filter pool, same composition as Arm C)	6,000 CrossDocked + 14,000 BindingNet, unfiltered	0.298, 0.254, 0.283	0.278	[0.223, 0.334]

Comparisons across arms are descriptive (per-arm point estimate and marginal bootstrap CI, n = 3 seeds); a formal target-clustered paired test on matched evaluation rows was not run and would be needed before a significance claim is made. The arms were not budget-matched, and the imbalance runs against the weaker arm: Arm C trained for 38 epochs over 24.6 h against Arm A's 27 epochs over 17.0 h at the same per-epoch throughput (Table A.1), so Arm C's lower R^2 cannot be attributed to under-training.

Before those numbers are interpreted, one confound has to be named, because it offers a competing explanation that the design cannot exclude. Matching the two arms on total row count does not match them on *domain*. The BindingNet rows are compounds aligned into a crystal template, whereas both the validation and the test partitions, and all of Arm A's training data, are CrossDocked docked-and-minimised poses. Arm C therefore trains on a mixture whose larger half is generated by a different procedure from the data it is evaluated on, while Arm A trains and is evaluated entirely in-domain. "Augmenting with BindingNet did not help at matched size" and "training partly out of domain hurt more than the extra data helped" both fit the result, and they are different claims with different implications: the first is about the value of that data, the second about the cost of the mismatch. Separating them would require an arm trained on BindingNet rows re-posed through the CrossDocked pipeline, or evaluation on a BindingNet-style held-out set, neither of which was run. The reading below is therefore stated as a bound on what augmentation achieved *under this protocol*, not as a property of the data source.

Two readings follow, both stated with the appropriate caveat. First, Arm A's CI95 [0.299, 0.425] comes close to, but does not clearly exceed, the ligand-only ceiling of 0.3531; Arm C's CI95 [0.224, 0.320] does not reach the ligand-only ceiling at all, and its lower bound sits below the point estimate of the class-level prior established in the next section (0.2468), so Arm C cannot be confidently distinguished from a predictor that has learned no more than the typical affinity of a target's protein class. Augmenting the training pool with BindingNet rows, even after removing the compound-level leak, did not close the distance to the ligand-only ceiling at the row

counts tested here, and the pure-CrossDocked arm of matched total size performed descriptively better. Second, the compound-level filter itself had little visible effect once applied (Arm C 0.272 against the unfiltered Arm B-match's 0.278, heavily overlapping CIs): the leak identified above does not appear to have been propping up the earlier Arm B result by a large margin, which is a reassuring, if secondary, methodological finding, even though removing the leak remained the correct decision on principle regardless of its measured impact.

IV.8 How Much of the Predictor's Performance Is a Class-Level Prior?

A model can appear to read a protein target's binding pocket while, in fact, reading only the target's class membership: a ligand's chemotype can correlate with the typical affinity range of the protein family it was designed against, without the model having learned anything about structure–activity relationships within that target. y-Randomisation, a standard QSAR control against chance correlation [42], was adapted here to isolate this effect specifically: labels are permuted *within* each target, so the per-target label multiset (and with it, the target's typical affinity range) is preserved, while which ligand carries which label is destroyed. A model that still performs well under this control is reading class membership, not within-target structure–activity relationship.

IV.8.1 Two baselines under within-target permutation

A global permutation, which destroys all structure in the labels, served as a first check: the 14-descriptor ligand-only ridge reached a mean R^2 of only 0.0037 (max 0.0141) under 20 global permutations, far below its real-label performance of 0.3531, confirming that the real-label result is not an artefact of an overly flexible 14-feature model. Under within-target permutation, however, the same ridge model

reached R^2 = 0.2468 (SD 0.0019 across 20 permutations): **70% of its real-label performance is reproducible with no within-target signal at all.** The same control was then run on the deployed EGNN architecture itself, not only on the descriptor pipeline, by retraining it under its exact Stage 0 hyperparameters and split with training labels permuted within target: it reached R^2 = 0.307 (Pearson 0.560), compared with 0.342 (Pearson 0.609) for the same architecture trained on real labels. **The EGNN reproduces 90% of its own real-label performance (0.307 / 0.342) without any within-target structure–activity signal**, a larger fraction than the descriptor baseline's 70%.

What this control measures, and what it does not, should be stated precisely, because the natural way to describe the result overstates it. The permutation destroys the association between a particular ligand and a particular label while preserving each target's label multiset. What survives is therefore everything predictable from target membership plus whatever the features reveal about it, and the measured quantity is a *control-based share of apparent predictive performance*: the fraction of a model's score that does not require within-target structure–activity signal. It is not an identification of the pathway by which the model achieves that share. A plausible reading of why the EGNN's share exceeds the descriptor model's is that the EGNN receives 3-D pocket geometry at its input and can therefore condition on target identity directly, whereas the descriptor model has no protein input and must reach the same prior indirectly, through chemotype–family correlation; but this thesis did not run the ablation that would establish it (for example, retraining the EGNN with the protein channel removed or randomised while keeping labels permuted). The 70% and

90% figures are measurements; the explanation of the gap between them is a hypothesis, and is labelled as one here and wherever it recurs.

IV.8.2 Consequence: a variance decomposition and a narrower window than it first appears

Table 4.9 assembles the resulting decomposition, using the descriptor pipeline (for which both the chance floor and the class-level prior could be measured without a second GPU retraining) as the reference scale, together with an independently estimated label-noise ceiling ($R^2 \approx 0.578$, from a repeat-measurement correlation of 0.76 reported for this assay class in the literature reviewed in Chapter II).

Table 4.9 Variance decomposition of apparent predictive performance

Quantity	R^2
Chance (labels globally permuted)	0.0037
Class-level prior (labels permuted within target)	0.2468
Ligand-only ceiling (real labels, 14 descriptors)	0.3531
Label-noise ceiling (estimated from repeat-measurement correlation)	0.5776
Signal available beyond the class prior (0.3531 − 0.2468)	0.3308 under best known conditions

Read from the correct zero point, that is, from the class-level prior rather than from $R^2 = 0$, the structure-aware ensemble of Chapter III's uncertainty suite accounts for 48.5% of the remaining headroom above the class prior, a 14-descriptor ligand-only ridge for 32.1%, and the original Vina docking score for only 4.0%. The window within which any of these models' true performance can be distinguished, however, is itself narrow: the gap between the floor (the ligand-only ceiling, 0.3531) and the label-noise ceiling (0.5776) spans only 0.2245 R^2, which is narrower than the average width of a single target-clustered bootstrap confidence interval measured for this task (0.3871; ratio 0.58). A single comparison reporting a gain of 0.02–0.05 R^2 against a

similarly sized benchmark, a magnitude common in the literature reviewed in Chapter II, cannot in principle resolve where within this window a model sits. This same ceiling is why the argument made earlier in this thesis, that the Stage 0 predictor's moderate correlation (Pearson 0.58–0.65) rules out a trivial explanation for the guidance failures of Chapters V to VII, needs qualification: a model with no protein input at all reaches a comparable correlation by the class-prior route alone, so correlation magnitude on its own does not establish that the predictor has learned pocket-specific structure–activity relationships. This qualification is returned to in Chapter IX.

A literature check run specifically against this decomposition, after it had been measured, did not find the within-target permutation control applied to protein–ligand affinity prediction elsewhere, so the control itself is reported as novel in this setting. It did, however, find contemporaneous and independent work arriving at a compatible conclusion by a different route: MIRAGE [60], a benchmark released while this thesis was being written, reports that apparent generalisation in affinity prediction tracks the amount of historical *protein-family support* available for a test target rather than ligand chemistry, that controls which cannot exploit the test family stay flat, and that model rankings reverse on novel families. Its proposed reporting standard, performance measured as the excess over a support-insensitive baseline, is the same quantity this section arrives at from the other direction, by constructing that baseline explicitly through label permutation rather than by stratifying on family support. The two do not measure the same thing and neither subsumes the other: MIRAGE localises the effect to family support across models and benchmarks, while the decomposition here localises it within a single deployed architecture and quantifies

what fraction of its own performance survives the destruction of within-target structure–activity signal. That two independent methods, applied to different data by different groups in the same period, both conclude that a large share of reported affinity-prediction performance does not come from learning binding chemistry is stronger evidence than either alone, and it is reported here as convergence rather than as priority for either side.

The labels underlying this entire decomposition are themselves heterogeneous: 37% Kd, 37% IC50 and 26% Ki measurements, and this composition shifts between splits (41% Kd / 33% IC50 in training against 46% IC50 / 26% Kd in test), with IC50 the most assay-condition-dependent of the three and over-represented in the test partition. This is disclosed as a further source of irreducible variance contributing to the label-noise ceiling above, not as a new finding.

IV.9 Where Does the Predictor Fail? Activity Cliffs, Matched Pairs and Protein Families

The sections above describe what the predictor reads on average. This section asks where it fails specifically, using three standard diagnostics from the QSAR literature that none of this project's earlier chapters had applied: activity cliffs, matched molecular pairs and stratification by protein family.

IV.9.1 Activity cliffs

Following the convention of the MoleculeACE benchmark [51] (ECFP4 Tanimoto similarity ≥ 0.9, |Δ log-affinity| ≥ 1.0, computed within target only, since the leakage-safe split is target-disjoint and cross-target pairs are not meaningful here), 2,856 activity-cliff pairs were found among 4,050,366 within-target pairs checked across the 11,855 test ligands and 127 test targets, involving 344 distinct compounds

(2.9% of the test set). On this cliff stratum, the ligand-only model's R^2 (0.3611) nominally exceeded the structure-aware model's (0.2904), the pattern reported for a comparable benchmark in the literature [51], but the paired, target-clustered difference ($\Delta R^2 = -0.0708$, 95% CI [−0.428, 0.065]) did not exclude zero ($p = 0.285$) and is not reported as a significant finding; on the much larger non-cliff stratum (11,511 pairs' worth of ligands, 126 targets), the structure-aware model's nominal advantage ($\Delta R^2 = +0.0594$) was likewise not significant (95% CI [−0.032, 0.150]). Neither stratum resolves a winner at this sample size; the honest reading is that activity cliffs do not provide additional evidence either way beyond the aggregate comparison of Section IV.8.3.

IV.9.2 Matched molecular pairs

Single-cut matched-molecular-pair fragmentation (the standard Hussain–Rea algorithm, applied within target only, requiring the shared core to be at least 60% of molecular weight) found 183 matched series, 375 pairs, 360 with a non-zero affinity difference. Sign agreement with the true direction of the affinity change, the most local and in principle the easiest structure–activity signal a model could capture [30], was 0.461 for the structure-aware ensemble and 0.483 for the ligand-only ridge, both at or below the 0.5 expected by chance (95% CI [0.399, 0.535] and [0.391, 0.596] respectively, both including 0.5). A paired bootstrap of the difference between the two models on the identical pairs (−0.0222, 95% CI [−0.102, 0.043], $p = 0.551$) found no significant difference between them. Neither model reliably predicts the direction of the simplest, most local structure–activity relationship this thesis tested, a finding consistent with, and more severe than, the class-prior decomposition of Section IV.8:

performance across this project's predictors does not appear to derive from structure–activity capture even at the most local level achievable.

IV.9.3 Protein-family stratification

Stratifying the same ligand-only-versus-structure comparison by protein family (Pfam annotation, available for all 127 test targets) found only two of ten family strata with the ten-target-cluster minimum this thesis requires before reporting a verdict: the kinase family PF00069 (11 targets, 3,306 ligands) and a combined "other family" stratum (83 targets, 6,751 ligands); the remaining seven families (3–7 targets each) are marked underpowered rather than given a verdict, because their point estimates varied wildly (one early draft of this analysis printed a spurious ΔR^2 of $+5.8 \times 10^{16}$ for a six-row family whose local label variance was close to, but not exactly, zero, a bug fixed with an explicit relative-variance guard before any result was reported from it). Within the two powered strata, neither showed a significant structure-versus-ligand-only difference after correction (kinase: $\Delta R^2 = +0.0747$, 95% CI $[-0.264, 0.365]$, $q = 0.652$; other families combined: $\Delta R^2 = +0.0459$, 95% CI $[-0.068, 0.154]$, $q = 0.652$). Even within the single largest, practically most consequential family tested, kinases, structure-aware prediction does not significantly outperform a ligand-only baseline, consistent with the aggregate finding of Section IV.8.3 holding down to the family level where it can be tested at all.

IV.10 Summary of the Chapter

Four results of this chapter carry forward. First, an apparent generalisation gap of the affinity predictor was in large part a split artefact: after moving to a target-disjoint split of adequate size, the same architecture explained 34% of test variance (Figure 4.2), with an interval that excluded zero and a nominal margin over a weak

ligand-only baseline (0.20). This margin did not survive a fuller audit: against a stronger 14-descriptor ligand-only model, the structure-aware predictor is statistically indistinguishable (0.3531 against 0.3420–0.3437), and no representation of the pocket tested in this thesis, including a chemically typed interaction representation, reliably beat ligand-only after correction for multiple comparisons. Second, the size shortcut identified early in this chapter (correlation 0.72 with heavy-atom count) generalises: it reappears, in a sharper form, in a chemically explicit interaction representation, where adding one size covariate to four typed interaction counts more than quadruples the explained variance. Third, 70% (descriptor pipeline) to 90% (the deployed EGNN itself) of the predictor's apparent performance is reproducible from a class-level prior alone, under a within-target label permutation that destroys all genuine structure–activity relationship; what remains after removing this prior is a narrow window, smaller than a single confidence interval on this task, and neither activity cliffs nor matched molecular pairs nor the largest single protein family tested provide evidence that any predictor in this thesis captures local structure–activity relationship within it. A dedicated attempt to close this gap by augmenting training data with a larger external affinity data set (BindingNet) did not succeed at matched training-set size, after a compound-level leak in the combined pool was found and removed. Fourth, and decisive for the rest of the thesis, repairing the predictor was necessary but not sufficient: with the repaired predictor as with the original one, guidance produced no detectable change in the independently computed docking score at any tested strength (a span of three orders of magnitude for the repaired predictor). Guidance also changed the predictor's own score only marginally, by 1 to 2%. Three questions followed: whether a differently constructed predictor or a differently constructed

guidance mechanism could recover a signal (Chapter V), whether the same predictor used without a gradient, as a ranker of finished molecules, would do better (Chapter VI), and, once both of those also failed, whether the failure traced to the signal or the mechanism, a question Chapter VI.5 answers directly with a positive control built from a genuine, non-surrogate affinity signal.

CHAPTER V

ARCHITECTURE AND MECHANISM ROUTES: TRACKS A AND B, AND THE GRADIENT DIAGNOSTIC

Chapter IV ended with a null result for the Stage 0 predictor, even after the predictor had been repaired. Two explanations remained that could not be excluded by that result alone. The first was that the *predictor* was the limiting factor, and that a differently constructed predictor would give a gradient that steers generation. The second was that the *mechanism* by which the gradient is applied, in its scale, timing or training objective, was the limiting factor. This chapter tests both explanations in separate routes, Track A and Track B, and then reports a diagnostic that examines the gradient itself.

V.1 Decision Rule for the Two Routes

A standing rule was declared before the routes were run: affinity guidance would be considered not viable only if both routes, independently, failed. A failure of the mechanism alone would not exclude a better predictor, and a failure of the predictor alone would not exclude a better mechanism. Every checkpoint was assessed with the dual-criterion protocol of Chapter III: a stability check of the guidance strength (fragmentation of the generated molecules), a gradient-informativeness test (KS test of the independent Vina Dock distribution, guided against unguided, with the same pocket and seed, corrected with the Benjamini–Hochberg procedure across all tests of the checkpoint), and a degradation check (ligand efficiency and PoseBusters validity). Both routes were run in two tiers: an

exploratory tier (2 to 3 pockets, 20 molecules per condition), and a full tier only where justified.

A remark on statistical power applies to everything in this chapter. The standard deviation of the Vina Dock score within one pocket was about 1.1 to 1.6 kcal/mol. With 30 molecules per condition, the standard error of the difference between two conditions is thus about 0.3 to 0.4 kcal/mol. By a normal approximation, a per-pocket test at the nominal 5% level therefore has 80% power to detect a shift of roughly 1 kcal/mol, and a test at the corrected level of 24 comparisons only for shifts of roughly 1.4 kcal/mol. A null result in this chapter thus means that no effect of this order of magnitude was observed at the pocket level; it does not exclude smaller effects. Pooling across pockets increases the power, and the pooled Mann–Whitney tests of Chapter IV ($p > 0.8$) suggest that pooled effects, if present, are small. These are approximations derived from the reported standard deviations, and not the outcomes of a formal power analysis.

V.2 Track A: A Physics-Anchored Predictor

V.2.1 Architecture and training

Track A replaced the black-box EGNN readout with a predictor whose output is a sum of physically interpretable terms. The backbone consists of three heterogeneous interaction layers of the GIGN type [59] with hidden size 256, which produce an embedding for each protein and ligand atom. The output is the sum of pairwise energy terms between all ligand and protein atoms, as in PIGNet2 [38]: a Morse-type van der Waals term, a hydrogen-bond term, a metal–ligand term and a hydrophobic term, in which the learned atom-pair embeddings set the well depth and width and the coefficients of the remaining terms are initialised at the values of the

released PIGNet2 configuration. Following that configuration, the ionic term and the correction of the van der Waals radii were omitted, the rotor penalty was omitted because rotatable-bond counts were not computed by the featuriser, and no learned output head was added, so that the prediction is the raw energy sum. This last choice preserves interpretability but has a known consequence: without a learned bias and scale, the R^2 of the predictions is expected to be poor even when their correlation with the labels is good. The data augmentation with additional decoy poses that belongs to the original PIGNet2 protocol was not used, in agreement with a scope reduction that was approved for the 4 GB GPU. Protein covalent edges, which are not stored in the data, were approximated with a distance cut-off.

The model was trained on the complete leakage-safe training set (46,964 entries), with Adam (learning rate 10^{-4}), batch size 1 and early stopping with a patience of 12 epochs on the validation loss. Training stopped at epoch 13, after 8,620 s, and the best checkpoint was the one of epoch 1 by both validation loss and test generalisation; a checkpoint selected by validation correlation at epoch 5 did not hold up on the test set. The comparison of these two saved checkpoints was inspected on the test set, but the retained checkpoint is also the one with the lowest validation loss. An earlier, partially trained version (5,000 iterations, about 1.7 epochs over a 3,000-entry subsample) had been used for the exploratory tier.

V.2.2 Predictive quality

The two versions were compared with the Stage 0 EGNN on the same 11,855 test entries (Table 5.1). The correlations of the physics-anchored model (0.583 partial, 0.581 full) were slightly lower than the EGNN's (0.609), and its R^2 was negative (−0.110 partial, −0.189 full). The predictive quality of the full model was almost

identical to that of the partial one, although the former had seen 15 times more data and had been trained to convergence; this suggests that the architecture was near its ceiling in this set-up, and was not limited by the amount of training. The hoped-for reduction of the heavy-atom-count correlation below 0.4 was not achieved (0.719 for the EGNN, 0.640 for the physics-anchored model).

Table 5.1 Track A: predictive quality on the leakage-safe test set (n = 11,855)

Model	Pearson	Spearman	R^2	Correlation with heavy-atom count
Stage 0 EGNN	0.609	n/a	0.342	0.719
GIGN + PIGNet2, partial (5,000 iterations)	0.583	0.623	−0.110	not measured
GIGN + PIGNet2, full training	0.581 (0.564, 0.596)	0.608 (0.595, 0.620)	−0.189 (−0.274, −0.114)	0.640 (0.621, 0.658)

Values in parentheses are bootstrap 95% confidence intervals. RMSE of the full model: 1.797 pK units.

V.2.3 Exploratory tier

The stability check of the partially trained model showed a much narrower usable range of guidance strength than for the EGNN: with a single pocket and four molecules per value, the single-fragment rate was 1.00 at λ = 0.1, 0.75 at 0.3 and 1.0, and 0.00 at 3.0 and 10, whereas the repaired EGNN remained intact up to λ = 30 (one of eight molecules fragmented at λ = 100). A plausible explanation, which was not tested, is that the physics head sums unbounded terms over all atom pairs, so that the same nominal λ corresponds to a much larger gradient on the positions. The exploratory checkpoint used λ = 0.1, 0.3 and 1.0 on three pockets with 20 molecules per condition (Table 5.2).

Table 5.2 Track A exploratory tier: 9 comparisons with the unguided baseline

Pocket : λ	KS p (raw)	KS p (BH)	Mean shift (kcal/mol)	PoseBusters pass rate (unguided → guided)
BSD_ASPTE : 0.1	0.336	0.431	−0.70	not degraded
BSD_ASPTE : 0.3	0.012	0.111	−1.41	not degraded
BSD_ASPTE : 1.0	0.079	0.182	−1.48	0.25 → 0.06
HDAC8 : 0.1	0.175	0.262	−0.32	not degraded
HDAC8 : 0.3	0.081	0.182	−0.54	not degraded
HDAC8 : 1.0	0.037	0.168	−0.54	0.60 → 0.21
CD38 : 0.1	0.571	0.630	−0.46	not degraded
CD38 : 0.3	0.175	0.262	−0.88	0.40 → 0.35
CD38 : 1.0	0.630	0.630	+0.15	0.40 → 0.00

Negative mean shifts indicate better (more negative) docking scores. n = 20 molecules per condition. No comparison survived the correction.

No comparison survived the correction (smallest corrected p = 0.111), so the formal verdict was *no signal*. The pattern was, however, qualitatively different from the earlier nulls: eight of nine comparisons had the correct direction, the raw shifts of up to 1.5 kcal/mol were about ten times larger than any seen with the EGNN, and two raw p-values were below 0.05. The apparent effect was confounded with a loss of physical validity, because the PoseBusters pass rate collapsed at λ = 1.0 in all three pockets (CD38: 0.40 to 0.00). Since part of an apparent docking gain can come from implausible poses that Vina still scores favourably, this exploratory tier was recorded as a yellow flag. It justified running the more expensive full tier, but it also had a selection risk that was recognised at the time: three pockets, and the largest of nine effects reported against the smaller rest.

V.2.4 Full tier

The full tier used the fully trained model, eight pockets (including the three above), a freshly derived stability check (single-fragment rate 0.88 at λ = 0.1 to 0.3, 0.62 at 1.0, complete fragmentation at 3.0 and above; the narrow range was therefore not an artefact of partial training), and 30 molecules per condition (Table 5.3 and Figure 5.1). The 24 comparisons (8 pockets × 3 strengths) were corrected together. None was significant: the smallest raw p was 0.0065 (HDAC8, λ = 1.0), and the smallest corrected p 0.157. Two raw p-values were below 0.05 (0.035 and 0.0065), against about 1.2 expected by chance among 24 tests under the null hypothesis.

Table 5.3 Track A full tier: summary over eight pockets by guidance strength

λ	Mean of pocket mean shifts (kcal/mol)	Pockets with negative mean shift	Pockets with negative median shift	Mean PoseBusters pass rate (unguided 0.561)	Pockets with PoseBusters degradation	Pockets with LE degradation
0.1	−0.154	4 / 8	4 / 8	0.583	1 / 8	4 / 8
0.3	−0.390	7 / 8	6 / 8	0.528	3 / 8	1 / 8
1.0	−0.175	5 / 8	7 / 8	0.165	8 / 8	3 / 8

24 KS tests with Benjamini–Hochberg correction; 0 significant. n = 29 to 30 molecules per condition. LE: ligand efficiency.

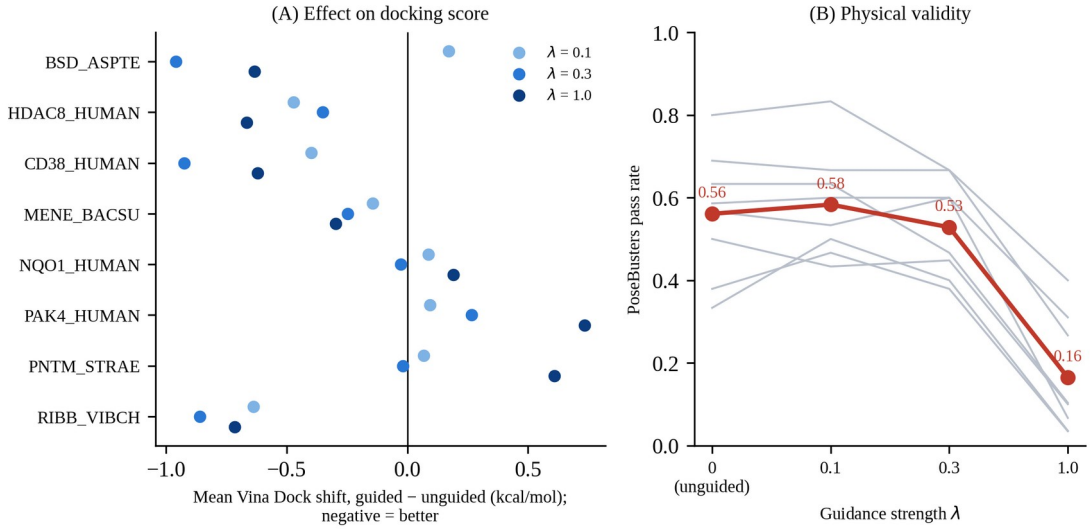


Figure 5.1 Track A, full tier (8 pockets, 29–30 molecules per condition, 24 comparisons corrected together with Benjamini–Hochberg). (A) Mean shift of the independent Vina Dock score for each pocket and guidance strength (negative is better; no shift was significant after correction). (B) PoseBusters pass rate of the unguided samples (λ = 0) and of the guided samples, for each pocket (grey) and on average (red).

Two features of the full tier are worth noting. First, the direction consistency did not strengthen when the sample was enlarged: 16 of 24 comparisons (67%) had a negative mean shift, against 8 of 9 (89%) in the exploratory tier, and the three exploratory pockets were also part of the full set, so this was not even an independent replication. This is the pattern expected if the exploratory impression was a small-sample selection effect. Second, the loss of physical validity was dose-dependent and, at λ = 1.0, universal: all eight pockets degraded and the mean pass rate fell from 0.56 to 0.17. Ligand efficiency degradation did not follow a dose–response. The validity result alone rules out λ = 1.0, and probably λ = 0.3, as operating points, regardless of the docking shift.

By the pre-declared bands the outcome was a clean null. It should be reported precisely as follows: no comparison was significant after correction, the direction consistency was weak and weakened with more data, and the only strength at which a

consistent trend of the median shift appeared (λ = 1.0, seven of eight pockets) coincided with the collapse of the PoseBusters pass rate, so that it cannot be interpreted as improved binding. As noted in Section V.1, effects below about 1 to 1.4 kcal/mol per pocket cannot be excluded.

V.3 Track B: Three Mechanism Variants

Track B kept the Stage 0 EGNN representation fixed and changed how its output was turned into a guidance signal. All three variants used the same evaluation harness (Table 5.4, Figure 5.2).

V.3.1 B3: timestep-windowed guidance

If the gradient carried usable signal only in a part of the diffusion trajectory, uniform guidance would dilute it. Guidance was therefore restricted to the first, the middle or the last third of the 1,000 steps, at λ = 10 (the safe operating point of the repaired predictor), for two pockets with 20 molecules per condition and the same seed within each pocket. All 8 comparisons were *no signal*. Raw KS p-values ranged from 0.57 to 0.98, and all corrected values were essentially 1. Six of eight mean shifts were in the correct direction, but they were tiny (for example −0.002 kcal/mol against a standard deviation of about 1.6), and no PoseBusters degradation was detected in any condition. The ligand-efficiency flag was raised in three of eight comparisons (the middle and late thirds at BSD_ASPTE and the late third at HDAC8), which is not consistent with the statement in the original finding document that no degradation was detected in any condition; the table values were used. Windowing therefore did not create a signal that uniform guidance lacked, which is expected if there was no signal at any time of the trajectory.

V.3.2 B1: gradient-norm normalisation

A possible cause of the null was a poor scale of the raw gradient (about 0.0012 per atom in a smoke test). B1 rescaled the gradient to a fixed norm per atom before applying λ. A three-way design compared unguided sampling, the raw gradient at its established stable strength (λ = 10) and the normalised gradient at a freshly derived stable strength (λ = 0.1, the largest value with a single-fragment rate of 1.00; 0.25 at λ = 0.3 and 0.00 at 1.0). The decisive comparison was normalised against raw, on three pockets with 20 molecules per condition. None of the nine tests survived the correction (smallest corrected p = 0.192). In two of three pockets the normalised gradient was worse than the raw one (medians +0.46 and +0.68 kcal/mol), and in the third it was numerically better (−0.61) but with degraded PoseBusters validity. The hypothesis that a badly scaled magnitude was the cause was thus not supported.

Two lessons of this checkpoint are recorded because they changed the method. First, one raw-gradient molecule of the BSD_ASPTE pocket had a Vina Dock score of +286.58 kcal/mol, against a normal range of −12 to −5. It was verified to be a genuine, if pathological, outcome of a severely malformed pose (the docking code contains no sentinel-value path, and Vina's score is not bounded for catastrophic clashes) and not an error code. This single value shifted the mean of that condition by +14.3 kcal/mol whereas the median shift was +0.06, and it is the reason why the KS test and the median are reported next to the mean. Second, λ = 0.1 had passed the fragmentation check but still reduced the PoseBusters pass rate (HDAC8: 0.60 to 0.30; CD38: 0.40 to 0.25), so whole-molecule connectivity and fine-grained physical validity are different failure modes and a stability check must include both.

V.3.3 B2: classifier-guidance reformulation

B2 replaced the regression gradient with the classical classifier-guidance target [9]. A linear head was trained on the frozen pooled embedding of the Stage 0 network to predict whether the pK is above the training median (7.03), reaching a validation AUROC of 0.783 and a test AUROC of 0.677, and its guidance gradient was that of log σ(logit). The gradient was more tolerant of large strengths (single-fragment rate 1.00 from λ = 1 to 100, and 0.00 at 300), and λ = 100 was used. The unguided and raw conditions were reused from B1, which is valid because unguided sampling does not depend on a guidance model and the raw condition is the same EGNN at the same strength; only the classifier condition was generated (3 pockets, 20 molecules per condition, 6 comparisons).

None of the six tests survived the correction (smallest corrected p = 0.263). By median, however, the classifier gradient made the Vina Dock score *worse* in all six comparisons (the original finding document counted five of six), and it caused the most severe collapse of physical validity of all Track A and B checkpoints (HDAC8: 0.60 to 0.05). The formal verdict was *no signal*, and not a demonstrated harm, since n = 20 is modest for the score variance, but no comparison looked favourable even before considering significance. The higher tolerance for large λ did not become a more effective signal: a larger displacement was applied before the structure broke, with no benefit to the independently computed docking score. One mean-based row (BSD_ASPTE, classifier against raw) had the correct direction only because of the +286.58 outlier of the raw condition; its median was +0.73.

Table 5.4 Track B: summary of the three mechanism variants (Stage 0 EGNN, 2 to 3 pockets, n = 20 per condition)

Variant	Mechanism	Strength used	Tests (significant after BH)	Median-shift direction	Physical validity
B3	Guidance in one third of the trajectory	λ = 10	8 (0)	Mixed, tiny	No degradation detected
B1	Unit-norm gradient per atom	λ = 0.1 (raw: 10)	9 (0)	Normalised worse than raw in 2 of 3 pockets	Degraded in 2 of 3 pockets
B2	Classifier head, gradient of log σ(logit)	λ = 100	6 (0)	Worse in 6 of 6	Degraded in 2 of 3 pockets, severe at HDAC8

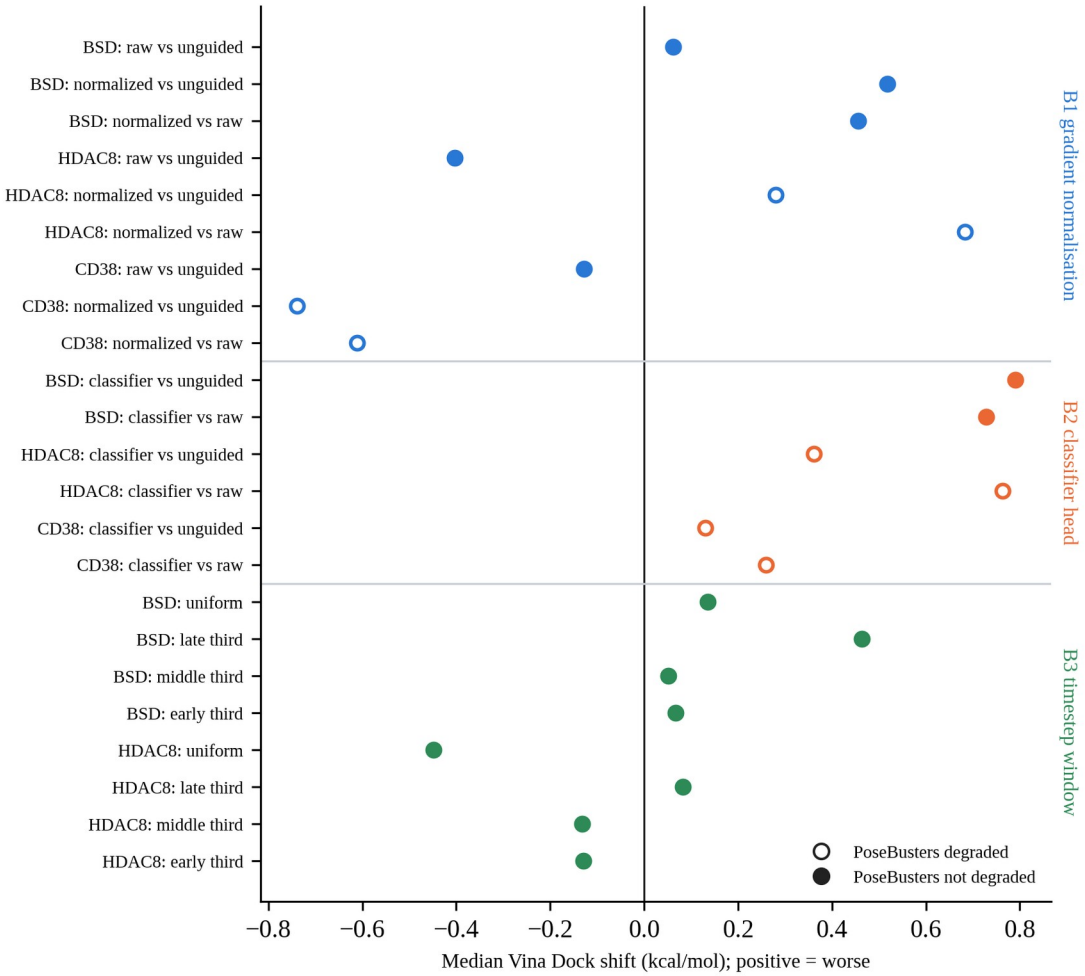


Figure 5.2 Track B: median shift of the independent Vina Dock score for every comparison (positive is worse). Open symbols mark comparisons in which the PoseBusters pass rate decreased. None of the 23 comparisons was significant after correction. Medians are shown because a single +286.58 kcal/mol outlier dominates the mean of one B1 and one B2 comparison.

V.4 Diagnostic of the Gradient Direction

The two routes above established that neither the mechanism variants nor a change of architecture produced an effect. A separate diagnostic asked what the Stage 0 gradient actually points at. It was a standalone diagnostic and not a guidance checkpoint, so no pass or fail gate applied.

V.4.1 Method

Clean-data estimates were captured from *unguided* trajectories at three timesteps (i = 150, 500 and 850, representing the late, middle and early thirds), for three pockets with 16 samples each (144 states), by a read-only hook that does not alter the sampled trajectory. At each state the raw gradient of the Stage 0 predictor was computed post hoc, and the per-atom gradient magnitude was correlated, sample by sample, with two atom properties: (i) the *heaviness* of the atom under its current soft type distribution (the expected atomic number under the softmax of the type logits), as a continuous proxy for the shortcut of preferring heavy elements, and (ii) the minimum distance of the atom to any pocket atom. Per-sample Pearson correlations were averaged and given a bootstrap 95% interval with 2,000 resamples.

V.4.2 Results

The gradient magnitude correlated weakly and negatively with heaviness (mean −0.127; 95% CI −0.178 to −0.075) and moderately and positively with the distance to the pocket (mean +0.240; 95% CI 0.194 to 0.285) (Figure 5.3). Both intervals exclude zero.

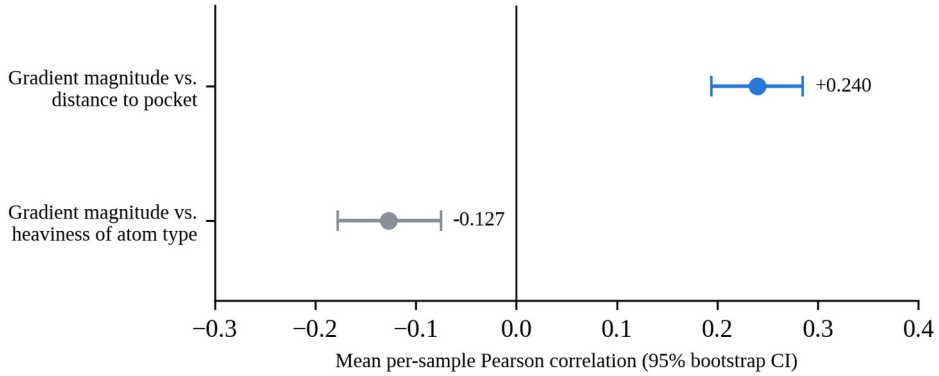


Figure 5.3 Correlation of the per-atom gradient magnitude of the Stage 0 predictor with the heaviness of the atom type and with its distance to the pocket, over 144 captured states (3 pockets, 3 timesteps, 16 samples). Points are means of per-sample Pearson correlations; bars are bootstrap 95% confidence intervals.

V.4.3 Interpretation and its limits

The first correlation does not support the hypothesis that the gradient direction is dominated by the same size shortcut that dominates the predictor's point predictions (Chapter IV): the gradient is, if anything, slightly larger on atoms leaning toward lighter types. This result also has a mechanical counterpart, since the number of ligand atoms is fixed before sampling begins and cannot be increased by guidance within a trajectory. It should be noted that this contradicts a statement found in some earlier project summaries, in which the failure of guidance was attributed to molecular size; the diagnostic data do not show a size effect on the *gradient*. Size acts on the *predictor's output* and on post-hoc *selection* (Chapter VI), which is a different mechanism.

The second correlation is the more informative one. If the gradient was concentrated on the contacts that determine binding, its magnitude would be expected to decrease with the distance to the pocket, that is, a negative correlation. The observed positive correlation means that the largest gradient *magnitudes* occur on the atoms that are on average farthest from the pocket.

One property of this measurement bounds every reading of it and is stated before the interpretation continues: the quantity correlated here is a **norm**, and a norm carries no direction. Knowing that an atom receives a large gradient says nothing about whether it is pushed toward a contact-forming position, away from one, or along it; a large, correctly aimed gradient on a distant atom and a large, misaimed one are indistinguishable in this diagnostic. Phrases such as "the strongest pull" should therefore be read as "the largest magnitude", not as a claim about where atoms are being sent. Establishing direction requires a signed quantity this project did not compute: the cosine similarity between the gradient and the vector from the atom toward its nearest pocket contact, or a finite-difference check of whether taking the step actually reduces the predicted affinity error. That measurement is the natural strengthening of this diagnostic and remains open. @sub The signed measurement, and what it changes

The measurement left open above was subsequently made, on the same predictor and the same kind of captured states, and it is reported here because it revises the reading of this section. For every ligand atom of 23,202 captured atom-states across 20 pockets (the pocket set of Section IV.5, 16 samples each), the cosine was taken between the gradient and the unit vector from that atom toward its nearest pocket atom. The sign convention was fixed against the sampler rather than assumed: `guided_sampling.py` adds the gradient to positions, so a positive cosine means the atom is pushed *toward* its nearest contact. Interpretation rules for every possible outcome were written down before the full run finished (`guidance/DIAG2_INTERPRETATION_PREREGISTRATION.md`).

Two controls had to pass before the result was read. A random-direction control, substituting an isotropic random vector for the gradient, gave a mean cosine of +0.003 with a 95% interval of [−0.004, +0.009], as an uninformative direction should. A finite-difference check stepped the atoms a short distance along +gradient and recomputed the predictor's own score: the mean change was positive (+0.0012) and 80.9% of states increased, confirming the sign convention empirically. That it is 80.9% rather than 100% is reported rather than smoothed: over a finite step in a non-linear function, a minority of states move the other way, which bounds how literally a single step can be read but leaves the convention itself unambiguous.

The result is that **the gradient's direction cannot be distinguished from random**. The mean cosine toward the nearest contact is −0.019, with a target-clustered 95% interval of [−0.046, +0.008] that includes zero, and 48.5% of atoms are pushed toward their nearest contact, which is chance. The pre-registered reading of this outcome, written before the number was known, is that it supports F1, a signal carrying no usable information about the interface, and that it specifically does *not* license the stronger claim that guidance pushes atoms away from contacts. Earlier phrasing in this project, including the first draft of this section, described the gradient as acting in the "wrong direction" on the basis of the magnitude–distance correlation alone; that phrasing is withdrawn here. A norm cannot carry direction, and when direction is measured directly it is absent rather than inverted.

One secondary pattern is reported for completeness, having been declared in advance as descriptive and not decisive: the direction is mildly distance-dependent, with atoms in the nearer half of the distance distribution averaging +0.034 and those in the farther half −0.072 (median nearest-contact distance 3.57 Å). This is consistent

with the magnitude finding of Section V.4.2, in that whatever the gradient does to distant atoms is both larger and, on average, slightly outward; but the aggregate direction remains indistinguishable from random, and no claim rests on the split halves. A single-pocket pilot of this same measurement had given +0.087, which the full 20-pocket run did not sustain, and that discrepancy is exactly why the pilot was never reported as a result.

A related caution applies in the other direction, and is demonstrated rather than hypothesised in this project: Appendix E reports a checkpoint trained specifically to improve this gradient–distance alignment, which did improve the diagnostic itself while leaving the independently computed docking outcome null. Improving the proxy is therefore known, in this project's own data, not to imply improving the outcome, and the diagnostic must not be treated as a surrogate endpoint. This is consistent with the persistence of the null result under changes of scale (B1), timing (B3) and training objective (B2), since none of these changes alters what the representation encodes. It should be read as a correlational finding with a plausible explanation, not as a demonstrated cause. Alternative explanations were not excluded: atoms far from the pocket have fewer neighbours within the cut-off of the network, and the predictor had been trained on clean structures with only 0.1 Å of noise, so it may respond unreliably to the noisy clean-data estimates of early timesteps, a mismatch that has been reported to degrade generation in a related setting [23]. The diagnostic was also run on the EGNN only. A physics-anchored model with distance-decaying pair terms could be expected to produce a negative distance correlation by construction, and re-running the diagnostic on the Track A checkpoint would be a cheap, natural follow-up that was not performed.

V.5 Conclusions of Tracks A and B

Both independent routes failed under the declared criteria. Changing the architecture to a physics-anchored predictor with comparable quality (Track A, 0 of 24 significant comparisons, with weakening direction consistency and universal loss of validity at the highest strength), and changing the scale, timing or objective of the guidance on the fixed representation (Track B, 0 of 23 significant comparisons, with two variants reducing validity) each gave no detectable improvement in the independently computed docking score. The gradient diagnostic offers a testable, if partial, account: the strongest gradient acts on atoms far from the pocket. The logical possibility that a combination of the two routes, for example normalisation applied to the Track A predictor, would work was considered and not pursued, because B1 gave no positive reason to expect that it would behave differently on another representation. The two routes did not address non-gradient use of the predictor, which is the subject of Chapter VI.

CHAPTER VI

REJECTION SAMPLING: THE AFFINITY PREDICTOR AS A POST-HOC RANKER (TRACK C)

The routes of Chapter V used the predictor through its gradient. A predictor with moderate accuracy might nevertheless be useful in a different role, as a *ranker* of finished molecules. In rejection sampling, or best-of-N selection, a large pool of molecules is generated without guidance and only the molecules with the highest predicted affinity are retained. This mechanism requires no gradient through the predictor, no change to the sampler, and no assumption about the noisy intermediate states. It is a different mechanism class from everything tested in Chapter V, and it had been explicitly left out of the conclusions of that chapter. This chapter tests it with the Stage 0 EGNN as the ranker.

VI.1 Design

VI.1.1 Pools, ranker and evaluation

Fifteen pockets of the 20-pocket set of Section IV.5 were used, with two seeds each, giving 30 *pools*. Each pool consisted of 300 molecules sampled without guidance ($\lambda = 0$), reconstructed, scored once by the frozen Stage 0 EGNN (higher score means a stronger predicted binder) and docked once with AutoDock Vina (exhaustiveness 8) and PoseBusters. Between 285 and 300 of the 300 attempted molecules per pool were reconstructed, and no pool failed. The seeds of each pocket were combined, giving pockets of 576 to 600 molecules. Top-k selection, the random

subset and the scrambled filter are all post-hoc selections from the same, already docked pool, so no additional generation or docking was necessary for the analysis.

The scope had been reduced during the run, after a review of the time budget: from 20 pockets and 3 seeds (60 pools) to 15 pockets and 2 seeds (30 pools), because a pool cost about 4.6 h after the optimisations of Section III.7 and the run had to fit in the time available for the thesis. Fifteen pockets remain above the number used in the earlier ablations, and two seeds retain a weaker seed-to-seed consistency check. Pools already completed for a third seed of two pockets were excluded from the analysis and not force-included.

VI.1.2 Statistics

The primary selection was the top 10% of each pocket's pool by predicted score (sensitivity analyses at 5, 20 and 30%). For each pocket, the following were computed:

(i) the shift of the mean Vina Dock score of the selected molecules against the full pool, with a 95% bootstrap interval (2,000 resamples); (ii) an empirical one-sided p-value against 2,000 random subsets of the same size; (iii) the negative control of Section III.6.2, a scrambled filter with 2,000 permutations of the predicted scores, together with a null-consistency check (KS test between the two null distributions), which no pocket failed, so no implementation error of the permutation was indicated; (iv) a KS test of the selected against the full-pool distribution, corrected with the Benjamini–Hochberg procedure across the 15 pockets; (v) three diagnostics of quality degradation: the shift of the heavy-atom count, the PoseBusters pass rate, and the ligand efficiency, which was KS-tested and corrected in the same way.

The comparison of (iv) and (v) is the selected subset against the pool it was drawn from, not against an independent sample or against the unselected remainder; the two groups therefore overlap by construction (every selected molecule is also a member of the full pool), which is a deliberate choice and not an oversight. It answers the practical question this chapter asks: does applying this ranker to a pool of otherwise identically generated molecules change what a practitioner ends up using, relative to not selecting at all. A two-sample test is not strictly valid on overlapping groups, so these p-values should be read as a description of how far the selected subset's distribution sits from the pool's, not as a textbook-exact significance level. This does not by itself explain the pattern reported below, because the random-subset and scrambled-filter controls of (ii) and (iii) are built from the identical pool-versus-subset structure and show no comparable shift; any bias contributed by the overlap alone would be expected to appear equally in those controls, and it does not.

VI.2 Results

VI.2.1 Raw docking score

The selection improved the raw Vina Dock score in most pockets. The mean shift was −1.31 kcal/mol (median −1.33; range −3.73 to +1.95), and in 14 of 15 pockets the KS test of the full selected-versus-pool distribution was significant after the correction. This is a different quantity from the bootstrap confidence interval of the mean shift reported alongside it in Table 6.1, and the two need not agree for every pocket: SQHC is the one case where they do not, with a KS test significant after correction but a mean-shift interval that crosses zero (−0.13, 95% CI −1.22 to +4.66), because a distributional test can detect a shift in shape or spread that a single point estimate of the mean does not resolve; its per-seed shifts (+0.78 and −1.07) were also

opposite in sign, which is the more direct evidence that this particular pocket's effect is weak and inconsistent. Read by themselves, these numbers would suggest a usable rejection-sampling signal.

VI.2.2 Ligand efficiency, size and validity

The signal disappeared when the outcome was normalised by molecular size (Table 6.1 and Figure 6.1). The ligand efficiency of the selected molecules was significantly worse than that of the pool in 14 of 15 pockets (BH-corrected $p < 0.0001$ in all these; mean shift +0.099 kcal/mol per heavy atom, where positive is worse); the one exception, PNTM_STRAE, had a small improvement (−0.030). The selected molecules were larger by 12.0 heavy atoms on average (median 12.7; range 7.4 to 15.1), and their PoseBusters pass rate was lower by 28.8 percentage points on average (median 31.6; range 5.7 to 52.7; mean pass rate 0.57 for the pool and 0.28 for the selection). This is the signature of Vina hacking [18]: larger molecules with poses that are often invalid can receive favourable raw scores without a corresponding gain of binding per atom. The raw improvement reflects, at least in part, that the ranker prefers larger molecules, a preference that is consistent with the correlation of 0.72 between the predictor's output and the heavy-atom count (Chapter IV).

Table 6.1 Track C: top 10% selection by the Stage 0 ranker, per pocket

Pocket	n (pool)	Vina Dock shift (95% CI)	Heavy atoms shift	Ligand-efficiency shift	PoseBusters pass rate, pool → selection	Verdict
NOS2	599	−3.73 (−4.07, −3.30)	+14.3	+0.091	0.79 → 0.43	Confounded by size
PNTM	596	−3.45 (−6.02, −1.96)	+9.5	−0.030	0.62 → 0.45	Confounded by size
PAK4	598	−2.83 (−3.11, −2.55)	+14.8	+0.116	0.77 → 0.48	Confounded by size

Pocket	n (pool)	Vina Dock shift (95% CI)	Heavy atoms shift	Ligand-efficiency shift	PoseBusters pass rate, pool → selection	Verdict
PYRD	597	−2.70 (−2.99, −2.36)	+13.6	+0.090	0.71 → 0.23	Confounded by size
XANLY	599	−2.54 (−2.92, −2.16)	+10.5	+0.054	0.68 → 0.52	Confounded by size
DPP2	596	−2.13 (−2.45, −1.70)	+15.1	+0.145	0.64 → 0.28	Confounded by size
CD38	591	−1.41 (−1.78, −0.97)	+8.2	+0.033	0.48 → 0.42	Confounded by size
IMA1	600	−1.33 (−1.56, −1.15)	+12.7	+0.161	0.80 → 0.38	Confounded by size
MENE	598	−1.31 (−1.61, −0.99)	+7.4	+0.021	0.48 → 0.40	Confounded by size
HDAC8	597	−1.00 (−2.04, −0.28)	+14.8	+0.165	0.55 → 0.12	Confounded by size
NQO1	597	−0.90 (−1.26, −0.62)	+10.1	+0.072	0.61 → 0.28	Confounded by size
SQHC	576	−0.13 (−1.22, +4.66)	+13.4	+0.077	0.27 → 0.10	Confounded by size
BSD	594	+0.21 (−0.14, +0.58)	+10.9	+0.094	0.35 → 0.03	No signal
P2Y12	578	+1.57 (+0.77, +2.46)	+10.8	+0.097	0.29 → 0.09	Wrong direction
RIBB	592	+1.95 (+1.28, +2.65)	+14.4	+0.304	0.54 → 0.02	Wrong direction

Negative Vina Dock shifts are improvements; positive ligand-efficiency shifts are degradations. Vina Dock in kcal/mol; ligand efficiency in kcal/mol per heavy atom. Pockets are sorted by Vina Dock shift; short names denote PNTM_STRAE, PYRD_TRYCC, XANLY_BACGL, MENE_BACSU, SQHC_ALIAD, BSD_ASPTE and RIBB_VIBCH. The verdict of SQHC rests on a weak, seed-inconsistent effect.

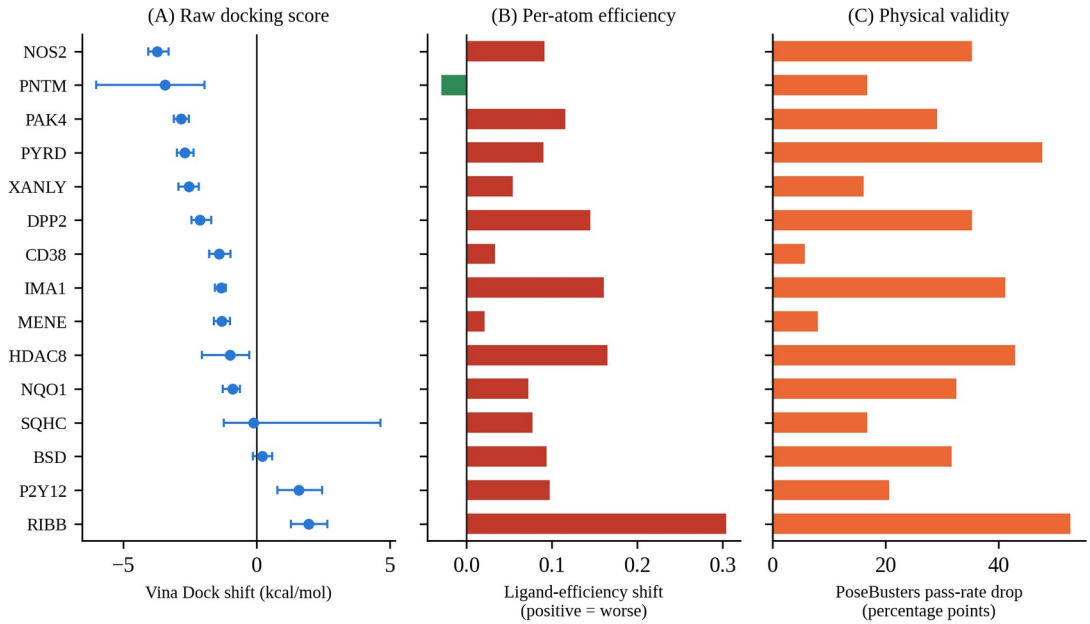


Figure 6.1 Track C, top 10% selection by the Stage 0 ranker in 15 pockets. (A) Shift of the raw Vina Dock score with 95% bootstrap intervals; (B) shift of ligand efficiency (red: worse, green: better); (C) drop of the PoseBusters pass rate in percentage points.

Figure 6.2 shows three concrete examples of what a high-scoring selected molecule from this analysis actually looks like: one top-ranked molecule from the pool of each of three pockets, re-docked with AutoDock Vina and rendered with its principal hydrogen-bond partners labelled. These are illustrative, not a random or representative sample of the 15 pockets' selections, and are included to make the quantitative pattern of Figure 6.1 concrete rather than to support a statistical claim.

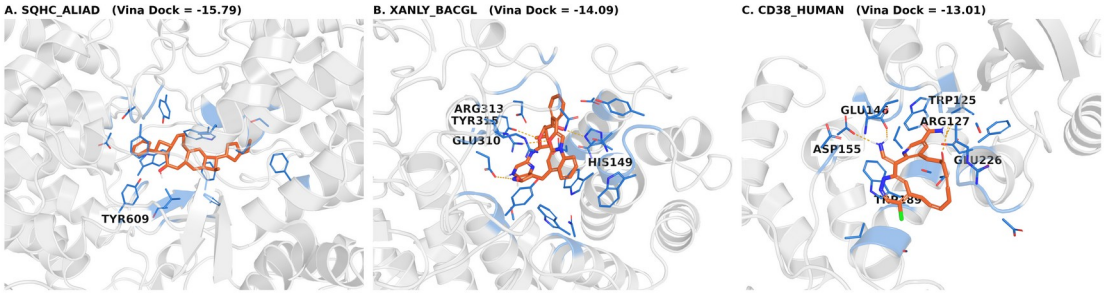


Figure 6.2 Three top-ranked molecules from Track C pools (SQHC_ALIAD, XANLY_BACGL, CD38_HUMAN), re-docked and rendered with hydrogen-bond partner residues labelled. The large, high-affinity-scoring poses illustrate the kind of molecule the ranker favours before the size normalisation of Figure 6.1 is applied.

The verdicts by the pre-declared bands were: 12 of 15 pockets *signal but confounded by size*, 2 of 15 *signal in the wrong direction*, 1 of 15 *no signal*, and none *clear signal*.

VI.2.3 Two pockets with the opposite behaviour

In two pockets, RIBB_VIBCH and P2Y12, the top-10% selection was significantly *worse* than a random subset of the same size on the raw score (empirical p_random = 1.0000 for RIBB_VIBCH, that is, none of 2,000 random draws did as poorly). This held at all four values of k tested for RIBB_VIBCH. In these pockets the ranker was anti-correlated with the independently computed docking outcome, and not just uninformative.

VI.2.4 Sensitivity to the selection fraction

The raw-score effect was not stable when the selected fraction was varied. Averaged over pockets, the shift of the raw Vina Dock score was −1.38, −1.31, −0.94 and −0.55 kcal/mol at k = 5, 10, 20 and 30%, and the number of pockets with a negative shift fell from 11 to 12, 10 and 10. Individual pockets reversed sign (HDAC8: −1.41 at 5% but +1.01 at 30%; PNTM_STRAE: −2.91 at 20% but +1.38 at

30%). The degradation of ligand efficiency was the more robust finding at all fractions. It should be read as a monotone weakening of the raw effect with increasing k, which is the expected behaviour of a ranker whose first-ranked molecules are the largest.

VI.3 Post-Hoc Analysis of the Validity Loss

The drop in the PoseBusters pass rate ranged widely across pockets (5.7 to 52.7 points), and it was asked, after seeing this variation, whether it could be explained. Two exploratory hypotheses were examined, both with n = 15 pockets, Spearman correlations and case-resampling bootstrap intervals, none of them pre-registered or corrected for multiple testing.

The pocket size (the number of protein atoms in the 10 Å crop) correlated *negatively* with the drop (Spearman r = −0.59; 95% CI −0.83 to −0.13; Pearson r = −0.54, p = 0.039; between −0.53 and −0.71 under leave-one-out). Larger pockets thus tolerated the selection better, the opposite of the naive expectation. As a plausibility check, the pocket size correlated with the mean size of the unguided molecules (r = +0.72; 95% CI +0.20 to +0.96). A possible reading is that molecules generated in a small, tight pocket already fit closely, so that a shift toward larger molecules has no room and produces clashes, whereas a large pocket can accommodate a somewhat larger molecule. The spread of heavy-atom counts within the pool predicted the drop independently of the pocket size (partial Spearman r = +0.70; 95% CI +0.17 to +0.91), while the diversity of the pool did not (r = +0.24; CI −0.26 to +0.59). An earlier version of this analysis had truncated each pool to its first 300 molecules, that is, to a single seed, and had produced a non-significant partial correlation; the error

was found by re-running the saved script on the complete pool, and the corrected result is the one reported here.

These correlations are hypotheses for a future, targeted study, for example a model of pocket volume against clash rate, and not findings. With n = 15 and post-hoc selection of the questions asked, they are at the limit of what can be supported.

VI.4 Interpretation

Three conclusions are supported. First, the same frozen predictor that failed as a gradient signal also failed as a post-hoc ranker, by a different route: the selection produced a raw-score gain that turned into a loss when normalised by molecular size, and it lowered physical validity substantially. Second, the negative control behaved as designed (the scrambled-filter and random-subset nulls agreed in all pockets), so the failure is not an implementation error of the selection. Third, the mechanism of failure that this chapter documents is specific: the *selection* is dominated by size. It differs from the mechanism diagnosed for the gradient in Chapter V, where the gradient magnitude was not correlated with atom heaviness in the direction that a size shortcut would predict, and was instead concentrated on atoms far from the pocket. Some earlier summaries of this project describe both failures as having the same root cause, molecular size. This is not what the data show. The two failures are compatible with a common upstream cause, a predictor that encodes ligand-side properties and not pocket-specific interaction, but the evidence for that common cause is indirect (Chapter IX).

Two limits should be kept in mind. The ranker was applied to *unguided* pools of the diffusion model, not to a pool with a different composition. A predictor trained with an explicit size control could behave differently as a ranker, and this was not

tested; in particular, the combination of the Track A predictor with rejection sampling was considered and not run, because the size correlation of that predictor (0.64) was only slightly lower. Finally, the result concerns *top-10% selection* by one specific ranker with a stated set of pockets and seeds; the significant effects of the raw-score improvement and of the degradation are large and consistent across pockets, whereas per-pocket verdicts, such as those of SQHC_ALIAD and BSD_ASPTE, rest on weaker evidence.

VI.5 A Positive Control: Reranking by a Genuine, Non-Surrogate Affinity Signal

Everything in this chapter so far has used the Stage 0 EGNN of Chapter IV as the ranker. A question left open at the end of Section VI.4 is whether the Track C failure and the Chapter V gradient failure share a common cause, a predictor that does not read pocket-specific interaction, or whether failure is simply what any selection-based or gradient-based mechanism produces on this generator regardless of the quality of the signal supplied to it. Neither of the two mechanisms examined so far ever varied the *signal* independently of the *mechanism*: both Chapter V and this chapter, up to this point, supplied only the same trained EGNN. This section removes that confound directly, by replacing the EGNN ranker with AutoDock Vina itself, applied to the identical generator, the identical 20-pocket, 3-seed set used for the baseline and affinity-guided ablations of Section IV.5, and evaluated by the identical independent docking pass used throughout this chapter and Chapter IV. Vina was never trained on this project's pK labels, supplies no gradient, and is a genuine, zero-shot measure of pose quality rather than a learned surrogate.

VI.5.1 Design

For each of the 60 pocket–seed trials, an unguided pool was oversampled at twice the usual yield (K = 2), reconstructed, and scored once with Vina's *score_only* mode, which has no conformational search and runs in about 0.57 s per molecule. The better-scoring half of each pool was kept. The kept molecules were then evaluated by an independent, full Vina *dock* pass (exhaustiveness 16, identical settings to the baseline and affinity-only conditions of Section IV.5). The number that decides which molecules survive selection (the cheap score-only pass) is therefore never the number later reported as the result (the independent dock pass), which rules out the strongest form of circularity, in which a selection criterion and its own evaluation metric are the same quantity and an improvement becomes definitional rather than informative. It does not, however, make the two measurements independent, and this limit is load-bearing for how the result may be read: both passes evaluate poses with the *same empirical scoring function*, differing in whether a conformational search is run. Any systematic bias of that scoring function, the preference for larger ligands documented in Section II.3.1 above all, is therefore shared by the selector and the evaluator rather than cancelled between them. The design establishes that the harness can detect a change driven by a genuinely informative signal; it does not establish that the selected molecules are better binders by a measurement independent of Vina. Section VI.5.3 reports exactly the size and synthesizability movement that this shared bias predicts, and the interpretation in Section VI.5.4 is framed accordingly. A 6-pocket pilot, stratified by the measured per-pocket wall time of the existing 20-pocket runs, was run first; because the pilot and the full run share an output directory and this project's

DONE-marker resumability (Section III.7), the pilot's results were reused, not recomputed, when the run was extended to the full 60 trials.

VI.5.2 Result

Table 6.2 reports the outcome. Reranked molecules reached a significantly better independently computed Vina Dock score than the unguided baseline: −8.068 ± 2.660 kcal/mol against −6.813 ± 10.079 (n = 600 reranked, 597 baseline molecules across the 20 pockets; Wilcoxon, paired by pocket, $p = 1.9 \times 10^{-6}$; Mann–Whitney $p = 3.1 \times 10^{-10}$). A target-clustered bootstrap of the per-pocket delta (B = 2,000) gave −1.267 kcal/mol, 95% CI [−2.149, −0.740], excluding zero.

Table 6.2 Reranking by a genuine Vina signal versus the unguided baseline, 20 pockets × 3 seeds (n = 597 baseline, 600 reranked molecules)

Metric	Baseline	Vina-reranked	Test	p
Vina Dock (kcal/mol)	−6.813 ± 10.079	−8.068 ± 2.660	Wilcoxon (paired by pocket)	1.9×10^{-6}
Per-pocket delta (target-clustered bootstrap, B = 2,000)	—	−1.267, 95% CI [−2.149, −0.740]	—	excludes zero
SA score	0.601	0.572	Wilcoxon (paired by pocket)	8.2×10^{-5}
RA-score	0.435	0.353	Wilcoxon (paired by pocket)	2.67×10^{-5}
QED	0.449	0.462	Wilcoxon (paired by pocket)	0.154
Ligand efficiency (kcal/mol per heavy atom)	−0.340	−0.369	Wilcoxon (paired by pocket)	0.756
PoseBusters pass rate	0.551	0.582	Wilcoxon (paired by pocket)	0.209 (held, did not fall)
Heavy atoms	21.32 ± 9.06	23.09 ± 8.36	Wilcoxon (paired by pocket)	0.0002

p-values are two-sided. The analysis computed a pocket-paired Wilcoxon signed-rank test and an unpaired Mann–Whitney test for every metric; the paired value is quoted here because the pocket is the unit of replication, and both are retained in the saved analysis output. Means, standard deviations and p-values are regenerated from the per-molecule result files by `guidance/rq2_table_from_raw.py` rather than assembled by hand (n = 597 baseline, 600 reranked molecules over 20 pockets). The confound in the last row is discussed in the same place it is reported, not held back for a later section.

VI.5.3 The confound, disclosed directly

The reranked molecules are not a free improvement. They are significantly larger (+1.8 heavy atoms on average, $p = 0.0002$) and significantly less synthetically accessible by both measures used elsewhere in this thesis (SA and RA-score both fall, $p < 10^{-4}$), the same size-correlated pattern that undermined the raw-score gain of Section VI.3.2's selection by the EGNN ranker. If this were the whole story, the result would collapse into the same conclusion as the rest of this chapter: a ranker that prefers larger molecules, producing an apparent improvement that a size-normalised metric erases. It is not the whole story, for two reasons visible in the same table. First, ligand efficiency, the metric in this thesis specifically constructed to detect a pure size-inflation artefact, is flat ($p = 0.756$) rather than degraded, unlike 14 of the 15 pockets of Section VI.3.2. Second, physical validity does not fall; the PoseBusters pass rate holds or rises slightly, the opposite of the pattern that accompanied the EGNN ranker's raw-score gain. The improvement is therefore partly genuine on a per-atom basis and is accompanied by, but not fully reducible to, a shift toward larger molecules; both halves of that sentence are supported by the table and neither should be dropped in summarising it.

VI.5.4 Interpretation: what this isolates

This result answers the question left open at the end of Section VI.4. The selection mechanism tested here, best-of-N selection from an oversampled, unguided pool, scored and evaluated by two separate passes of the same docking program, is mechanistically identical to this chapter's earlier procedure; only the *signal* changed, from a trained EGNN surrogate to Vina's own zero-shot score. Reranking by the genuine signal succeeds, on the same generator, the same 20 pockets and the same

independent evaluation protocol on which reranking by the trained surrogate failed by over-selecting on size alone. This localises the cause of the Track A, B and C failures toward the *signal* rather than the *mechanism*: it supports F1, the hypothesis that the guidance predictor does not carry usable information about the true binding interface (Section V.4's gradient-field diagnostic and the representation-ladder audit of Chapter IV point the same way), and it weakens F2, a correctly informative signal misapplied by a flawed mechanism, as the explanation for the EGNN-based failures specifically, because the same family of mechanism, best-of-N selection, succeeds once supplied with a signal that is actually informative. This does not rule out F2 for the gradient-guidance routes of Chapter V on its own terms, because Vina is not differentiable and this section never tested a gradient mechanism with the genuine signal; Chapter IX returns to this limit and to the particle-guidance extension (SVDD, paired with a genuine value function rather than a trained surrogate) that this result motivates as future work but that was not run for this thesis.

This section's own novelty should not be overstated. Best-of-N and best-of-K selection applied to diffusion-based structure-based drug design, ranked by a combined QED, SA and Vina score, was proposed independently as a generation technique by BoKDiff [29], which explicitly describes itself as the first application of this family of techniques to this problem; Reinforcement-learning-based molecule generation guided directly by docking scores has also been reported to succeed in an unrelated, 2-D setting [49]. The selection mechanism used in this section is therefore prior art, not a new technique. Its contribution is narrower and different in kind: using that established mechanism as a *diagnostic instrument*, a signal-swapped positive control run on the same generator and pockets as this thesis's own gradient-guidance

and rejection-sampling failures, and read against the F1/F2/F3 mechanism framework of Section III.1 in a way the original study does not attempt.

CHAPTER VII

SYNTHESIZABILITY GUIDANCE (TRACK D)

The title of the thesis names two guidance signals, affinity and synthesizability. The chapters so far concerned the first. This chapter applies the same standards to the second: whether the gradient of a learned predictor of synthesizability can steer the frozen generator toward molecules that are more synthesizable according to an independent measure. Synthesizability is a categorically different target from affinity: it is a property of the ligand alone, it is derived from a fingerprint-based classifier, and its predictor was expected to be easier to learn.

VII.1 Target, Predictor and Guidance Design

The independent measure was the RAscore [48], a neural classifier that estimates whether a retrosynthesis planner would find a route for a molecule; its value lies in [0, 1]. The guidance predictor, SynthPredNet, is a ligand-only EGNN with a sigmoid output (0.43 M parameters) trained to regress the RAscore of a ligand from its 3D graph. Since it has no protein input, the guidance depends on the ligand's own geometry only. A guidance function of this kind faces the problem of Section III.3.2: the predictor was trained on features derived from bonds (degree, number of hydrogens and hybridisation), which do not exist during sampling. Two designs were therefore compared: a *placeholder* design in which the three bond-derived fields were set to a fixed "unknown" value, and a *bond-aware* design in which they were replaced by a parameter-free, differentiable estimate computed from the interatomic distances of the current diffusion state (a soft version of the bond-length rule used in the

stability evaluation). Two errors were found and corrected in this estimator with a synthetic-geometry test before it was used: element pairs with no such bond order were scored as if they had the single-bond value, and the steepness of the sigmoid saturated within a fraction of a picometre, which made the indicator a hard step with almost zero gradient on realistic geometries.

VII.2 The Original Design: Divergence between the Own Score and the Independently Computed Score

The first experiment used the placeholder design on the single bundled example pocket, with 8 molecules per value of λ_synth and a single seed, without any statistical test. The predictor's own score increased with λ while the independently computed RAscore of the finished molecules decreased (Table 7.1 and Figure 7.1A). At λ = 0.03 the own score was 29% higher and the independently computed score 53% lower than in the unguided run. Per molecule, 6 of 8 independently computed scores were lower and none was meaningfully higher at λ = 0.03, whereas at λ = 0.01 six were unchanged, one higher and one lower, so no safe and beneficial window was found. Molecules fragmented at λ ≥ 1.

Table 7.1 Original (placeholder) design, one pocket, n = 8 per value

λ_synth	Single-fragment rate	Predictor's own score	Independently computed RA-score
0 (unguided)	100%	0.458	0.361
0.01	100%	0.487 (+6%)	0.365 (+1%)
0.03	100%	0.591 (+29%)	0.168 (−53%)
0.1	100%	0.645 (+41%)	0.154 (−48%)
0.3	100%	0.735 (+61%)	0.126 (−58%)
1.0	12.5%	0.864	one valid molecule
≥ 3.0	0%	n/a	none scoreable

Two independent unguided baselines with the same seed gave independently computed RA-scores of 0.361 and 0.299, because the two runs consumed a different number of random draws before the baseline point. This indicates the size of the run-to-run noise at n = 8.

The suggested root cause was the train–inference mismatch of the bond-derived features. The bond-aware design was the test of this hypothesis, and it did not resolve the divergence. At λ = 0.1 (5 of 8 molecules intact) the own score rose by 12% while the independently computed score fell by 16%, and fragmentation started earlier than in the placeholder design (single-fragment rate 0.625 at λ = 0.1, 0.125 at 0.3, 0 at 1.0). A plausible explanation is that the bond-aware features add a second gradient path (position to estimated bonds to features to score), so that the same nominal λ corresponds to a larger effective step. Time-dependent scheduling and a richer guidance architecture were also considered and were explicitly not attempted, because the result of the bond-aware design did not point to timing or capacity as a suspect, and continuing without a specific hypothesis would have been a search for a fix rather than a test of one.

The precision of these statements is limited. They come from one pocket, n = 8 and a single seed, with no test and no confidence intervals, and a re-examination of the work in the next section confirmed this limitation.

VII.3 Re-Examination: Convergence and Leakage of the Synthesizability Predictor

Before repeating the guidance experiment at higher power, the state of the predictor itself was audited, as had been done for the affinity predictor. Two problems were found. (i) The predictor had been trained with a fixed ceiling of 40 epochs and stopped with its early-stopping counter at 1 of 6, so it had not converged. (ii) Its data were divided by a random 90/10 split of the 15,000 labelled entries with no held-out test set and no leakage check, so the reported quality (validation $R^2 = 0.769$, Pearson

0.878) rested on a split in which the same pocket or a near-identical ligand could occur in both subsets.

The 15,000 entries were re-partitioned by target identity with the leakage-safe assignment of Section III.2.4, without recomputing any label, giving 8,835 training, 1,015 validation and 1,940 test entries (3,210 entries were discarded because their target was not covered by the assignment). The same architecture was retrained without an epoch ceiling (early stopping at epoch 20 with a patience of 15; best checkpoint at epoch 5). On the held-out test set it reached $R^2 = 0.414$, Pearson 0.647, Spearman 0.649 and RMSE 0.287 (Table 7.2 and Figure 4.2). Its quality was therefore clearly better than chance, and higher than the affinity predictors' correlations (0.58 to 0.61). The drop from the earlier values is consistent with the removal of leakage-driven inflation, but the two evaluations differ in more than the split (they also differ in the evaluation set and in convergence), so the difference is not a controlled estimate of the size of the leakage.

Table 7.2 SynthPredNet: original and leakage-safe evaluation

Evaluation	R^2	Pearson	Spearman	RMSE
Original (random split, validation only, not converged)	0.769	0.878	0.871	0.192
Leakage-safe (target-disjoint test set, n = 1,940, converged)	0.414	0.647	0.649	0.287

VII.4 Guidance with the Leakage-Safe Predictor

With the retrained predictor and the bond-aware design, the strength sweep was repeated with the same single-pocket design as Section VII.2, now on pocket

BSD_ASPTE rather than the bundled example pocket used there, with 30 molecules per value, the full 1,000 steps and a grid from 0 to 30 (Table 7.3). A KS test of the independently computed RAscore, guided against unguided, was applied to each of the five values with enough scoreable molecules, and the five tests were corrected with the Benjamini–Hochberg procedure.

Table 7.3 Leakage-safe predictor, bond-aware design, pocket BSD_ASPTE, n = 30 per value

λ_synth	**Single-fragment rate**	**Scoreable (n)**	**Own score**	**Independently computed RA-score**	**Shift of independently computed score (95% CI)**	**KS p (BH)**
0 (unguided)	0.90	27	0.5605	0.2581	n/a	n/a
0.01	0.90	27	0.5668	0.2515	−0.009 (−0.161, 0.140)	0.997
0.03	0.90	26	0.5922	0.1916	−0.066 (−0.207, 0.079)	0.997
0.1	0.77	23	0.5784	0.1863	−0.071 (−0.212, 0.072)	0.997
0.3	0.21	6	0.5577	0.0659	−0.192 (−0.315, −0.078)	0.997
1.0	0.07	2	0.7925	0.2391	−0.018 (−0.194, 0.148)	0.997
≥ 3.0	0.00	0	n/a	n/a	n/a	n/a

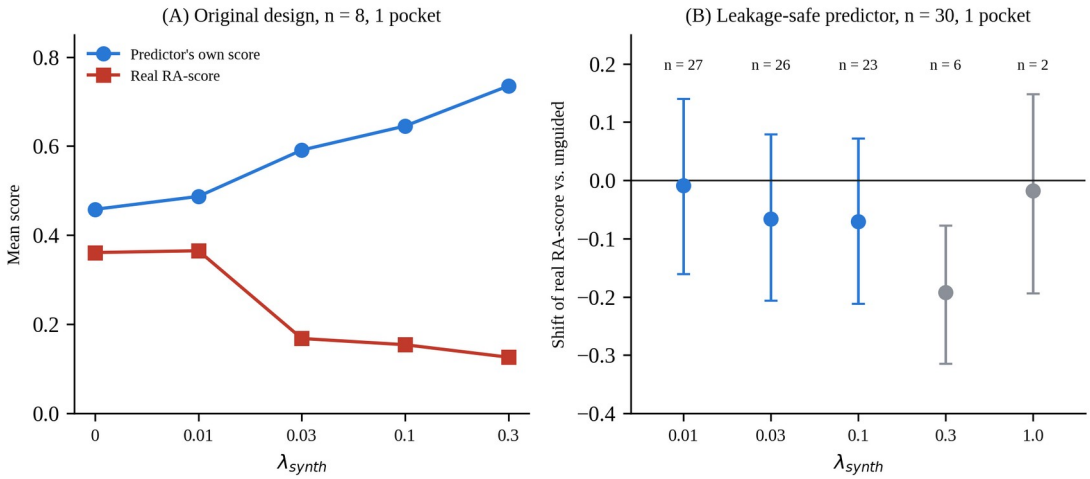


Figure 7.1 Track D. (A) Original design (placeholder features, n = 8, one pocket): the predictor's own score and the independently computed RA-score as a function of guidance strength. (B) Leakage-safe predictor with bond-aware features (n = 30, same pocket): shift of the independently computed RA-score with 95% bootstrap intervals; grey points rest on 6 and 2 scoreable molecules.

No comparison survived the correction (corrected p = 0.997 for all five). The mean shift of the independently computed score was negative at every non-zero strength, and never positive, but all intervals except that of λ = 0.3 included zero, and the interval at λ = 0.3 rests on only 6 scoreable molecules because of fragmentation (single-fragment rate 0.21), which is precisely the kind of large point estimate from a tiny sample that the standards of this work forbid to report as a finding. Two further cautions apply to the consistent sign. The conditions share their initial noise (one seed), so the five shifts are not independent observations. A third caution concerns which molecules enter the comparison at all, and it is structural rather than statistical. The RA-score is computed only on molecules that reconstruct into a single scoreable fragment, so every mean reported here is a mean over survivors. That would be harmless if the survival rate were constant across conditions, but it is not: fragmentation increases with guidance strength, from none at λ = 0 to a single-fragment rate of 0.21 at λ = 0.3. The subsets being compared are therefore selected

differently by the very variable under test, and the selection plausibly runs in a specific direction, since the molecules that survive strong guidance are those that resisted being pulled apart by it. A shift in mean RA-score across λ can accordingly arise from a change in which molecules are scored rather than from a change in the molecules themselves, and this design cannot separate the two. Reporting the shift alongside the survival rate, as done here, is the minimum; an analysis that assigned unscoreable molecules a worst-case value, or that compared only molecules scoreable in every condition, would bound the effect properly and was not run. And the run-to-run noise of the unguided baseline was already about 0.06 to 0.1 RAscore units between nominally equivalent baselines of the same unguided sampler (0.361 and 0.299 at n = 8, Table 7.1; 0.258 at n = 30), which is as large as most shifts in the table. The predictor's own score rose by at most 5.7% at the largest well-powered strength (λ = 0.03: 0.5605 to 0.5922, with an independently computed score change of −25.8% in the mean) and was *not* monotone (0.5577 at λ = 0.3), and no test was applied to it. The divergence between the two scores that was striking in the original design is thus reproduced in sign, but not established statistically, in the leakage-safe design.

What the data support is therefore limited to the following statements: there was no strength at which a meaningful or statistically supported increase of the independently computed RAscore was observed (the only higher mean, +1% at λ = 0.01 in the original design, was within the run-to-run noise); the guidance destroyed the connectivity of the molecules at λ ≥ 0.3; and in the exploratory design, a divergence of the own score and the independently computed score was observed, with a magnitude that the small sample could not confirm. The multi-pocket confirmation (step D.4) was skipped, by a gating rule that had been declared in

advance: if the divergence persisted even partially, the investigation should proceed to the mechanistic diagnostic and not spend multi-pocket effort on an unresolved single-pocket result. The consequence is that the whole of Track D rests on one pocket, and this is a real limitation.

VII.5 Diagnostic of the Synthesizability Gradient

The diagnostic of Section V.4 was adapted to the ligand-only predictor. Clean-data estimates were captured from unguided trajectories at three timesteps, for three pockets with 16 samples each (144 states), and the per-atom gradient magnitude of the bond-aware guidance was correlated with three proxies that could plausibly relate to the classifier's behaviour (Table 7.4). The distance to the pocket does not apply, since the predictor has no protein input.

Table 7.4 Track D diagnostic: gradient magnitude against three atom-level proxies (n = 144 states)

Proxy	Mean correlation	95% CI	Reading
Heaviness of the atom type	0.032	(−0.019, 0.083)	Not distinguishable from 0
Aromatic fraction	−0.014	(−0.061, 0.036)	Not distinguishable from 0
Heteroatom fraction (N, O, S, P, halogens)	0.042	(−0.008, 0.091)	Not distinguishable from 0

None of the three proxies explained the gradient. In contrast with the affinity gradient, for which a concrete correlate (distance to the pocket) was found, no correlate was found here. The intervals are narrow, so correlations above about 0.09 in magnitude with these three proxies are excluded, but the proxies are simple, scalar and per-atom; they do not exclude that the gradient follows a more complex direction. Two candidate explanations were not tested. The first is the known failure mode of classifier guidance in which the gradient exploits directions in the input space that

raise the classifier's output without changing the property that it was trained to recognise [9]. The second is that a fingerprint-based target may have a structure that a smooth, continuous 3D gradient cannot approximate. The asymmetry between the two halves of the thesis, a diagnosed mechanism for affinity and an undiagnosed one for synthesizability, is a real difference in explanatory power and is reported as such.

VII.6 Conclusion of Track D

The gradient of a moderately accurate, leakage-safe synthesizability predictor (test $R^2 = 0.41$) gave no evidence of improving the independently computed RAscore in the single pocket that was examined at adequate sample size, destroyed the connectivity of the molecules at strengths of 0.3 or more, and produced, in the exploratory design, a divergence of the predictor's own score from the independently computed score that was not confirmed statistically after the predictor was retrained. Two fixes had been attempted (bond-aware features and a leakage-safe, converged retrain) and three diagnostic proxies had been excluded. The conclusion is that no benefit was demonstrated, not that harm was demonstrated. The finding also has a methodological corollary that is important for the rest of the thesis: the quality that had been reported for the original predictor came from an unaudited split, and the audit changed it substantially.

CHAPTER VIII

REDESIGN OF THE TRAINING TARGET: DELTA-LEARNING AGAINST THE DOCKING SCORE (TRACK E)

Chapters IV to VII showed that guidance failed for predictors of different architecture, of different objective and of a different target, and that a size shortcut and a misdirected gradient were the best available explanations. Both explanations point to the *training signal*, not to the capacity of the network: a predictor trained to reproduce the experimental affinity can achieve a respectable correlation by learning that larger ligands bind more strongly (Section IV.4), and the gradient of such a predictor need not concentrate on the contacts that matter (Section V.4). Track E tested two targeted modifications of the training signal, chosen after the failures had been diagnosed and pre-registered before any model was trained. Only the first was executed, because it failed its gate; the second was designed, its infrastructure was verified, and it was not run. This chapter reports the design, the pre-registered criteria, the infrastructure and the result.

VIII.1 Rationale and Premise Checks

VIII.1.1 The two modifications

E1 (delta-learning). Instead of the absolute pK, the network is trained to predict the residual $\Delta = pK_{exp} - pK_{Vina}$, where $pK_{Vina} = \max(-E_{Vina}, 0)/1.364$ is the docking score converted to a pK scale, and the prediction is reconstructed as $\hat{y} = pK_{Vina} + \hat{\Delta}$. The idea, known from the Δ-machine-learning literature, is that a physics-based baseline captures the coarse trend and that the

network only needs to learn a correction [40], [54], [58]. The residual is almost uncorrelated with ligand size (Pearson correlation of Δ with the heavy-atom count 0.052, against 0.497 for the absolute pK on the unclipped anchor), so the target formulation removes the size component that the earlier predictors exploited. *E2 (interaction supervision).* Auxiliary heads would predict, for every ligand atom, whether it participates in a hydrogen bond, a hydrophobic contact or a salt bridge, with labels computed from the docked pose by PLIP [43], as an inductive bias that pushes the shared encoder toward the contacts that matter for binding.

VIII.1.2 Premise checks that changed the plan

Four measurements made before training changed the design and the bar for success.

First, "the baseline" was not one number. The old EGNN had $R^2 = -0.537$ on the old split, the Track A model −0.189 on the leakage-safe test set, and the Stage 0 EGNN +0.342. Exceeding a negative R^2 would have been trivial, so the criterion was set against the strongest reference.

Second, simple baselines were as good as the network, and the interval of the earlier comparisons was too narrow. The confidence intervals reported in Chapter IV resampled *complexes*. The independent unit for a new-target claim is the *target*: the test set has 127 targets, with a median of 14 complexes per target and a maximum of 1,392. With a target-clustered bootstrap the interval of the Stage 0 EGNN R^2 widened from 0.322–0.361 to 0.160–0.465, about eight times wider, and the network became statistically indistinguishable from a two-feature linear model of the Vina score and the heavy-atom count ($R^2 = 0.362$) (Table 8.1). Within a pocket, which is where guidance acts, the raw Vina score ranked complexes better than the EGNN (mean

within-target Spearman correlation 0.248 against 0.195), which supports the premise of delta-learning directly.

Third, Δ is size-neutral but not lower in variance (training SD 1.753 against 1.801 for pK). The residual can be learnt only if it is a function of the structure, which is an assumption to be tested, not a given.

Fourth, the gradient of the delta model would not be the gradient of affinity. Vina is an external program without a differentiable path in this pipeline, so at generation time only the gradient of the *correction* $\hat{\Delta}$ would be available. Scoring improvement therefore does not imply guidance improvement, and a gradient diagnostic (criterion C4) was declared as the gate to generative use.

Table 8.1 Track E: reference predictors on the leakage-safe test set (127 targets; target-clustered 95% CI)

Reference	R^2 (95% CI)	Within-target Spearman (95% CI)	Correlation of prediction with heavy-atom count
Constant (training mean)	0.000	n/a	n/a
Vina only, raw (−E/1.364, clipped at 0)	0.076 (−0.213, 0.270)	0.248 (0.156, 0.334)	0.622
Vina only, linearly calibrated	0.273 (0.152, 0.341)	0.248	0.622
Heavy-atom count only (linear)	0.307 (0.135, 0.416)	0.191 (0.051, 0.331)	1.000
Vina + heavy-atom count (linear)	0.362 (0.200, 0.470)	0.266 (0.162, 0.368)	0.940
Stage 0 EGNN	0.342 (0.160, 0.465)	0.195 (0.078, 0.307)	0.719
Track A GIGN + PIGNet2 (no bias term)	−0.189 (−0.768, 0.162)	0.233 (0.142, 0.325)	0.640

Within-target Spearman correlation: mean over the 73 test targets with at least 10 complexes. The ground-truth pK has a correlation of 0.555 with the heavy-atom count on the test set. The anchor is clipped at zero because 79 poses have a positive Vina score (clashes).

VIII.2 Design

VIII.2.1 Architecture and target

The backbone was the Track A model (Section V.2), reused unchanged in its interaction layers with hidden size 256 and 0.494 M parameters. The E1 head kept the physics-sum readout and added a learned bias b and a learned scale s of free sign, $\hat{\Delta} = b + s \cdot (-\Sigma E_phys)$: the bias remedies the missing offset that was identified as the cause of the negative R^2 of Track A, and the scale is not sign-constrained because a positive residual does not need to mean more attraction; it may mean that Vina under-rewards a contact. Since a physics-sum readout could under-represent an arbitrary residual, a scalar MLP on pooled embeddings was a pre-registered *readout ablation* (E1-MLP). The absolute-target control (A0′) used exactly the same backbone and head with the absolute pK as target, so that the only difference between A0′ and E1 was the target (Figure 8.1). The residual was standardised on the training set (mean 0.518, SD 1.753), and the loss was the mean squared error on the standardised residual, matching the loss of the earlier models and the R^2 metric. Huber loss was not used in the primary arms, because a different loss would have confounded the effect of the target.

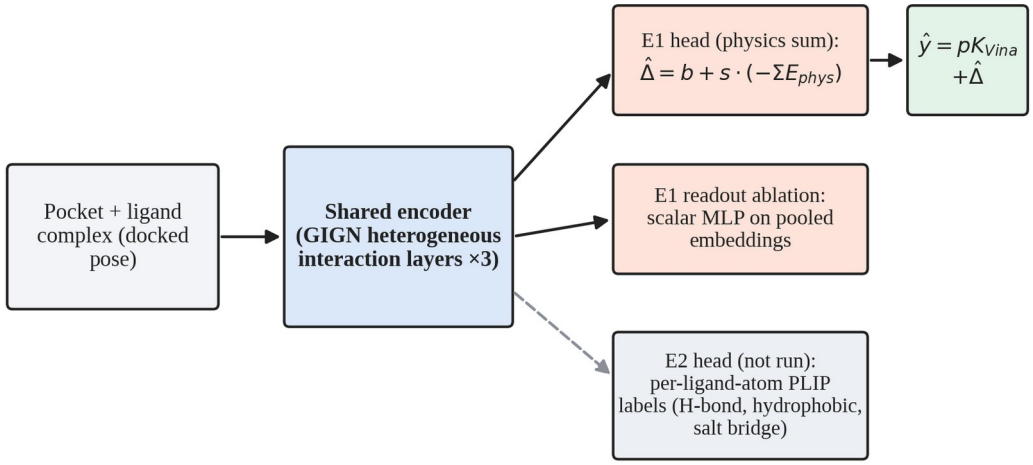


Figure 8.1 Design of Track E. A shared GIGN encoder feeds the E1 delta head (physics-sum readout with learned bias and free-sign scale, primary) and a scalar-MLP readout (ablation); the prediction is reconstructed as the Vina anchor plus the predicted correction. The E2 auxiliary head with PLIP labels was designed but not run.

VIII.2.2 Anchor and its semantics

The Vina scores were taken from the annotation file of the data set, available for all 64,888 entries of the leakage-safe split (none missing). The semantics of that column (score-only against locally minimised) are not documented. They were examined before the target was frozen, by recomputing Vina 1.2.6 scores with this project's preparation for 100 test-set reference poses for which the receptor and the ligand file are on disk. The stored value is a Vina-type score of approximately the stored pose, but it could not be reproduced exactly: the mean difference (stored minus recomputed) was −0.79 kcal/mol for score-only and −0.44 for minimised evaluation, the mean absolute error 0.82 and 0.79 kcal/mol and the Pearson correlation 0.86 and 0.89 (Figure 8.2). The two interpretations could not be distinguished, because for 90 of the 100 poses they differ by less than 0.2 kcal/mol under this preparation (the poses

are already at a local minimum). The stored anchor was therefore treated as an opaque, fixed score attached to the pose, with a preparation noise of about 0.8 kcal/mol (about 0.6 pK units) relative to a recomputation. This does not affect the scoring comparisons on CrossDocked poses, since training and test use the same anchor, but an anchor recomputed for *generated* molecules would carry a systematic offset of about 0.3 to 0.6 pK units that would have to be calibrated before any guidance use.

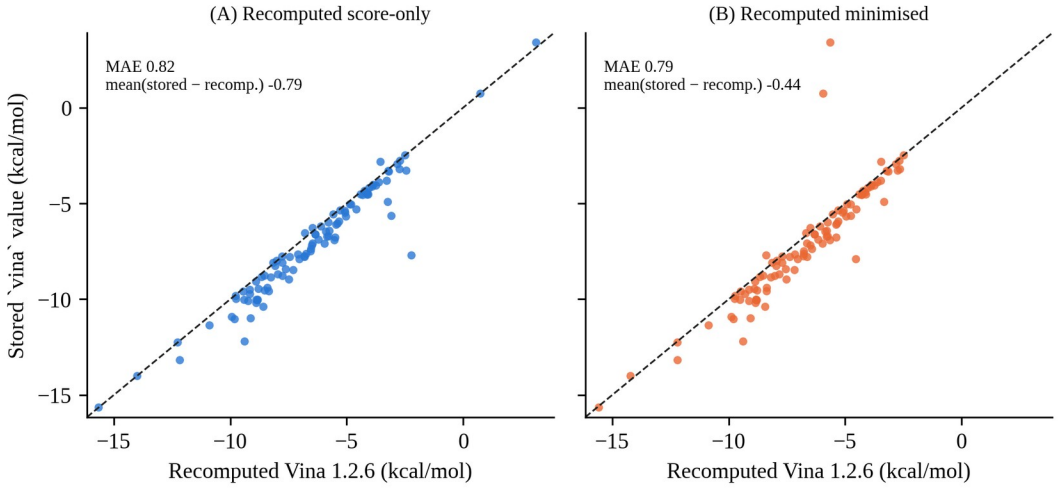


Figure 8.2 The `vina` column of the annotation file against Vina 1.2.6 scores recomputed with this project's preparation for 100 reference poses. (A) Score-only evaluation; (B) local minimisation. The dashed line is the identity.

VIII.2.3 The auxiliary head and label reliability (E2, designed)

The auxiliary head was designed as a per-ligand-atom multi-label classifier with masked, class-balanced binary cross-entropy (positive weights clipped at 10) and a fixed weight of 0.3 chosen a priori, because adaptive weighting schemes each add hyper-parameters and none has been validated for a residual main task with geometry-derived labels. Scrambled labels, permuted across complexes with the same number of ligand atoms so that prevalence and loss scale are preserved while the mapping

from geometry to label is destroyed, were the mandatory negative control. Labels were to be computed once, offline, from pockets rebuilt from the data base (13.3 complexes per second; 81 minutes for all 64,888 complexes on two workers; zero failures on a 200-complex sample; interaction labels of 8 reconstructed complexes reproduced those of real receptors in 7 of 8 cases, with one hydrogen bond differing).

The labels were, however, tested for stability under a 0.2 Å Gaussian perturbation of the ligand coordinates (100 complexes). Hydrogen-bond, hydrophobic and salt-bridge labels were reasonably stable (atom-level Jaccard 0.81, 0.69 and 0.81), but π-stacking merged with π-cation was not (Jaccard 0.16; complex-level presence agreement 0.69) (Figure 8.3). This is a stress test of the whole pipeline and not an error rate against a ground truth, which does not exist, but it showed that π-stacking could not be used as a supervision target. It was therefore excluded from the loss (the labels were stored), and E2 would have been limited to three interaction classes. Even the stable classes churn by 20 to 30% at this perturbation, and labels on diffusion-generated poses, which are noisier than 0.2 Å, would be less reliable still.

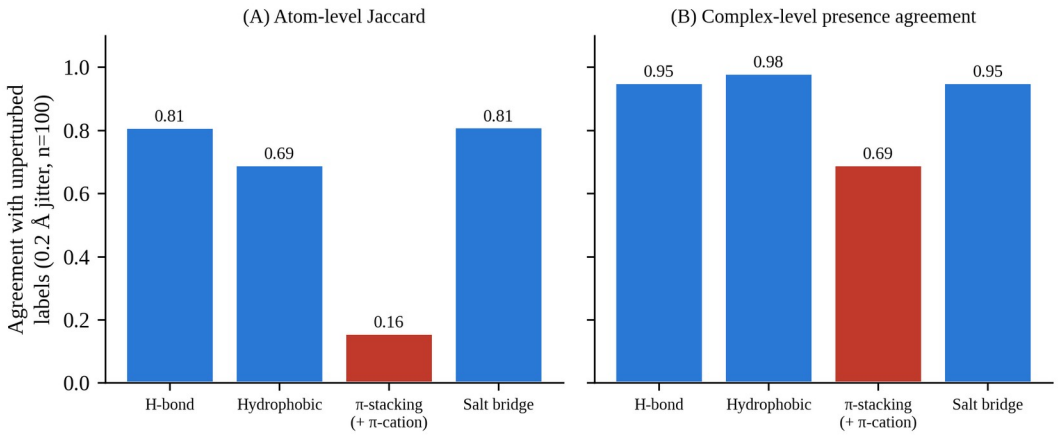


Figure 8.3 Stability of PLIP interaction labels under a 0.2 Å Gaussian perturbation of the ligand coordinates (100 complexes): (A) agreement at the level of atoms (Jaccard index) and (B) agreement at the level of complexes (presence of at least one interaction of the class).

VIII.3 Pre-Registration

VIII.3.1 Arms, metrics and decision rules

The design document was committed before any model was trained, and its criteria were frozen at that commit; later changes were made only as dated amendments (eleven amendments, A1 to A11, none of which changed a threshold; the first eight preceded any training, the next two concerned the audit of earlier work and a literature quotation, and the last, a literature detail, was added after the runs had finished). The arms were the matched control A0′, E1, and E1-MLP, each with three seeds, and, in the second stage, E1+E2 and a scrambled-label control. The reference predictors of Table 8.1 required no training, and two more seeds of the Stage 0 EGNN (seeds 2022 and 2023) were trained to bound the seed variance of the strongest reference. Three metrics were defined: pooled R^2 on the test set (M1), within-target mean Spearman correlation over the 73 targets with at least 10 complexes, which is the guidance-relevant metric (M2; the threshold is necessary, since a within-target rank correlation over three or four complexes is not a meaningful quantity, but it does restrict the metric to a subgroup, and how that subgroup differs from the full test set is quantified below), and the correlation of the prediction with the heavy-atom count (M3). M2's subgroup was characterised rather than assumed representative, because a metric computed on 73 of 127 targets invites the question of what the other 54 would have contributed. The included targets hold 11,652 of the 11,855 test complexes, so the metric covers 98.3% of the evaluation data despite covering 57% of its targets; the excluded targets are small by construction (median 3 complexes each, against 39 for the included ones). They are not, however, a random sample of the rest: their mean experimental affinity is 6.02 pK against 6.81 for the included targets, a systematic

difference of about 0.79 pK units, with comparable spread (SD 1.75 against 1.64). M2 therefore measures within-target ranking on the better-populated and, on average, higher-affinity part of the test set, and a reader should not read it as a statement about sparsely measured or weak-binding targets. The direction of the bias is worth noting: excluding the weakest-binding targets removes part of the affinity range over which a ranking metric would have the most room to fail, so M2 is, if anything, measured under conditions mildly favourable to the models being compared. The resampling unit was the target (paired cluster bootstrap, 2,000 resamples, seed 20260925), an arm's value being the mean over its seeds inside each resample, with Benjamini–Hochberg correction across the tests of a stage.

The pass criteria of the first stage were the following.

(C1a) E1 exceeds the Stage 0 EGNN in R^2 by at least 0.05, with the lower bound of the paired interval above 0. (C1b) E1 exceeds the linear model of Vina score and heavy-atom count, with the lower bound above 0; the decision D3 at the review gate made the joint passing of C1a and C1b the definition of a positive result. (C1c) E1 exceeds the raw Vina score in within-target Spearman correlation, with the lower bound above 0. (C1d) The correlation of the prediction with the heavy-atom count is at most 0.605 (the ground truth 0.555 plus 0.05). (C1e) E1 exceeds A0′ in R^2, which attributes any gain to the delta target and not to the backbone or the bias.

If the point estimates met a margin but the interval included zero, the outcome was to be reported as *inconclusive* and never as a positive or a null result. If C1a failed, E1 would be considered falsified at this scale. The second stage (C2a to C2d and the scrambled-label control C3) and the gradient gate C4 (the correlation of the gradient magnitude with the distance to the pocket must have an upper 95% bound

below +0.10, against +0.240 in Section V.4) would apply only to arms that passed. The order was a hard rule: after the first stage the work stopped for review, and the second stage was not to start automatically.

VIII.3.2 Power, disclosed in advance and recomputed

The paired standard deviation of the R^2 difference between two predictors on the 127 target clusters was measured at 0.044 in the planning phase, for single models, and the disclosed detection probability for a true gain of +0.05 was about 21%; the margins were declared a floor for practical relevance and not a promise of detectability. After the runs, the standard deviation was recomputed for the seed-averaged arms actually used and was 0.075 to 0.085, so that the probability of detecting a true +0.05 gain was only about 8 to 10%. This means that the design could not have *confirmed* a small positive effect. It could, and did, exclude the required one, as shown below.

VIII.4 Infrastructure and Verification

Long runs on a single 4 GB laptop GPU with interruptions require infrastructure that can be trusted. The trainer wrote atomic checkpoints every fixed number of steps and at the end of each epoch, handled termination signals (a SIGTERM mid-epoch produced an atomic checkpoint and an exit code that signals resumption, and a SIGKILL was recovered from the last periodic checkpoint), and resumed exactly. Testing this revealed a real bug: the model was built before the random generators were seeded, so the initial weights were not a function of the seed. After the fix, an interrupted and resumed run was bitwise identical to an uninterrupted run on the CPU (differences exactly 0.0, interrupted mid-epoch 2), and on the GPU the first epoch was bit-identical while later epochs differed at the level of atomic non-

determinism (maximum difference of the predictions 0.048 between a resumed and an uninterrupted run, against 0.059 between two uninterrupted runs), so resuming was indistinguishable from re-running. Ten unit tests covered the delta target, its reconstruction and the label scrambling. The analysis pipeline reproduced the recorded R^2 and Pearson correlation of the Track A model through the new inference path (−0.189 and 0.581), and the fast bootstrap reproduced the frozen reference intervals.

Checkpoint selection was determined by validation data only: the best checkpoint, the learning-rate scheduler, the early stopping (patience 12, at most 30 epochs) and the choice of the readout for the second stage. Test predictions were computed once, from the best checkpoint, after training. This is enforced by an automated static test (five tests: the test split is referenced only at data set construction and inside the final evaluation; the run loop never touches it; the best checkpoint is written only in the branch of a validation improvement), which serves as a positive control against selection on the test set, a practice that has been observed in the field [18]. The audit of the earlier trackers (Section III.8) is the counterpart for the previous work. No interruption occurred in the production runs (zero pause, resume or failure events), so the equivalence of resumed and uninterrupted runs was verified in the dedicated tests, not on the production runs. The first stage cost about 27 GPU-hours (per run 2.2 to 2.6 h for A0′, 2.2 to 2.4 h for E1, 1.5 to 2.8 h for E1-MLP, and 3.3 to 3.5 h for each extra EGNN seed).

VIII.5 Results of the First Stage

VIII.5.1 Predictive quality

Table 8.2 and Figure 8.4 show the test-set results. The delta model with the physics head (E1) reached a mean R^2 of 0.117 (target-clustered 95% CI −0.197 to 0.316; per seed 0.102, 0.176 and 0.074). The absolute-target control on the same backbone (A0′) reached 0.259 (0.099, 0.360), the Stage 0 EGNN 0.342 (0.160, 0.465), and the linear model of Vina score and heavy-atom count 0.362 (0.200, 0.470). The MLP readout was worse and unstable (mean −0.212; per seed −0.391, −0.368 and +0.123). The within-target Spearman correlation of every arm was between 0.18 and 0.27 with strongly overlapping intervals (Figure 8.5); no model demonstrated a within-pocket ranking clearly above the raw Vina score (0.248).

Table 8.2 Track E, first stage: test-set results (127 targets; seed-averaged; target-clustered 95% CI)

Model	Per-seed R^2	Mean R^2 (95% CI)	Within-target Spearman	Correlation with heavy atoms
E1 (Δ target, physics head)	0.102 / 0.176 / 0.074	0.117 (−0.197, 0.316)	0.258	0.677
E1-MLP (Δ target, MLP head)	−0.391 / −0.368 / 0.123	−0.212 (−0.806, 0.130)	0.269	0.576
A0′ (absolute target, same backbone)	0.275 / 0.259 / 0.242	0.259 (0.099, 0.360)	0.225	0.611
Stage 0 EGNN (pre-registered reference)	0.342	0.342 (0.160, 0.465)	0.195	0.719
EGNN, three seeds	0.342 / 0.347 / 0.406	0.365 (0.196, 0.477)	0.180	n/a
Linear: Vina + heavy atoms	n/a	0.362 (0.200, 0.470)	0.266	0.940

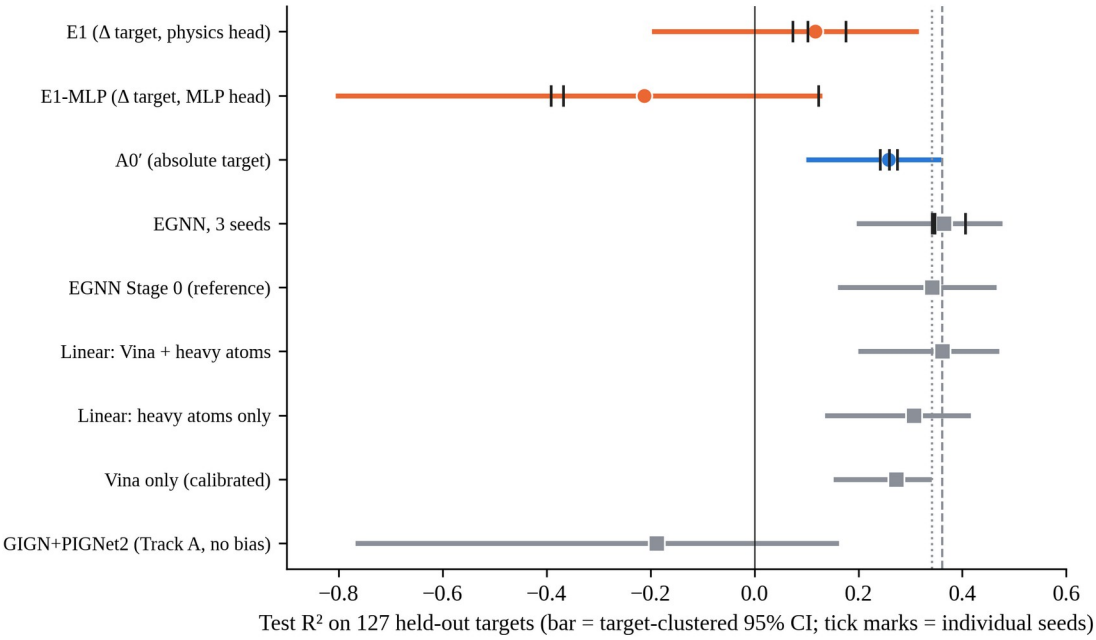


Figure 8.4 Test R^2 of the Track E arms and the reference predictors on 127 held-out targets. Bars: target-clustered 95% confidence intervals; ticks: individual seeds; dotted and dashed vertical lines: the Stage 0 EGNN and the linear Vina + heavy-atom baseline.

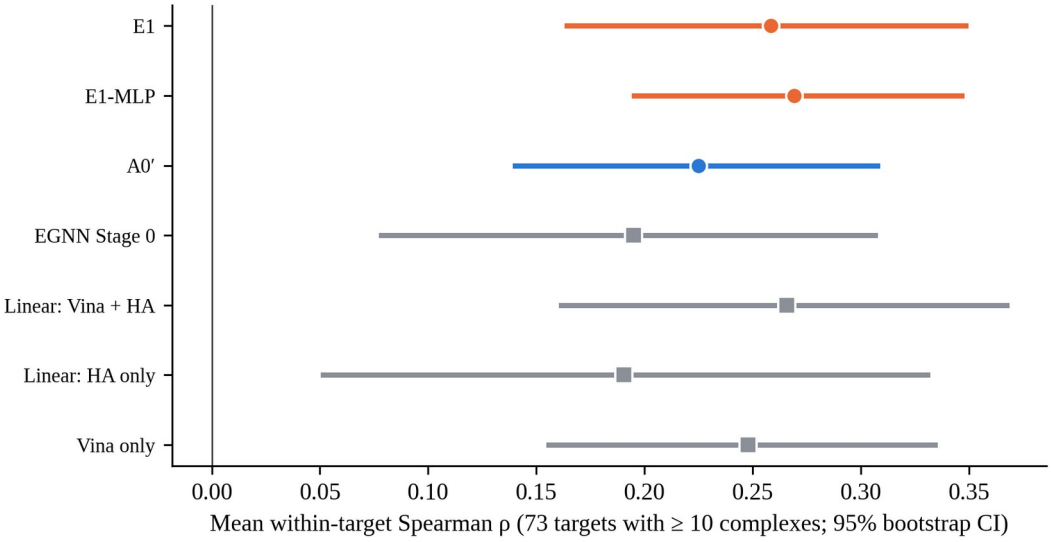


Figure 8.5 Mean within-target Spearman correlation over the 73 test targets with at least 10 complexes (95% bootstrap intervals). All arms overlap with raw Vina.

VIII.5.2 Pre-registered criteria

E1 failed every criterion (Table 8.3). The interval of the difference from the Stage 0 EGNN (−0.224; 95% CI −0.422 to −0.091) lay entirely below the required +0.05 and below zero, and the same was true for the difference from the linear baseline (−0.244; −0.444 to −0.121). The comparison that isolates the delta target, against A0′ on the same backbone, gave −0.141 (−0.318 to −0.026), so the delta target did not help and was worse at the 95% level (the 99% interval, −0.379 to +0.007, just includes zero). E1 was not different from raw Vina in within-target ranking (+0.010; −0.018 to +0.038). The heavy-atom guard was also failed (0.677 against the limit 0.605). E1-MLP failed as well (difference from the EGNN −0.553; −1.085 to −0.249), and it passed the size guard only because it was a poorer predictor overall. All BH-adjusted one-sided p-values for the hypothesis that E1 is better were 1.0 (unadjusted 0.25 to 1.00). Effect sizes were z = −2.63 (C1a), −2.97 (C1b) and −1.83 (C1e), and the paired Cohen's d_z of the per-target squared error against the linear baseline was −0.33.

Table 8.3 Pre-registered criteria of the first stage, applied to E1 (seed-averaged; difference = E1 minus reference)

ID	Test	Difference (95% CI)	99% CI	Verdict
C1a	R^2 against the Stage 0 EGNN (needs at least +0.05 and lower bound > 0)	−0.224 (−0.422, −0.091)	(−0.499, −0.063)	Fail; significantly worse
C1b	R^2 against Vina + heavy-atom linear (lower bound > 0)	−0.244 (−0.444, −0.121)	(−0.513, −0.092)	Fail
C1c	Within-target Spearman against raw Vina (lower bound > 0)	+0.010 (−0.018, +0.038)	(−0.028, +0.048)	Fail; not different

ID	Test	Difference (95% CI)	99% CI	Verdict
C1d	Correlation with heavy atoms at most 0.605	0.677	n/a	Fail
C1e	R^2 against A0′ (lower bound > 0)	−0.141 (−0.318, −0.026)	(−0.379, +0.007)	Fail; the delta target did not help
Sensitivity	R^2 against the three-seed EGNN mean	−0.247 (−0.426, −0.134)	(−0.486, −0.108)	Fail; same conclusion

The outcome is not *inconclusive* in the sense of the pre-registered rule. The design was underpowered to confirm a small gain, but the upper bounds of the three decisive intervals (−0.091, −0.121 and −0.026) lie below zero, so a positive difference of the required size is excluded at this scale. The verdict of the frozen rule was that E1 was falsified at this scale: the tested delta-learning configuration (this backbone, this anchor, this loss, three seeds, the 127-target leakage-safe test set) was falsified against its pre-registered criterion at the evaluated scale, not delta-learning as a general methodology, which other backbones and feature sets have been reported to use successfully elsewhere (Section II.6) and which Section VIII.7 returns to explicitly. The second stage was not run, and neither were the conditional gradient-surgery arm, the sensitivity to the auxiliary weight, or the gradient gate C4, since no arm passed the first stage.

VIII.6 Why the Delta Target Failed

Four pieces of evidence, all measured, explain the result, and they are separated here from the hypotheses that were not tested.

(i) The residual was almost unlearnable from structure by this model (Figure 8.6). The best validation loss of E1 (0.87 to 0.96 in standardised units) was about that of a predictor that always outputs the mean residual (0.92 on the validation set),

whereas the absolute-target control genuinely learnt (0.68 to 0.73). The MLP readout had validation losses of 1.02, 1.02 and 0.87 for its three seeds, that is, worse than the constant for two of them.

(ii) The constant residual already scored $R^2 = 0.193$ on the test set ($\hat{y}$ = pK_Vina plus the mean residual), and E1 (0.117) did not exceed it, so the learnt correction added nothing measurable beyond the Vina anchor. This establishes that *this model did not learn the residual*, which is the claim made; it does not establish that the residual is unlearnable in principle, and at least three explanations remain unseparated by this design. The residual inherits the noise of both its terms, the experimental label and the Vina anchor, so its signal-to-noise ratio is lower than that of the absolute target the same backbone does learn. Its numerical range and variance differ from the absolute target's, so a loss and learning rate tuned for one are not necessarily appropriate for the other, and they were not re-tuned per target type. And the backbone itself is the weaker of the two used in this thesis (Section VIII.2). Auditing these directly, by reporting the residual's own variance against the label noise floor, by re-tuning the loss scale for the residual target, and by repeating on the stronger backbone, is what would convert "not learnt here" into "not learnable"; none of it was run, and the conclusion is worded accordingly throughout.

(iii) Both delta arms had their best checkpoint at epoch 1 or 2 (E1-MLP once at epoch 13) and then overfitted, with the training loss falling to about 0.25 while the validation loss increased, as the absolute-target arm and Track A did.

(iv) The reconstruction inherits the size behaviour of the anchor (the correlation of the prediction with the heavy-atom count was 0.68, against 0.56 for the

ground truth), so the delta target did not remove the size shortcut from the *predictions* although it removed it from the *target*.

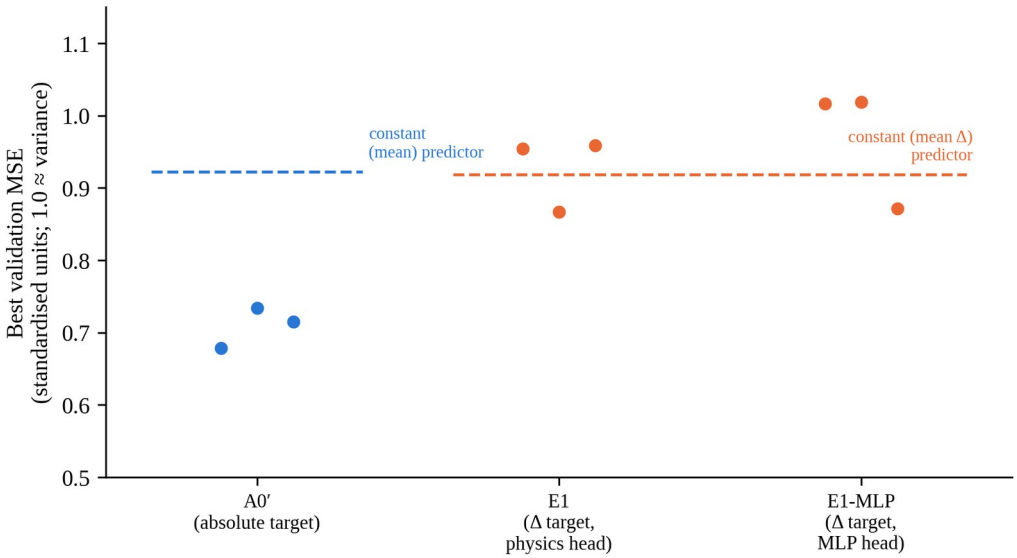


Figure 8.6 Best validation error (standardised units; 1.0 corresponds to the variance of the target) of the three arms for each seed. Dashed lines: error of a constant predictor. The absolute target is learnt; the delta target is not learnt better than by a constant.

A fifth observation is a caveat and not a finding: the GIGN + PIGNet2 backbone was itself weaker than the EGNN under this budget (A0′ 0.259 against 0.34 to 0.41), so E1 was tested on the weaker of the two backbones. The comparison C1e is independent of the backbone, and it failed too. Three hypotheses were *not* tested and must not be stated as findings: heterogeneity of the pK labels (Ki, Kd and IC50 are mixed), the cross-docking provenance of the data set (the receptor and the ligand of a pose can come from different structures, so that the label refers to the ligand's measured binding and not necessarily to that receptor conformation, a source of label noise that affects all arms equally and limits the absolute R^2), and the 0.8 kcal/mol preparation noise of the anchor, which may cap how much of Δ any structure model can recover.

VIII.7 Limits of the Conclusion

The conclusion is limited to this architecture and training budget, three seeds, a 127-target test set, the cross-docked label regime and a fixed, unweighted squared-error loss. It does not show that delta-learning fails in general: tree-ensemble models of a residual on hand-built features have been reported to succeed [54], [58], and a different backbone could behave differently. It does not show that E2 cannot help, since E2 was defined as an addition to a *working* E1 and E1 did not work. E1 was tested for scoring only. The results of this stage also show that its own design could not have confirmed a small effect (detection probability 8 to 10%), and that the seed variance of the EGNN reference (0.342, 0.347, 0.406) is smaller than the shortfall of E1 (0.22 to 0.25).

Three optional analyses that would need no retraining were proposed and not carried out: a robustness test that perturbs the anchor at test time, a re-scoring of previously generated molecules with early-recognition metrics, and any new hypothesis such as E2 on the absolute target or a different backbone, which would need a separate scoping decision. The E2 budget of about 28 GPU-hours was left unspent.

VIII.8 Position in the Literature

To the best of the search that could be performed (with the stated limits: a bibliographic search with the tools available, a verification of the most relevant sources by their abstracts or texts, and one paper, DiffInt, whose training loss could not be confirmed), no study was found that combines (a) a delta-learning target against a docking score, (b) a typed per-atom interaction classification loss from PLIP as auxiliary supervision of a separate guidance predictor, and (c) the use of that

predictor's gradient to steer a frozen 3D diffusion generator. Typed interactions enter diffusion pipelines in three different ways, none identical to E2: as inference-time conditioning (DeepICL [65]), as co-generated diffusion outputs (BInD [32]), and as an auxiliary classification loss inside the generator's own denoiser (InterDiff [56]). The claim of novelty is therefore restricted to the *combination*, formulated as "we did not find", and Track E does not introduce a new architecture. The name of the "Delta Score" of the evaluation literature [18], which is a specificity metric computed against a randomly chosen off-target pocket, is unrelated to the residual target used here. Because the second modification was not run, the combination was designed and pre-registered but not tested; what was tested is the first component.

VIII.9 Summary of the Chapter

A pre-registered, verified and adequately reported test of delta-learning against a Vina anchor did not improve the affinity predictor. It was significantly worse than the Stage 0 EGNN and a two-feature linear baseline, it did not beat an absolute-target control on the same backbone, its best validation error did not beat that of a constant predictor, and its within-pocket ranking was not better than that of raw Vina. The negative result closes the last of the routes tested in this thesis and, with Chapters V to VII, shows that neither a change of the predictor architecture, of the guidance mechanism, of the target of guidance, nor of the training target rescued gradient guidance in this setting.

CHAPTER IX

SYNTHESIS AND DISCUSSION

This chapter draws the experiments of Chapters IV to VIII together. It states what the evidence supports and with what strength, discusses alternative explanations and threats to validity, and relates the findings to the literature. Statements are graded: a *measured* fact is a number that was computed in this work; a *consistent* interpretation is one that the measurements are compatible with and that no measurement contradicts; and a *hypothesis* is an explanation that was not tested.

IX.1 Summary of the Evidence

Table 9.1 summarises the routes. Each row states what was changed relative to the previous routes, the outcome on the pre-declared criteria, and the main limitation of that route. Every route either failed its pre-declared correction (Tracks A, B and D; the baseline ablation) or produced an apparent gain that a size- or validity-normalised re-analysis invalidated (Track C) or a result significantly worse than a simple baseline (Track E).

Table 9.1 Summary of the experimental routes

Route	What was varied	Outcome	Main limitation
Baseline ablation (IV)	Guidance strength λ, original and repaired EGNN predictor	No effect on independent Vina Dock at any tested strength (20 pockets; 8 pockets, 3 strengths); fragmentation at large λ	Predictor is size-correlated (0.72); position-only guidance

Route	What was varied	Outcome	Main limitation
Track A (V)	Predictor architecture (physics-anchored GIGN + PIGNet2)	0 of 24 comparisons significant; direction consistency 89% to 67% with more data; PoseBusters collapse at $\lambda = 1.0$	8 pockets, n = 30; per-pocket detectable shift about 1 to 1.4 kcal/mol
Track B (V)	Scale (B1), objective (B2) and timing (B3) of the gradient	0 of 23 comparisons significant; B2 worse by median in 6 of 6 with severe validity loss	2 to 3 pockets, n = 20
Diagnostic (V)	Gradient direction of the EGNN	Gradient magnitude correlates with distance to pocket (+0.240) and not with heaviness (−0.127)	One predictor; correlational
Track C (VI)	Non-gradient use as a ranker (top 10%)	Raw docking gain in 12 of 15 pockets turns into worse ligand efficiency in 14 of 15, +12 heavy atoms, −29 points PoseBusters	Unguided pools; one ranker
Positive control (VI.5)	Signal swapped: Vina's own score replaces the trained surrogate, identical mechanism and pockets	Significant gain ($p = 1.9 \times 10^{-6}$, 20 pockets × 3 seeds); ligand efficiency flat; PoseBusters held; size and synthesizability both worsen significantly	Only the selection mechanism was tested with a genuine signal; a gradient-based version was not, since Vina is not differentiable
Track D (VII)	Target of guidance: synthesizability	No supported gain; divergence between own and independently computed score in the exploratory design, not confirmed statistically	One pocket for the powered test; three proxies tested for the mechanism
Track E (VIII)	Training target: delta against Vina	Significantly worse than the Stage 0 EGNN and a two-feature linear baseline; not better than a constant in validation	Scoring only; E2 not run; weaker backbone

Figure 9.1 shows the same pattern from a different angle: the fraction of pre-registered comparisons that survived Benjamini–Hochberg correction, for every checkpoint that used the shared KS-test protocol of Section III.6.1. Every gradient-based checkpoint (Tracks A, B and the powered Track D comparison) has a fraction

of exactly zero; the one checkpoint with a large surviving fraction is the rejection-sampling ranker of Chapter VI, and that checkpoint is precisely the one whose apparent raw-score gain did not survive the size-normalised re-analysis of Chapter VI. A high fraction of significant comparisons is, in this one case, a measure of a confound surviving the correction, not a measure of successful molecular design; Figure 9.1's own caption states this explicitly for the same reason.

Table 9.2 restates the five main-thesis routes as a claim, its measured evidence, and the exact scope within which the claim holds, so that no claim in this thesis is read more broadly than its evidence supports.

Table 9.2 Claim, evidence and scope, Tracks A to E

Route	Claim	Measured evidence	Scope of the claim
Track A (V)	A physics-anchored predictor of comparable quality does not rescue gradient guidance	0/24 KS tests significant after BH correction; direction consistency fell from 89% to 67% between tiers	Holds for this predictor, this pocket set (8 pockets, n=29–30) and λ up to the validity ceiling; does not exclude a smaller effect below the ~1–1.4 kcal/mol detectable shift (Section V.1, Figure 9.2)
Track B (V)	Rescaling, reformulating or time-windowing the gradient on a fixed representation does not rescue guidance	0/23 KS tests significant; one variant (B2) worse by median in 6/6 comparisons	Holds for the Stage 0 EGNN representation, 2–3 pockets, n=20; does not test a representation change (that is Track A)
Track C (VI)	Non-gradient, post-hoc top-k selection does not produce a defensible improvement	Raw Vina Dock gain in 12/15 pockets; ligand efficiency worse in 14/15; PoseBusters pass rate down 29 points on average	Holds for top-10% selection from unguided pools by this ranker, 15 pockets; sensitivity analysis shows the raw effect weakens further at larger k; a size-controlled ranker was not tested

Route	Claim	Measured evidence	Scope of the claim
Positive control (VI.5)	The same selection mechanism succeeds once supplied with a genuine, non-surrogate affinity signal, localising the Track A–C failures to signal quality	Significant Vina Dock improvement ($p = 1.9 \times 10^{-6}$), per-pocket bootstrap CI excluding zero; ligand efficiency flat ($p = 0.756$); PoseBusters pass rate did not fall	Holds for Vina-score reranking, K = 2 oversampling, the same 20 pockets × 3 seeds as the main ablations; the gain is accompanied by, but not reducible to, a significant size increase (+1.8 heavy atoms) and lower synthesizability; a gradient analogue was not tested because Vina is not differentiable
Track D (VII)	Gradient guidance toward synthesizability gives no supported gain	0/5 KS tests significant at the powered scale; own-score/independent-score divergence reproduced in sign but not confirmed statistically	Rests on one pocket at adequate power (the powered tier); the exploratory-design divergence is suggestive, not established, and the mechanism of the divergence is undiagnosed
Track E (VIII)	A residual (delta) training target, reconstructed against a Vina anchor, is worse than simple baselines on this backbone	E1 R^2=0.117 (95% CI −0.197, 0.316), significantly below the Stage 0 EGNN, the linear baseline and the absolute-target control on the same backbone, by a pre-registered test	Bounded to this architecture, training budget, three seeds, the 127-target leakage-safe test set and the cross-docked label regime (Section VIII.7); does not generalise to delta-learning with other backbones or hand-built features

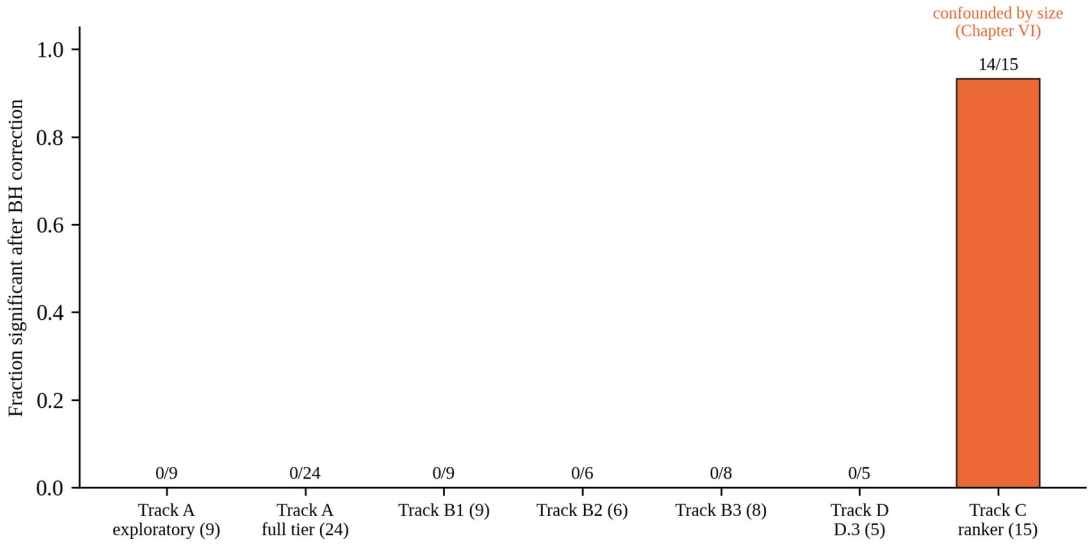


Figure 9.1 Fraction of comparisons significant after Benjamini–Hochberg correction, for every checkpoint that used the KS-test protocol of Section III.6.1. Counts in parentheses are the number of comparisons in that checkpoint. This fraction is a measure of statistical significance, not of successful molecular design: the large surviving fraction corresponds to the Track C ranker, whose raw docking improvement did not survive the size-normalised evaluation of Chapter VI and was accompanied by reduced physical validity.

IX.2 What the Evidence Supports

IX.2.1 Main finding

Under the conditions tested (a frozen TargetDiff model, position-only guidance on the clean-data estimate, predictors trained on clean docked poses, evaluation with Vina docking, PoseBusters, ligand efficiency and the RAscore, and pocket sets of 1 to 20 targets) neither the gradient of a learned affinity predictor, nor its use as a ranker, nor the gradient of a learned synthesizability predictor produced a defensible improvement. The result was reproduced for predictors of different quality (Pearson correlation 0.58 to 0.65 on leakage-safe test sets), architecture and target, and it was not explained by low predictor accuracy or by a bad choice of strength, since the strength was varied over three orders of magnitude and the structural integrity of the molecules limited it from above.

IX.2.2 Predictor quality is not guidance usefulness

A conceptual point runs through every route of this thesis and is stated here explicitly because it is easy to miss when each result is read on its own: a predictor that is accurate on a held-out test set does not thereby imply that its gradient, or its score, is useful for steering a generator. The two quantities were measured separately throughout, and they dissociate. The clearest demonstration is post-synthesis: of the eight further checkpoints of Appendix E, the one with the best predictive quality of any variant tried (ESM2-augmented EGNN, Pearson 0.731, Appendix E.2.3) and the one trained with the entire target protein family held out of training ($R^2 = 0.507$ on that unseen family, Appendix E.3.2) both still show a null, independently computed docking effect under the same correction as every other checkpoint. Within the main thesis, the same dissociation is visible in the other direction: the two-feature linear baseline and the repaired EGNN were statistically indistinguishable on the LP test set (Chapter IV), yet neither produced a detectable gradient effect, and the ranker of Chapter VI produced its one large, statistically decisive effect precisely because its moderately accurate predictor's score was dominated by a ligand-side property (size) that a generator can trivially exploit, which is a case of a predictor being *too* usable for the wrong reason. Accuracy on the held-out prediction task and usefulness as a steering signal are different properties, tested by different protocols (Section III.6.1 and III.6.3), and this thesis finds them uncorrelated across every architecture, training recipe and representation tried.

This dissociation has a sharper explanation than "accuracy and usefulness are different properties" in the abstract: Chapter IV.8 shows that accuracy itself, on this task, is not what it appears to be. A predictor with no protein input at all reproduces

70% (a 14-descriptor ridge model) to 90% (the deployed EGNN, retrained under a within-target label permutation) of a structure-aware model's real-label performance, and the correlation figures quoted above (Pearson 0.58 to 0.65) sit well inside the range a class-level prior alone can reach. The repeated observation in this thesis that "the predictor is reasonably accurate, yet guidance still fails" is therefore not as puzzling as it first appears: a predictor can be *reasonably accurate by a class-prior route* while still carrying little or no usable information about the specific interface a generator would need to exploit, and the representation-ladder audit of Chapter IV.6, which found no pocket representation, including a chemically typed interaction representation, reliably beating a ligand-only baseline after correction, is the direct structural counterpart of this statistical finding.

IX.2.3 Grading the explanations

Measured. (i) The affinity predictors' outputs are strongly correlated with the ligand size (0.64 to 0.72), and a chemically typed interaction representation shows the same pattern in sharper form (Chapter IV.6). (ii) Selection by the trained-surrogate ranker enriches larger molecules by about 12 heavy atoms and lowers physical validity by about 29 percentage points, and the raw docking gain does not survive normalisation by size. (iii) The magnitude of the affinity gradient is positively correlated with the distance of an atom from the pocket (+0.240; 95% CI 0.194 to 0.285) and not with its heaviness; its signed direction, measured separately over 23,202 atom-states in 20 pockets, is indistinguishable from random (mean cosine toward the nearest contact −0.019, 95% CI [−0.046, +0.008]), so the gradient is uninformative about the interface rather than pointed away from it. (iv) The delta target is not learnt better than a constant by the model used. (v) Replacing the trained

surrogate with a genuine, non-surrogate affinity signal (Vina's own score) in the identical selection mechanism, pockets and evaluation protocol produces a significant, largely size-independent improvement (Chapter VI.5).

Consistent. The failure of the gradient routes and the failure of the trained-surrogate ranker are compatible with a common upstream cause: a point predictor trained to minimise the error of the experimental affinity encodes ligand-side properties (size), which are informative for the *average* complex, and not the pocket-specific interactions that a generator would need to exploit. The gradient diagnostic (iii) is compatible with this reading, and so is the null result of every mechanism that left the representation unchanged (B1 to B3), and so is the representation-ladder finding that no pocket representation reliably beats ligand-only (Chapter IV.6). Earlier drafts of this synthesis described the evidence for a *common* cause as indirect, since the two failures have different measured signatures: the gradient is not correlated with heaviness in the direction a size shortcut would predict (Section V.4), while the ranker's selection is dominated by size (Chapter VI). Measurement (v) changes this. It does not merge the two signatures into one, but it supplies a *direct* test of the shared hypothesis that both signatures point toward, by holding the mechanism fixed and varying only the signal: when the EGNN surrogate is replaced by a genuinely informative, non-surrogate signal, the selection mechanism that failed in Chapter VI succeeds. The strength of this evidence differs by route, and the distinction should not be blurred. For the ranking route it is direct: mechanism held fixed, signal swapped, outcome reversed, which is as clean an isolation of F1 (the learned predictor's information content) as this design can produce. For the gradient routes it is indirect and inferential: Vina is not differentiable, so the same swap cannot be performed

inside a gradient mechanism, and what supports F1 there is convergence between separate lines of evidence, the gradient-field diagnostic, the representation ladder and the class-prior decomposition, rather than a controlled substitution. F2 is correspondingly narrowed but not excluded for Chapter V's routes, and any summary of this thesis that states the failure "traces to the signal rather than the mechanism" without that qualifier is claiming more than the gradient evidence carries.

Hypotheses, not tested. (a) The mismatch between the clean, docked poses on which the predictors were trained and the noisy clean-data estimates on which they are evaluated during sampling may make the gradient unreliable at early steps. A related study reports that an interaction prior pretrained on accurate poses degraded generation when queried on noisy intermediate structures [23]. (b) The evaluation target was Vina, whereas the predictors were trained on experimental affinities; a predictor of the Vina energy itself, evaluated by Vina, would ask a different question from the one tested here. (c) Position-only guidance may be too weak or too blunt: the gradient of atom types was set to zero, and the gradient was not propagated through the denoiser. (d) For synthesizability, a smooth 3D gradient may be a poor way to steer a fingerprint-based target.

Hypotheses (a) and (c) were, in fact, tested directly in a follow-up investigation of 8 further independently-trained checkpoints, run after this synthesis and reported in full in Appendix E: a noise-matched training curriculum (testing (a)) and a direct, step-by-step trace of the guidance nudge through the reverse-diffusion trajectory of one molecule (testing (c)). Neither rescues a real guidance effect, and the trace gives a specific mechanistic account of why: the nudge is small enough, for most of the trajectory, to be statistically buried in the sampler's own intrinsic step

noise, and only compounds to a sub-Ångström, late-trajectory effect too small to move the docking outcome in a corrected, reproducible way (Section E.5). Hypothesis (b), by contrast, was addressed only partially: a Vina-derived training target was tried (Appendix E.2.1) and also did not change the outcome, though a predictor of the Vina energy architecture itself, rather than just its label, remains untested.

IX.2.4 The role of statistical power

Most of the results are null results, and a null result is only as informative as the design that produced it. For the per-pocket gradient tests the detectable shift was about 1 kcal/mol at the nominal level and 1.4 kcal/mol after correction (Section V.1). Effects smaller than that cannot be excluded at the pocket level, and pooled tests suggest that pooled effects are small (Chapter IV). The routes with a strong, consistent effect, the ranker of Chapter VI (a shift of −1.31 kcal/mol on average, with 14 of 15 pockets significant), were detected, and the loss of physical validity in the gradient routes at large λ was detected and was dose-dependent. The design could therefore detect large effects of both signs. It could not detect small beneficial effects, and it was not designed to. The stated conclusion is thus "no effect of practical size was found", and not "no effect exists". For Track E the power calculation was reported in advance and again after the runs, and the result excluded the required effect.

Figure 9.2 makes this quantitative. Using the standard two-sample power formula and the within-pocket Vina Dock standard deviations actually measured in this project's sweeps (1.1 to 1.6 kcal/mol; Chapters IV and V), the minimum shift detectable at 80% power with 20 molecules per condition (the size used in most of Track B) is about 1.0 to 1.4 kcal/mol at the nominal 5% level, and about 1.4 to 2.0

kcal/mol once a Bonferroni-equivalent correction for 24 simultaneous tests is applied; at 30 molecules per condition (Track A's full tier and the Task F ablation) the corresponding range is about 0.8 to 1.6 kcal/mol. This is a standard calculation from a textbook formula, not a new measurement; it is shown to make the qualitative statements above precise and falsifiable.

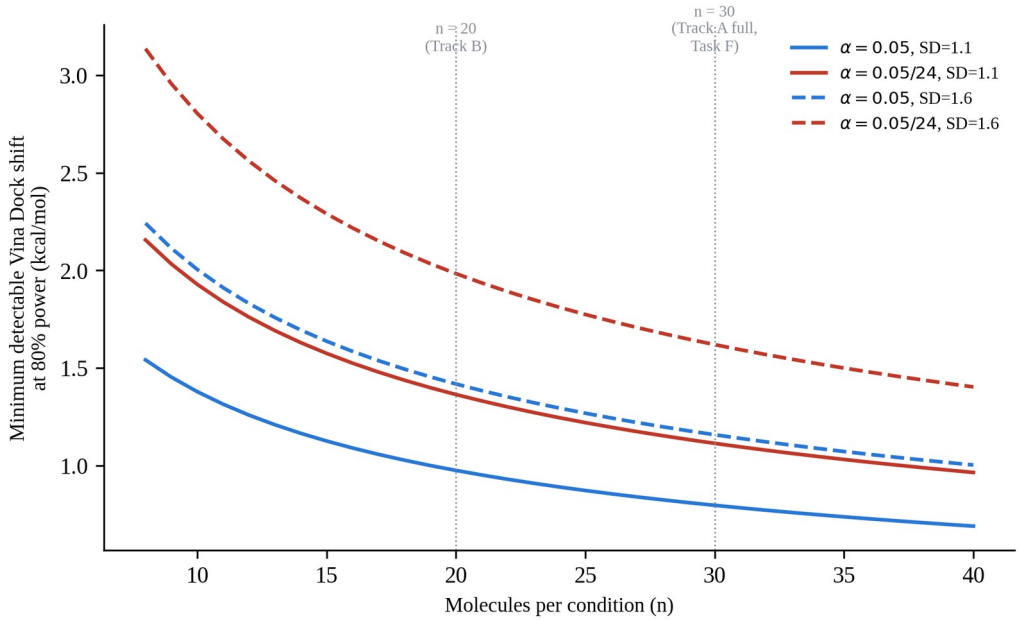


Figure 9.2 Minimum Vina Dock shift detectable at 80% power as a function of the number of molecules per condition, computed from the two-sample normal-approximation formula at the range of within-pocket standard deviations measured in this project (1.1 to 1.6 kcal/mol), at the nominal 5% level and at a Bonferroni-equivalent level for 24 simultaneous tests. Vertical lines mark the sample sizes actually used in Track B (20) and in Track A's full tier (30); the Task F ablation used 10 molecules per pocket per seed, which pools to about 30 per pocket across its three seeds and is shown at the same mark.

IX.3 Methodological Findings

Several results of this work concern the *evaluation* of learned guidance and predictors, and they are, together with the negative outcome, the main transferable findings.

The split can create a generalisation gap. A validation set that shared 74% of its targets with the training set and a test set of 27 pockets gave a picture of a poor and unfixable predictor ($R^2 = -0.54$; a ligand-only baseline that matched the network). A target-disjoint split with 127 test targets showed $R^2 = 0.34$ and a nominal margin over the ligand-only baseline for the same architecture and hyper-parameters (Chapter IV). The earlier diagnostics were correct measurements of the wrong split, and a causal story built on them was revised.

The unit of resampling matters. Resampling complexes instead of targets gave intervals about eight times too narrow (Chapter VIII). The margin over simple baselines that the narrow interval suggested was not supported by the target-clustered interval: the EGNN was statistically indistinguishable from a linear model of two scalar features.

Leakage can go unnoticed on either side of the coupled title. The reported quality of the synthesizability predictor came from a random split without a held-out test set, and dropped when re-evaluated on a target-disjoint set (although the comparison was not controlled).

Summary statistics can mislead. One malformed pose (+286.58 kcal/mol) shifted a mean by 14 kcal/mol; medians and rank-based tests were reported next to means. A fragmentation check did not detect the loss of physical validity that PoseBusters detected. A raw docking gain of 1.3 kcal/mol was an artefact of size, visible only after normalisation. A three-pocket exploratory tier with eight of nine effects in the correct direction became 16 of 24 in a properly sized replication that included the same pockets.

Negative controls and gates worked. The scrambled-filter control found no implementation error in the ranker analysis; the pre-declared gates stopped work on unresolved results (D.4 in Track D, E2 in Track E) and made the decision to stop transparent; the automated test of test-set isolation gives a positive control for a practice that is documented to be common in the field [18].

Even the most local structure–activity test available failed. Activity cliffs and matched molecular pairs are, by construction, the stress tests least forgiving of a model that has learned only coarse, class-level trends (Chapter IV.9). Neither this thesis's structure-aware predictors nor its ligand-only baselines beat chance sign-agreement on matched-pair direction, and neither significantly outperformed the other on the activity-cliff stratum. This null result at the most local level is consistent with, and strengthens, the class-prior decomposition of Chapter IV.8: it is not only that a large share of apparent performance is explained by class membership, but that the residual, within-target signal that remains is not demonstrably captured even by the measure best suited to detect it.

A leakage audit run before, not after, a production experiment. The compound-level leak identified in the combined CrossDocked–BindingNet pool (Chapter IV.7) was found and a filter applied before the production Arm A and Arm C runs were trained, rather than discovered after the fact and used to discount an already-reported result, as happened with the sequence-level leak of Section IV.2. This ordering is itself a transferable practice: a leakage audit is most useful, and least liable to be read as after-the-fact rationalisation, when it precedes the experiment whose validity it protects.

Errors were found by the process and reported. They include a receptor-file mirror with a different coordinate frame, a truncation of pools to a single seed in an exploratory analysis, the seeding order bug of the trainer, two errors of the differentiable bond estimator, and inconsistent counts in earlier finding documents (for example, the Track B2 count of comparisons worse by median, and a statement about ligand-efficiency degradation in Track B3). The numbers of this thesis were re-derived from the raw result files.

IX.4 Threats to Validity

IX.4.1 Internal validity

The predictors were each trained once for the guidance experiments (the Track E arms and the EGNN reference had three seeds); the seed variance of the guidance predictors was therefore not measured. The pocket sets differ between routes (1, 3, 8, 15 and 20 pockets), and the exploratory tiers used 2 to 3 pockets and 20 molecules per condition, so their power was low. Figure 9.3 shows this variation directly: the evidentiary base of Track C's ranker result (15 pockets, pools of about 590 molecules) and of Track E's scoring result (127 targets) is substantially larger than that of any single gradient-guidance checkpoint (1 to 20 pockets, 8 to 30 molecules per condition), which is why the former two produced statistically decisive results (a confounded positive and a confirmed negative, respectively) while most gradient checkpoints produced only a bounded null. Docking used Vina with exhaustiveness 8 for most gradient routes and 16 for the ablations, and a single docking program was used as the main outcome. Statistical tests were corrected within a checkpoint but not across checkpoints, and several analyses were post hoc and flagged as such (the

pocket-size analysis of Chapter VI, for example). The Track D result in the powered design rests on a single pocket.

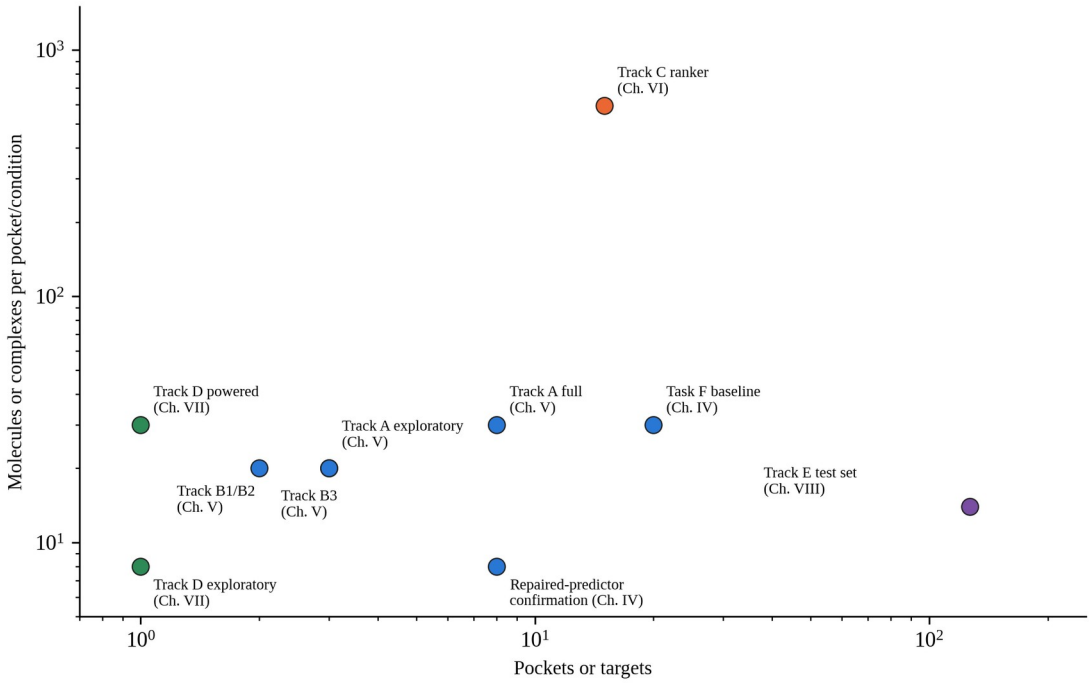


Figure 9.3 Number of pockets or targets against the number of molecules or complexes evaluated per pocket or condition, for every checkpoint of Chapters IV to VIII (logarithmic scales). The pocket/molecule counts are those already stated in the corresponding chapter.

IX.4.2 External validity

All results concern one generator (TargetDiff), one data set (CrossDocked2020, whose poses are cross-docking outputs whose selection is known to be biased toward ligands favoured by docking software [18]), guidance on the clean-data estimate, and position-only steering. Nothing in this work bears on guidance that is trained on noisy inputs, that propagates the gradient through the denoiser, that also steers atom types, or that uses a generator with a different conditioning scheme. The reference frame for the outcome was docking, which is itself an approximation of binding; no experimental validation was possible.

IX.4.3 Construct validity

The RAscore is a proxy of synthesizability. Vina Dock is a proxy of affinity that is size-biased [57]; ligand efficiency and PoseBusters were used to control this, but they are proxies too. The experimental pK labels are heterogeneous (Ki, Kd, IC50) and attached to docked poses whose receptor may come from a different structure, which limits the absolute quality of any predictor and affects all arms equally.

IX.5 Relation to the Literature

The results agree with three lines of published work. First, docking scores of generated molecules are known to be inflated by the size and chemistry of the ligand, and it has been observed that some models use trained Vina predictors to guide sampling at the cost of hacking the score [18], [57]; Chapter VI shows the mechanism in an isolated setting, a ranker whose selection is dominated by size, although the observation of that study concerned evaluation and not gradient guidance. Second, learned affinity predictors are sensitive to leakage and to ligand-only shortcuts [19], [34], [53]; the size correlation of 0.64 to 0.72 of the predictors and the match between a ligand-only baseline and the network on the old split are instances. Third, the mismatch between clean training inputs and noisy intermediate inputs has been reported to degrade guidance-like priors [23].

The results do not contradict studies that report improved docking scores with interaction-aware generation [32], [56], [65], because those studies place the interaction information in the *generator* (as conditioning, as a co-generated output or inside the denoiser's loss), and not in a separate guidance predictor, which is the

design that was tested here. Whether interaction supervision would rescue a guidance predictor (the E2 hypothesis) remains untested in this work.

One result does sit in direct tension with this thesis's null finding and is reported as such rather than explained away. BADGER's classifier-guidance variant applies the gradient of a separately trained affinity predictor to a frozen diffusion SBDD generator, essentially the same arrangement as Chapters IV to VI, and reports up to a 60% improvement in binding affinity [24]. This thesis's own Chapter IV to VI results offer three candidate, untested explanations for the discrepancy, stated as hypotheses and not as findings about that paper, since its full methods were not read for this thesis (Appendix B). First, an apparent improvement measured on a predictor's own score, on a single model or seed, or without target-level resampling and multiple-comparison correction, has in this thesis repeatedly looked larger than the same comparison does once these checks are applied (Section IX.3); whether BADGER's evaluation includes a leakage-safe split, corrected significance testing or a negative control was not stated in the materials reviewed. Second, a different base generator, predictor architecture or training-data regime could plausibly produce a gradient with a different relationship to the pocket than the one diagnosed in Section V.4, which was specific to the predictors trained in this project. Third, the two works report different quantities: a 60% improvement is a relative figure of an unstated baseline and metric, whereas this thesis reports absolute shifts in kcal/mol together with their confidence intervals and a validity check, and the two are not directly comparable without access to BADGER's raw results. The discrepancy is not resolved here; it is a motivated direction for a focused follow-up (Section IX.6) and a reason the conclusions of this thesis are bounded to the generator, predictors and evaluation

protocol actually tested (Section X.1), not extended to gradient-guidance methods in general.

The positive control of Chapter VI.5 narrows, without closing, one specific branch of this discrepancy: the question of whether this thesis's own evaluation harness, rather than gradient guidance in general, is responsible for the null result. Because the identical generator, pocket set and independent evaluation protocol detect a significant, largely size-independent improvement once a genuinely informative signal is substituted for the trained surrogate, an insensitive harness cannot by itself explain the gradient-guidance and rejection-sampling nulls reported in Chapters IV to VI; whatever accounts for the discrepancy with BADGER, it is not that this project's statistics are too conservative to detect a real effect of the size BADGER reports. This leaves the remaining candidate explanations, a different predictor's information content and a different gradient mechanism, as the more probable sources of the discrepancy, which is consistent with, though it does not prove, the first and second explanations above.

IX.6 Implications and Recommendations

For practitioners who want to steer a frozen generator with a learned predictor, the results suggest five checks that can be made cheaply before a costly sampling study: (i) evaluate the predictor with a target-disjoint split and target-clustered intervals; (ii) test its output against ligand size and against ligand-only baselines, including a within-target label-permutation control to establish how much of its apparent accuracy is a class-level prior rather than within-target structure–activity signal (Chapter IV.8); (iii) measure the direction of its gradient against pocket-contact proximity on captured mid-trajectory states, a diagnostic that took about 20 minutes in

this work; (iv) evaluate any docking gain after normalising by size and with a validity check, since both the raw gain and the loss of validity were visible in the same data; and (v), where a cheap genuine signal exists even if it cannot itself be used for production guidance, run it through the same selection or gradient mechanism as a positive control before concluding that a null result reflects the signal rather than an insensitive evaluation harness, since this was the single check that converted an otherwise indirect inference into a direct one in this thesis (Chapter VI.5).

For future research, the results point to experiments that follow from the diagnosed failures and are not covered here: predictors trained on noised inputs at every diffusion time, with an explicit size control or contact-aligned gradient objective; a matched-size comparison of rankers, in which the candidates are selected within strata of heavy-atom count; a powered multi-pocket repetition of the synthesizability experiment with a different, rule-based synthesizability score to test whether the divergence is specific to the learned RAscore; the auxiliary interaction supervision (E2) on the absolute target; evaluation with an orthogonal scoring method; and larger pocket sets sized by an explicit power calculation. A direct priority follows from Section IX.5's unresolved discrepancy with BADGER [24]: reading that paper's full methods, and, where feasible, applying this thesis's leakage-safe split, target-clustered statistics and negative controls to its reported setting, would establish whether the two results disagree because of evaluation rigor or because of a genuine difference in what the two guidance predictors learned.

CHAPTER X
CONCLUSIONS AND RECOMMENDATIONS

X.1 Conclusions

The thesis asked whether learned guidance signals improve the molecules of a frozen, pocket-conditioned diffusion model on independently computed evaluation outcomes, that is, on quantities that are computed independently of the guidance model. The question was tested with a frozen TargetDiff generator, position-only guidance on the clean-data estimate, two families of affinity predictors and one synthesizability predictor, and an evaluation harness consisting of AutoDock Vina docking, PoseBusters validity, ligand efficiency and the independently computed RAscore, with corrected statistics, bootstrap intervals and negative controls.

The scope of what follows should be stated before the individual conclusions are: this thesis does not establish that learned guidance fails in general. It establishes that the tested configurations did not produce defensible improvements under the specified generator, data regime, guidance mechanism, predictors, evaluation proxies and computational budget. A different generator, a noise-aware predictor, a gradient that is propagated through the denoiser, or a larger compute budget are all outside what was tested, and Appendix E reports which of these untested directions were in fact pursued afterward and what they changed (none of the eight further checkpoints altered the conclusion below, which is additional evidence for it, not a revision of it). The following conclusions are drawn.

(1) *No route built on a trained surrogate produced a defensible improvement.* Gradient guidance with an EGNN affinity predictor, with a physics-anchored GIGN + PIGNet2 predictor and with three variants of the mechanism (gradient normalisation, classifier reformulation, timestep windowing), rejection sampling with the EGNN as a ranker, gradient guidance toward synthesizability, and a redesigned training target (delta-learning against the docking score) each failed their pre-declared criteria. None of the 56 pocket-level gradient comparisons of Tracks A and B (9 and 24 in the two tiers of Track A, 23 in Track B) was significant after correction within its checkpoint. The predictors reached Pearson correlations of 0.58 to 0.65 on leakage-safe test sets, and this was originally read as ruling out low predictor accuracy as the explanation. Conclusion (4) below revises this reading: a predictor with no protein input at all reaches a comparable correlation through a class-level prior alone, so correlation magnitude on its own does not establish that these predictors had learned pocket-specific structure–activity relationship.

(2) *The apparent successes of the trained surrogate were artefacts, and a positive control localises the cause to the signal.* The only large and consistent effect obtained with the trained EGNN, a gain of 1.3 kcal/mol in the raw docking score with rejection sampling in 12 of 15 pockets, was accompanied by worse ligand efficiency in 14 of 15 pockets, molecules that were larger by 12 heavy atoms on average, and a drop of 29 percentage points in PoseBusters validity. The consistent direction of the first, exploratory tier of the physics-anchored route weakened from 89% to 67% when the design was properly sized. A pre-registered positive control then replaced the trained surrogate with AutoDock Vina's own zero-shot score in the identical selection mechanism, on the identical 20 pockets, 3 seeds and evaluation protocol: this

succeeded, improving the *predicted docking score* (Vina Dock −8.068 against −6.813 kcal/mol, Wilcoxon $p = 1.9 \times 10^{-6}$) in a way that survived size normalisation (ligand efficiency flat, $p = 0.756$; PoseBusters pass rate held), while significantly increasing molecular size (+1.8 heavy atoms) and significantly reducing both synthesizability scores. It is therefore a validated harness and a multi-objective trade-off, not a demonstration that better molecules were designed, and the improvement is in a computational score rather than in measured binding.

This is the single most important result of this thesis for interpreting everything that preceded it, and its reach should be stated exactly. It shows that the *selection* harness can detect a change driven by a genuinely informative signal, so an insensitive evaluation design cannot explain the null results of Chapter VI. For the rejection-sampling route it localises the failure directly to the *information content of the trained predictor*, because the mechanism was held fixed while only the signal changed. For the gradient routes of Chapter V it is indirect evidence only: Vina is not differentiable, so the same signal could not be sent through a gradient mechanism, and a mechanism-level contribution to those failures is narrowed but not excluded. The control is also not an independent measurement of binding, since the selecting and evaluating scores are two modes of one scoring function and share its biases (Section VI.5.1).

(3) *Guidance had a cost at the strengths at which an effect could be expected.* The physical validity of the molecules decreased dose-dependently with guidance strength (PoseBusters pass rate from 0.56 to 0.17 at $\lambda = 1.0$ for the physics-anchored predictor, in all eight pockets), and the molecules fragmented at larger strengths.

(4) *Measured properties of the predictors and their gradients were consistent with the failures, and a fuller audit identifies what the predictors actually read.* The predictor outputs are strongly correlated with ligand size (0.64 to 0.72), and the selection by the ranker is dominated by size. The magnitude of the EGNN gradient was positively correlated with the distance of an atom from the pocket (+0.240; 95% CI 0.194 to 0.285), that is, gradient magnitude increased, on average, with distance from the atoms most likely to form binding contacts. A later, signed measurement over 23,202 atom-states in 20 pockets found the gradient's *direction* indistinguishable from random (mean cosine toward the nearest contact −0.019, 95% CI [−0.046, +0.008]; 48.5% of atoms pushed toward their nearest contact, which is chance), with its own random-direction and finite-difference controls passing. The two results together say that the gradient is uninformative about the interface, not that it actively opposes it, and earlier phrasing in this project describing a "wrong direction" is withdrawn on that evidence (Section V.4). The synthesizability gradient was not explained by three simple proxies. An eight-block information-content audit (Chapter IV.6) extended this into a systematic comparison: across 40 pairwise, Benjamini–Hochberg-corrected tests, a structure-aware representation significantly beat a ligand-only one in 12 cases, tied in 28, and never lost; a chemically explicit, typed interaction representation (hydrogen bond, hydrophobic, π-stacking and salt-bridge counts) explained almost no variance on its own (R^2 0.07–0.08) until a single size covariate was added, at which point it explained four times as much. No representation of the binding pocket tested in this thesis reliably outperformed a model of the ligand alone.

(5) *An apparent generalisation gap of the affinity predictor was largely an artefact of the split, and a fuller audit shows the margin this revealed did not survive*

a stronger baseline. A validation set that overlapped the training set by 74% of its targets and a test set of 27 pockets suggested that the predictor did not generalise ($R^2 = -0.54$). On a target-disjoint split with 127 test targets the same architecture reached $R^2 = 0.34$, with a nominal margin over a weak ligand-only baseline (R^2 0.20) that was not tested pairwise with target-level resampling. Against a stronger, 14-descriptor ligand-only model built later in the project ($R^2 = 0.3531$), this margin disappeared: the structure-aware predictor was statistically indistinguishable from it (0.3420–0.3437 across three seeds). Similarly, the reported quality of the synthesizability predictor was re-evaluated on a target-disjoint set and was lower (R^2 0.41 against 0.77).

(5a) *Most of the predictors' apparent performance is a class-level prior, not structure–activity learning, and this holds for the deployed architecture itself.* A within-target label permutation, which preserves each target's typical affinity range while destroying which ligand carries which label, reproduced 70% of the 14-descriptor model's real-label performance and, when the identical control was run on the deployed EGNN architecture by full retraining, 90% of the EGNN's own real-label performance. The resolvable window beyond this prior, bounded below by the ligand-only ceiling and above by an estimated label-noise ceiling, spans only 0.22 R^2, narrower than the average width of a single confidence interval measured for this task. Neither activity cliffs nor matched molecular pairs, the two standard diagnostics most sensitive to genuine local structure–activity capture, provide evidence that any predictor in this thesis captures it: sign agreement on the direction of a matched pair's affinity change was at or below chance for both a structure-aware ensemble and a ligand-only model.

(5b) *Augmenting training data with a larger external data set did not close the gap, after a compound-level leak was found and removed before, not after, the production run.* A combined CrossDocked–BindingNet training pool was found to share 0.34% of its rows, but 10.12% of evaluation records by ligand identity, with the validation and test partitions; a filter removing these rows was built and enforced in code before three further training arms were run. A predictor trained on 20,000 CrossDocked rows alone (validation $R^2 = 0.362$ averaged over 3 seeds; coincidentally the same figure as, but a different quantity from, the two-feature linear baseline of conclusion (6) below) performed descriptively better than one trained on a matched total of CrossDocked plus leakage-filtered BindingNet rows ($R^2 = 0.272$), and the filtered and unfiltered BindingNet-augmented arms performed nearly identically (0.272 against 0.278), indicating the leak identified had not substantially inflated the earlier result even though removing it remained the correct decision in principle.

(6) *The pre-registered delta-learning experiment was significantly worse than the baselines.* The delta model reached $R^2 = 0.117$ against 0.342 for the EGNN and 0.362 for a linear model with two features, it did not beat an absolute-target control on the same backbone, and its best validation error was not below that of a constant predictor. The residual was practically not learnable from structure by this model. The second modification (auxiliary interaction supervision) was designed and its infrastructure verified, but it was not run because the gate was not passed.

X.2 Contributions

The contributions of the thesis are the following.

(i) A documented, statistically disciplined mapping of where a widely used class of methods, gradient guidance and rejection sampling with learned predictors,

fails for structure-based molecular diffusion, with quality-degradation checks and negative controls applied to every apparent positive result.

(ii) A pre-registered positive control, built from a genuine, non-surrogate affinity signal applied through the identical selection mechanism, pockets and evaluation protocol as the thesis's own null results, that validates the evaluation harness directly and localises the repeated guidance failures to the information content of the trained predictors rather than to an insensitive design or to the selection and gradient mechanisms themselves.

(iii) A leakage-safe, target-level split of the labelled CrossDocked2020 subset, and the demonstration that the choice of split and of the resampling unit can reverse conclusions about predictor quality.

(iv) A systematic, eight-block information-content audit of what representation predicts binding affinity on this split, extending a single ligand-only comparison into a corrected, pairwise test across ligand, pocket, raw and size-normalised contact fingerprints and a chemically typed interaction representation, together with a class-prior decomposition, via within-target label permutation, that was run not only for a descriptor pipeline but for the deployed 3-D equivariant architecture itself.

(v) Two mechanistic diagnostics of the guidance gradient that can be reproduced in about twenty minutes, and their results.

(vi) A pre-registered, verified and completed negative test of delta-learning as a target for a guidance predictor, with a reusable, resumable training infrastructure whose isolation of the test set is enforced by an automated test.

(vii) A training-data-composition study (BindingNet augmentation) that found and removed a compound-level leak before, rather than after, its production runs, and that found augmentation did not close the gap to a ligand-only baseline at matched training-set size.

(viii) An audit trail: numbers in this thesis were re-derived from raw results, and discrepancies found in earlier summaries were corrected.

X.3 Limitations

The conclusions are bounded by the design. They concern one generator, one data set with cross-docked poses, guidance on the clean-data estimate with position-only gradients, predictors trained on clean poses, and evaluation by docking, whose proxies for affinity and synthesizability have known limitations. The pocket sets were small (1 to 20), the exploratory tiers had low power, per-pocket effects smaller than about 1 to 1.4 kcal/mol could not be excluded, the seed variance of most guidance predictors was not measured, the powered synthesizability test rests on one pocket, and no experimental validation was possible. This thesis is computational throughout: no molecule discussed in it was synthesised, assayed or otherwise tested in a laboratory, and every statement about a molecule "binding better" is a statement about a predicted score. The evaluation criteria used here, predicted docking score, ligand efficiency, physical-pose validity, drug-likeness and retrosynthetic accessibility, are also an incomplete screen even on their own computational terms: they say nothing about absorption, distribution, metabolism, excretion or toxicity, nothing about selectivity against off-targets, and nothing about whether a proposed route would succeed in practice. A molecule that passes every check reported in this thesis is a candidate for further computational triage, not a drug-like candidate in the sense a

medicinal chemist would use the phrase, and an ADMET and off-target assessment would be a prerequisite for any claim of practical usability. The target-disjoint split used throughout is easier than a leave-superfamily-out split would be, and 40.1% of test-set ligands sit on a scaffold that also occurs somewhere in training, so "held-out target" should not be read as "held-out chemistry" when interpreting any of the predictor-quality results. The experimental labels mix Kd, Ki and IC50 measurements whose composition differs between training and test, a further source of irreducible variance folded into the label-noise ceiling of Chapter IV.8. The positive control of Chapter VI.5 is itself bounded: it tests only a selection-based mechanism, since Vina is not differentiable and a gradient-based analogue with a genuine signal could not be constructed; it therefore directly supports F1 (the trained predictors are uninformative) without directly excluding F2 (a correctly informative signal misapplied by the specific gradient mechanisms of Chapter V) for the gradient-guidance routes considered on their own. Absence of a detected effect is not evidence that no effect exists, and this thesis does not claim equivalence anywhere. The distinction is not rhetorical: a non-significant comparison bounds an effect only as tightly as the design's resolution allows, and no formal equivalence test (two one-sided tests against a pre-declared margin) was run for any route, because no margin of practical indifference was pre-declared. What the design does support is an upper bound from its own power calculation: effects at or above roughly 1.0 to 1.4 kcal/mol per pocket, depending on the sample size of the checkpoint (Section IX.2.4), would have been detected with 80% power, and none was. Effects below that bound are compatible with every null reported here. Readers who need a specific minimum effect of

interest, for instance the shift a medicinal chemist would consider worth a sampling run, should compare it against that bound rather than against the p-values.

X.4 Recommendations for Future Work

The following work follows from the diagnosed failures. First, predictors should be trained on noised inputs at every diffusion time, with an explicit size control or with an objective that aligns the gradient with pocket contacts, and evaluated on the diagnostics of this thesis before any costly sampling study. Second, rankers should be compared within strata of heavy-atom count. Third, the synthesizability experiment should be repeated on multiple pockets and with a rule-based score, to determine whether the divergence is specific to the learned RAscore. Fourth, the auxiliary interaction supervision of Track E, if it is pursued, should be tested first on the absolute target, where the base predictor is known to learn. Fifth, evaluation should include an orthogonal scoring method and, where possible, experimental data, and the pocket sets should be sized by a prior power calculation. Sixth, and directly motivated by the positive control of Chapter VI.5, a gradient-free particle-guidance mechanism such as SVDD, which requires only an evaluable rather than a differentiable value function, could be paired with a genuine signal such as Vina in place of a trained surrogate, to test whether a more sophisticated mechanism adds value on top of simple reranking once the signal itself is no longer the limiting factor; this was considered and deliberately not run for this thesis, both because of its implementation cost and because the positive control already supplies strong indirect evidence for what such an experiment would show when paired with the trained surrogates examined here.

The thesis reports a negative result. It is offered as a result and not as a failure: it locates where a plausible and popular idea does not hold, states the conditions of that statement, and provides the evaluation framework with which the next, better-designed attempt can be judged.

X.4.1 A proposed extension: uncertainty-aware acquisition (not tested)

The extension described here has not been implemented or tested in this thesis, and no result is claimed for it. It combines a predictor with an uncertainty estimate, an acquisition rule that uses that estimate, and a conflict-aware treatment of the two guidance gradients. Each component is stated with a hypothesis that could fail.

The surrogate would be an affinity predictor with Monte Carlo dropout, giving a mean μ and an uncertainty σ for each complex. Before σ is used for acquisition, its calibration must be shown: σ should correlate with the absolute error on held-out targets of a leakage-safe split. If it does not, an upper-confidence or expected-improvement rule has no basis. This calibration test is the first hypothesis of the extension, and it is to be run before any sampling study.

An upper confidence bound ($\mu + \kappa\sigma$) or an expected-improvement criterion would enter the reverse trajectory as a correction to the guidance gradient, with a time-dependent weight λ_t that favours exploration early and exploitation late. The weight is written λ_t rather than β_t to avoid a clash with the noise schedule of the diffusion model. The second hypothesis is that the acquisition-guided sampler gives no significant gain on independently computed scores over a random-acquisition control with the same number of steps.

Projecting conflicting gradients (PCGrad) between the affinity and the RA-score gradient would be tested as a remedy for the divergence reported in Chapter

VII. The third hypothesis is that the frequency of opposing gradient directions falls without a gain in RA-score.

Any such extension would be judged by the protocol of this thesis: Kolmogorov–Smirnov tests with Benjamini–Hochberg correction, target-clustered bootstrap intervals, and negative controls (random acquisition and shuffled σ). Before any improvement is claimed, ligand efficiency, heavy-atom count and PoseBusters validity must be checked; a gain in docking score alone would not be accepted, for the reasons given in Chapter VI.

A closed-loop simulation, in which docking scores retrain the surrogate and the acquisition function screens generated molecules, would be a simulation only. It would not establish synthesizability or biological activity. Its principal risk is that it optimises the docking score, which Chapter VI showed can be raised by increasing molecular size. The fourth hypothesis is that under such a loop the heavy-atom count rises across iterations while ligand efficiency falls. If the gates above are not passed, the output of the loop is not evidence of design.

These hypotheses are stated so that they can fail, and a null result would be reported in the same way as the results of this thesis.

REFERENCES

[1] M. F. Adasme, K. L. Linnemann, S. N. Bolz, F. Kaiser, S. Salentin, V. J. Haupt, and M. Schroeder, “PLIP 2021: Expanding the scope of the protein–ligand interaction profiler to DNA and RNA,” *Nucleic Acids Res.*, vol. 49, no. W1, pp. W530–W534, 2021, doi: 10.1093/nar/gkab294.

[2] S. Allenspach, J. A. Hiss, and G. Schneider, “Neural multi-task learning in drug design,” *Nat. Mach. Intell.*, vol. 6, pp. 124–137, 2024.

[3] M. A. Balcı, E. Çetin, G. Calibasi-Kocal, and Ö. Akgüller, “Pocket-surface discrete differential geometry as a leakage-robust feature class for protein–ligand binding affinity prediction,” *Molecules*, vol. 31, no. 11, 2026, doi: 10.3390/molecules31111899.

[4] Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate: A practical and powerful approach to multiple testing,” *J. R. Stat. Soc. B*, vol. 57, no. 1, pp. 289–300, 1995.

[5] C. Bouysset and S. Fiorucci, “ProLIF: A library to encode molecular interactions as fingerprints,” *J. Cheminform.*, vol. 13, art. 72, 2021, doi: 10.1186/s13321-021-00548-6.

[6] M. Buttenschoen, G. M. Morris, and C. M. Deane, “PoseBusters: AI-based docking methods fail to generate physically valid poses or generalise to novel sequences,” *Chem. Sci.*, vol. 15, no. 9, pp. 3130–3139, 2024, doi: 10.1039/d3sc04185a.

[7] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, “GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks,” in *Proc. 35th Int. Conf. Machine Learning (ICML)*, PMLR vol. 80, 2018, pp. 794–803.

[8] V. Dey and X. Ning, “Enhancing molecular property prediction with auxiliary learning and task-specific adaptation,” *J. Cheminform.*, vol. 16, art. 85, 2024, doi: 10.1186/s13321-024-00880-7.

[9] P. Dhariwal and A. Nichol, “Diffusion models beat GANs on image synthesis,” in *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021, pp. 8780–8794.

[10] J. Eberhardt, D. Santos-Martins, A. F. Tillack, and S. Forli, “AutoDock Vina 1.2.0: New docking methods, expanded force field, and Python bindings,” *J.*

Chem. Inf. Model., vol. 61, no. 8, pp. 3891–3898, 2021, doi: 10.1021/acs.jcim.1c00203.

[11] B. Efron, “Bootstrap methods: Another look at the jackknife,” *Ann. Statist.*, vol. 7, no. 1, pp. 1–26, 1979.

[12] D. Errington, C. Schneider, C. Bouysset, and F. A. Dreyer, “Assessing interaction recovery of predicted protein–ligand poses,” *J. Cheminform.*, vol. 17, art. 76, 2025, doi: 10.1186/s13321-025-01011-6.

[13] P. Ertl and A. Schuffenhauer, “Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions,” *J. Cheminform.*, vol. 1, art. 8, 2009, doi: 10.1186/1758-2946-1-8.

[14] A. V. Fassio et al., “Prioritizing virtual screening with interpretable interaction fingerprints,” *J. Chem. Inf. Model.*, vol. 62, no. 18, pp. 4300–4318, 2022, doi: 10.1021/acs.jcim.2c00695.

[15] P. G. Francoeur, T. Masuda, J. Sunseri, A. Jia, R. B. Iovanisci, I. Snyder, and D. R. Koes, “Three-dimensional convolutional neural networks and a cross-docked data set for structure-based drug design,” *J. Chem. Inf. Model.*, vol. 60, no. 9, pp. 4200–4215, 2020, doi: 10.1021/acs.jcim.0c00411.

[16] B. Gao, H. Tan, Y. Huang, M. Ren, X. Huang, W.-Y. Ma, Y.-Q. Zhang, and Y. Lan, “From theory to therapy: Reframing SBDD model evaluation via practical metrics,” arXiv:2406.08980, 2024 (preprint).

[17] B. Gao, M. Ren, Y. Ni, Y. Huang, B. Qiang, Z.-M. Ma, W.-Y. Ma, and Y. Lan, “Rethinking specificity in SBDD: Leveraging delta score and energy-guided diffusion,” arXiv:2403.12987, 2024 (preprint).

[18] B. Gao, H. Tan, Y. Huang, M. Ren, X. Huang, W.-Y. Ma, Y.-Q. Zhang, and Y. Lan, “Reframing structure-based drug design model evaluation via metrics correlated to practical needs,” in *Proc. 13th Int. Conf. Learning Representations (ICLR)*, 2025.

[19] D. Graber, P. Stockinger, F. Meyer, S. Mishra, C. Horn, and R. Buller, “Resolving data bias improves generalization in binding affinity prediction,” *Nat. Mach. Intell.*, vol. 7, no. 10, pp. 1713–1725, 2025, doi: 10.1038/s42256-025-01124-5.

[20] J. Guan, W. W. Qian, X. Peng, Y. Su, J. Peng, and J. Ma, “3D equivariant diffusion for target-aware molecule generation and affinity prediction,” in *Proc. 11th Int. Conf. Learning Representations (ICLR)*, 2023.

[21] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020, pp. 6840–6851.

[22] Q. Hu, C. Sun, H. He, J. Xu, D. Liu, W. Zhang, S. Shi, K. Zhang, and H. Li, “Target-aware 3D molecular generation based on guided equivariant diffusion,” *Nat. Commun.*, vol. 16, art. 7928, 2025, doi: 10.1038/s41467-025-63245-0.

[23] Z. Huang, L. Yang, C. Qin, Y. Xing, H. Yu, X. Zhou, B. Zheng, Y. Wang, X. Gao, and W. Yang, “Disentangled diffusion model for 3D molecular generation with protein–ligand interaction priors,” *Bioinformatics*, vol. 42, no. 6, p. btag165, 2026, doi: 10.1093/bioinformatics/btag165.

[24] Y. Jian, C. Wu, D. Reidenbach, and A. S. Krishnapriyan, “General binding affinity guidance for diffusion models in structure-based drug design,” *J. Chem. Inf. Model.*, vol. 66, no. 3, pp. 1342–1352, 2026, doi: 10.1021/acs.jcim.5c01166.

[25] J. Jiang et al., “ForkMerge: Mitigating negative transfer in auxiliary-task learning,” in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023.

[26] J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature*, vol. 596, no. 7873, pp. 583–589, 2021, doi: 10.1038/s41586-021-03819-2.

[27] A. Kautish, A. Sivakumar, and V. K. Prasad, “Quantum kernel-based delta (Δ)-learning for correcting a semiempirical method in the prediction of reaction barrier heights,” *J. Chem. Phys.*, vol. 165, no. 3, 2026, doi: 10.1063/5.0336022.

[28] A. Kendall, Y. Gal, and R. Cipolla, “Multi-task learning using uncertainty to weigh losses for scene geometry and semantics,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7482–7491.

[29] A. Khodabandeh Yalabadi, M. Yazdani-Jahromi, and O. Ozmen Garibay, “BoKDiff: Best-of-K diffusion alignment for target-specific 3D molecule generation,” arXiv:2501.15631, 2025 (preprint).

[30] K. Kwapien, E. Nittinger, J. He, C. Margreitter, A. Voronov, and C. Tyrchan, “Implications of additivity and nonadditivity for machine learning and deep learning models in drug design,” *ACS Omega*, vol. 7, no. 30, pp. 26573–26581, 2022, doi: 10.1021/acsomega.2c02738.

[31] H. Lai, L. Wang, R. Qian, J. Huang, P. Zhou, G. Ye, F. Wu, F. Wu, X. Zeng, and W. Liu, “Interformer: An interaction-aware model for protein–ligand docking and affinity prediction,” *Nat. Commun.*, vol. 15, art. 10223, 2024, doi: 10.1038/s41467-024-54440-6.

[32] J. Lee, W. Zhung, J. Seo, and W. Y. Kim, “BInD: Bond and interaction-generating diffusion model for multi-objective structure-based drug design,” *Adv. Sci.*, vol. 12, no. 35, p. e02702, 2025, doi: 10.1002/advs.202502702.

[33] H. Li, K.-S. Leung, M.-H. Wong, and P. J. Ballester, “Improving AutoDock Vina using random forest: The growing accuracy of binding affinity prediction by the effective exploitation of larger data sets,” *Mol. Inform.*, vol. 34, no. 2–3, pp. 115–126, 2015, doi: 10.1002/minf.201400132.

[34] J. Li, X. Guan, O. Zhang, K. Sun, Y. Wang, D. Bagni, and T. Head-Gordon, “Leak proof PDBBind: A reorganized data set of protein–ligand complexes for more generalizable binding affinity prediction,” *J. Phys. Chem. B*, vol. 130, no. 2, pp. 730–740, 2026, doi: 10.1021/acs.jpcb.5c08598.

[35] Z. Lin et al., “Evolutionary-scale prediction of atomic-level protein structure with a language model,” *Science*, vol. 379, no. 6637, pp. 1123–1130, 2023, doi: 10.1126/science.ade2574.

[36] B. Lin, W. Jiang, F. Ye, Y. Zhang, P. Chen, Y.-C. Chen, S. Liu, I. W. Tsang, and J. T. Kwok, “Dual-balancing for multi-task learning,” *Neural Netw.*, vol. 195, p. 108317, 2025, doi: 10.1016/j.neunet.2025.108317.

[37] J. Mistry et al., “Pfam: The protein families database in 2021,” *Nucleic Acids Res.*, vol. 49, no. D1, pp. D412–D419, 2021, doi: 10.1093/nar/gkaa913.

[38] S. Moon, W. Zhung, S. Yang, J. Lim, and W. Y. Kim, “PIGNet: A physics-informed deep learning model toward generalized drug–target interaction predictions,” *Chem. Sci.*, vol. 13, no. 13, pp. 3661–3673, 2022, doi: 10.1039/d1sc06946b.

[39] R. Quiroga and M. A. Villarreal, “Vinardo: A scoring function based on Autodock Vina improves scoring, docking, and virtual screening,” *PLoS ONE*, vol. 11, no. 5, p. e0155183, 2016, doi: 10.1371/journal.pone.0155183.

[40] R. Ramakrishnan, P. O. Dral, M. Rupp, and O. A. von Lilienfeld, “Big data meets quantum chemistry approximations: The Δ-machine learning approach,” *J. Chem. Theory Comput.*, vol. 11, no. 5, pp. 2087–2096, 2015, doi: 10.1021/acs.jctc.5b00099.

[41] M. Ren, B. Gao, B. Qiang, and Y. Lan, “Delta score: Improving the binding assessment of structure-based drug design methods,” arXiv:2311.12035, 2023 (preprint).

[42] C. Rücker, G. Rücker, and M. Meringer, “y-Randomization and its variants in QSPR/QSAR,” *J. Chem. Inf. Model.*, vol. 47, no. 6, pp. 2345–2357, 2007, doi: 10.1021/ci700157b.

[43] S. Salentin, S. Schreiber, V. J. Haupt, M. F. Adasme, and M. Schroeder, “PLIP: Fully automated protein–ligand interaction profiler,” *Nucleic Acids Res.*, vol. 43, no. W1, pp. W443–W447, 2015, doi: 10.1093/nar/gkv315.

[44] V. G. Satorras, E. Hoogeboom, and M. Welling, “E(n) equivariant graph neural networks,” in *Proc. 38th Int. Conf. Machine Learning (ICML)*, PMLR vol. 139, 2021, pp. 9323–9332.

[45] J. Scantlebury et al., “A small step toward generalizability: Training a machine learning scoring function for structure-based virtual screening,” *J. Chem. Inf. Model.*, vol. 63, no. 10, pp. 2960–2974, 2023, doi: 10.1021/acs.jcim.3c00322.

[46] P. Schake, S. N. Bolz, K. Linnemann, and M. Schroeder, “PLIP 2025: Introducing protein–protein interactions to the protein–ligand interaction profiler,” *Nucleic Acids Res.*, vol. 53, no. W1, pp. W463–W465, 2025, doi: 10.1093/nar/gkaf361.

[47] C. Shen, X. Zhang, C.-Y. Hsieh, Y. Deng, D. Wang, L. Xu, J. Wu, D. Li, Y. Kang, T. Hou, and P. Pan, “A generalized protein–ligand scoring framework with balanced scoring, docking, ranking and screening powers,” *Chem. Sci.*, vol. 14, no. 30, pp. 8129–8146, 2023, doi: 10.1039/d3sc02044d.

[48] A. Thakkar, V. Chadimová, E. J. Bjerrum, O. Engkvist, and J.-L. Reymond, “Retrosynthetic accessibility score (RAscore) – rapid machine learned synthesizability classification from AI driven retrosynthetic planning,” *Chem. Sci.*, vol. 12, no. 9, pp. 3339–3349, 2021, doi: 10.1039/d0sc05401a.

[49] M. Thomas, R. T. Smith, N. M. O’Boyle, C. de Graaf, and A. Bender, “Comparison of structure- and ligand-based scoring functions for deep generative models: a GPCR case study,” *J. Cheminform.*, vol. 13, no. 1, 2021, doi: 10.1186/s13321-021-00516-0.

[50] O. Trott and A. J. Olson, “AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading,” *J. Comput. Chem.*, vol. 31, no. 2, pp. 455–461, 2010, doi: 10.1002/jcc.21334.

[51] D. van Tilborg, A. Alenicheva, and F. Grisoni, “Exposing the limitations of molecular machine learning with activity cliffs,” *J. Chem. Inf. Model.*, vol. 62, no. 23, pp. 5938–5951, 2022, doi: 10.1021/acs.jcim.2c01073.

[52] M. Varadi et al., “AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models,” *Nucleic Acids Res.*, vol. 50, no. D1, pp. D439–D444, 2022, doi: 10.1093/nar/gkab1061.

[53] M. Volkov, J.-A. Turk, N. Drizard, N. Martin, B. Hoffmann, Y. Gaston-Mathé, and D. Rognan, “On the frustration to predict binding affinities from protein–ligand structures with deep neural networks,” *J. Med. Chem.*, vol. 65, no. 11, pp. 7946–7958, 2022, doi: 10.1021/acs.jmedchem.2c00487.

[54] C. Wang and Y. Zhang, “Improving scoring-docking-screening powers of protein–ligand scoring functions using random forest,” *J. Comput. Chem.*, vol. 38, no. 3, pp. 169–177, 2017, doi: 10.1002/jcc.24667.

[55] M. Wójcikowski, M. Kukiełka, M. M. Stepniewska-Dziubinska, and P. Siedlecki, “Development of a protein–ligand extended connectivity (PLEC) fingerprint and its application for binding affinity predictions,” *Bioinformatics*, vol. 35, no. 8, pp. 1334–1341, 2019, doi: 10.1093/bioinformatics/bty757.

[56] P. Wu, H. Du, Y. Yan, T.-Y. Lee, C. Bai, and S. Wu, “Guided diffusion for molecular generation with interaction prompt,” *Brief. Bioinform.*, vol. 25, no. 3, p. bbae174, 2024, doi: 10.1093/bib/bbae174.

[57] M. Xu, C. Shen, J. Yang, Q. Wang, and N. Huang, “Systematic investigation of docking failures in large-scale structure-based virtual screening,” *ACS Omega*, vol. 7, no. 43, pp. 39417–39428, 2022, doi: 10.1021/acsomega.2c05826.

[58] C. Yang and Y. Zhang, “Delta machine learning to improve scoring-ranking-screening performances of protein–ligand scoring functions,” *J. Chem. Inf. Model.*, vol. 62, no. 11, pp. 2696–2712, 2022, doi: 10.1021/acs.jcim.2c00485.

[59] Z. Yang, W. Zhong, Q. Lv, T. Dong, and C. Yu-Chian Chen, “Geometric interaction graph neural network for predicting protein–ligand binding affinities from 3D structures (GIGN),” *J. Phys. Chem. Lett.*, vol. 14, no. 8, pp. 2020–2033, 2023, doi: 10.1021/acs.jpclett.2c03906.

[60] M. Yazdani-Jahromi, S. Padhi, and I. Garibay, “MIRAGE: Measuring interpolation and redundancy in affinity generalization,” arXiv:2609.14491, 2026 (preprint).

[61] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, “Gradient surgery for multi-task learning,” in *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020, pp. 5824–5836.

[62] B. Zdrazil et al., “The ChEMBL database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods,” *Nucleic Acids Res.*, vol. 52, no. D1, pp. D1180–D1192, 2024, doi: 10.1093/nar/gkad1004.

[63] J. Zhang and H. Chen, “Molecule design using molecular generative models constrained by ligand–protein interactions,” *J. Chem. Inf. Model.*, vol. 62, no. 14, pp. 3291–3306, 2022, doi: 10.1021/acs.jcim.2c00177.

[64] X. Zhang, H. Gao, H. Wang, Z. Chen, Z. Zhang, X. Chen, Y. Li, Y. Qi, and R. Wang, “PLANET: A multi-objective graph neural network model for protein–ligand binding affinity prediction,” *J. Chem. Inf. Model.*, vol. 64, no. 7, pp. 2205–2220, 2024, doi: 10.1021/acs.jcim.3c00253.

[65] W. Zhung, H. Kim, and W. Y. Kim, “3D molecular generative framework for interaction-guided drug design,” *Nat. Commun.*, vol. 15, art. 2688, 2024, doi: 10.1038/s41467-024-47011-2.

APPENDIX A

REPRODUCIBILITY AND REPOSITORY

A.1 Hardware and Software

All experiments ran on one laptop with an NVIDIA GeForce RTX 3050 Laptop GPU (4,096 MiB; driver 595.91.07), a 20-thread CPU and 31 GB of RAM, under Linux. The guidance experiments were run in a separate conda environment (`drugdisc`), pinned in Table A.2. The actual guidance environment is captured in `guidance/thesis/environment_actual.yml` (a full `conda env export`, 166 conda-level packages) and `guidance/thesis/requirements-lock.txt` (`pip freeze`, 130 pip-installed packages, including the 119 third-party packages conda cannot itself lock); both were generated directly from the running `drugdisc` environment during this audit (2026-10-03T16:33:24Z UTC) and are tracked in the repository alongside this document. This supersedes the originally tracked `environment.yaml`, which documents the original TargetDiff repository's own setup (Python 3.8, PyTorch 1.13.1) and does not reproduce the environment the guidance code actually ran in; that file is retained for provenance, not as a build instruction for the guidance code. No environment specification was captured at the time any individual experiment was run, so `environment_actual.yml` and `requirements-lock.txt` document the environment retrospectively, as of the audit date above, rather than at each experiment's own run time; the version pins in Table A.2 were, however, stable across this project's working period and no dependency was knowingly upgraded between experiments. The interaction profiler PLIP was used in the Track E feasibility checks and label

extraction. Random generators were seeded for every sampling and training run; the seeds used are recorded in the run logs and result files listed below.

Table A.1 reports the measured training budget of the six production surrogate arms of Chapter IV.7, so that the comparison between them can be read against the cost that produced it rather than assumed to be matched. Two points follow. First, per-epoch throughput is nearly identical across the arms (0.63 to 0.65 h per epoch), as expected since both train the same architecture on the same total row count. Second, and more consequential for the interpretation in Chapter IV.7, the arms did *not* receive equal budgets: Arm C ran longer in every sense, 38 epochs and 24.6 hours against Arm A's 27 epochs and 17.0 hours, because its early-stopping criterion was satisfied later. The arm that performed worse is therefore the arm that trained longer, which removes under-training as an explanation for Arm C's lower R^2 and makes that comparison more, not less, interpretable. Wall-clock figures include thermal suspension by the queue's watchdog (Section A.1), which is reported separately because it inflates elapsed time without consuming compute.

Table A.1 Measured training budget of the six production surrogate arms (Chapter IV.7)

Arm (seed)	Epochs to early stop	Wall clock (h)	of which thermally suspended (h)	Hours per epoch
Arm A (2021)	9	5.72	0.97	0.64
Arm A (2022)	9	5.64	0.92	0.63
Arm A (2023)	9	5.66	0.93	0.63
Arm C (2021)	9	5.88	0.97	0.65
Arm C (2022)	14	9.07	1.57	0.65
Arm C (2023)	15	9.64	1.60	0.64
Arm A total	**27**	**17.02**	**2.82**	**0.63**
Arm C total	**38**	**24.59**	**4.14**	**0.65**

Regenerated by `guidance/surrogate_data/compute_budget_table.py` from the queue logs of the runs actually reported, not estimated or retyped. All six ran on the single GPU of Section A.1 at batch size 1 without mixed precision. Thermal suspension pauses the process when the CPU package exceeds its

threshold; it lengthens wall clock but does not add computation, so the per-epoch column is the fairer throughput comparison.

Table A.2 Exact package versions of the `drugdisc` environment, as actually installed

Package	Version
Python	3.10.21
PyTorch	2.5.1 (CUDA 12.1)
PyTorch Geometric	2.8.0.post1
RDKit	2026.03.5
NumPy	2.2.6
SciPy	1.15.3
scikit-learn	1.7.2
pandas	2.3.3
LMDB	2.3.0
PoseBusters	0.6.5
AutoDock Vina	1.2.6 (binary reports "7ac2999-mod"; the Python module attribute reports 1.2.5)
Meeko	0.1.dev3
PLIP	3.0.1

Captured by directly querying the installed packages (`import x; x.__version__`) in the running environment at the time of this audit, not transcribed from documentation or memory.

A.2 Exact Identifiers for Reproduction

Table A.3 records the identifiers needed to recover the exact code, data and frozen model used, as SHA-256 checksums computed directly from the files on disk. The checkpoint-selection timestamps embedded in each training run's directory name (for example `crossdocked_affinity_egnn_2026_10_01__23_54_19_vinatarget`) serve as the identifier for the 8 further checkpoints of Appendix E and are not repeated individually here; they are listed in full in each checkpoint's own finding document (Appendix E's artefact lists).

Table A.3 Checksums and identifiers of the principal frozen artefacts

Artefact	Identifier
Repository state at build time	commit `7e543224b0d2`, built from a working tree with 28 uncommitted change(s)

Artefact	Identifier
Frozen TargetDiff diffusion checkpoint (`pretrained_models/pretrained_diffusion.pt`)	SHA-256 `8977821339ba31b7103b46835ef4ddbf7f15f3f2efc5cb1f2a405e7905b09db6`
Reference EGNN checkpoint, PDBBind v2016 (`pretrained_models/egnn_pdbbind_v2016.pt`)	SHA-256 `1cd74eb55043bf788ac29186e56905b6a645a9d4722b03b97482ffb6f36970df`
Stage 0 EGNN, original split (`guidance_models/affinity_egnn.pt`)	SHA-256 `006808e6461f72156e2021cbbe034ac1cb9568942b857d8e9d1c414530753447`
Stage 0 EGNN, leakage-safe split (`guidance_models/affinity_egnn_lpsplit.pt`)	SHA-256 `a76d254a921c9e00d0a6409cad5006952e2308059ce3a0a602bd86ec1bb297a7`
Classifier head, B2 (`guidance_models/classifier_head.pt`)	SHA-256 `46caa0d864b99e8e7968baec50293638ee447d69e9279c5b6d31502b26aa9fd9`
SynthPredNet (`guidance_models/synth_ra_score.pt`)	SHA-256 `e79c12fa2696a5e99df835e35f11ab59407338ef2d94f521ecddb591f7fa844a`
Pocket-cropped CrossDocked2020 LMDB (`data/crossdocked_v1.1_rmsd1.0_pocket10_processed_final.lmdb`, 3.8 GB)	SHA-256 `376679dd1516e855f1cdd89bef2e25995958b77695cd63a88a26ec02a1d3b591`
Affinity annotation file (`data/affinity_info.pkl`)	SHA-256 `2f99f361f77f41a6e0252aa2692470a4d433eeef169aac5ff129caf79e050cf9`

All checksums were computed directly from the files on disk at the time of this audit (`sha256sum`), not copied from an external record.

A.3 Data Provenance and What a Reproducer Needs

A checksum proves that two copies of a file are identical; it does not tell anyone where the file came from or how to obtain it. Table A.4 closes that gap for the derived data this thesis depends on, naming for each artefact the upstream source and the exact script in this repository that produced it, so that a reader can regenerate it rather than having to request it.

Table A.4 Provenance of the derived data artefacts, with the script that generates each

Artefact	Upstream source	Generated by
Pocket-cropped CrossDocked2020 LMDB	CrossDocked2020 [15], via the TargetDiff preprocessing pipeline [20]	Upstream TargetDiff data scripts, unmodified
Leakage-safe (LP) target-level split	The pocket-cropped LMDB above	`guidance/lp_split/build_split.py`

Artefact	Upstream source	Generated by
Family-holdout split (PF00069 removed)	The LP split, plus Pfam annotations [37]	`guidance/lp_split/ build_family_holdout_split.py`
Sequence-level leakage filter (19 sequences, 28 templates, 265 rows removed)	BindingNet index	`guidance/surrogate_data/ apply_sequence_leakage_filter. py`
Compound-level leakage filter (423 rows, 123 compounds removed)	Sequence-filtered pool	`guidance/surrogate_data/ apply_compound_leakage_filte r.py`
BindingNet training index	BindingNet	`guidance/surrogate_data/ build_bindingnet_index.py`

Each script writes a summary file alongside its output recording the counts it removed; the leakage guards of Table A.6 re-derive those counts independently and fail if they disagree, so the table above is enforced by tests rather than only documented.

Two limits of this provenance record are stated rather than left for a reader to discover. First, redistribution: this thesis does not restate the licence terms of CrossDocked2020, BindingNet or the PDB-derived structures it builds on, because those terms were not independently verified during this work, and a reproducer must consult each source's own terms before redistributing any derived artefact. The scripts and the split definitions in this repository are the author's own; the underlying data are not. Second, scale: the pocket-cropped LMDB alone is 3.8 GB, and regenerating it requires the upstream CrossDocked2020 download, so an end-to-end reproduction from raw data is a substantial undertaking rather than a single command. A reproducer wanting to check a specific number rather than the whole pipeline is better served by the per-result scripts named throughout this appendix, each of which reads a cached intermediate and runs in seconds to minutes.

A.4 Test-Set Isolation, Per Training Run

A reader has to be able to check, without taking the author's word for it, that the held-out test set did not participate in choosing any model reported here. Table A.5 states for each family of training runs what the checkpoint was selected on and

whether the test set was evaluated at all during training, with the evidence that establishes it. Two distinctions matter and are kept separate in the table, because conflating them is how this kind of claim usually goes wrong. *Selection* means the quantity whose improvement caused a checkpoint to be kept or training to continue; in every trainer used here that quantity is the validation loss, which is read directly from the code (`best_val_loss` drives both checkpoint retention and early stopping in `train_egnn_stage0.py` and `train_surrogate_arm.py`). *Exposure* means whether the test set was evaluated at all, even only to write a number into a log. Exposure without selection does not invalidate a result, but it is weaker than no exposure, so it is reported rather than glossed.

Table A.5 Test-set isolation by training run, with the evidence for each claim

Training run(s)	Checkpoint selected on	Test set evaluated during training?	Evidence
Stage 0 EGNN, the deployed guidance predictor (Ch. IV)	Lowest validation loss	Yes, for logging only	Its training log contains 8 `[Test]` evaluations and 18 `[Validate]` evaluations; selection follows `best_val_loss` in the trainer
Hyper-parameter sweep of Section IV.3	**Test metrics** — the one deliberate exception	Yes	Disclosed in Table 4.2's own note: the configuration was labelled best on its test metrics, so its test performance is optimistic and it was never deployed as a guidance model
A1e heteroscedastic runs (5 runs)	Lowest validation loss	No	Launched with `--skip_test_logging`, which bypasses the test loader entirely (`scripts/queue_steps.sh`)
Arm A and Arm C production surrogates (6 runs, Section IV.7)	Lowest validation loss, with early stopping on patience	No	All six run logs contain zero `[Test]` lines

Training run(s)	Checkpoint selected on	Test set evaluated during training?	Evidence
Track E arms (Ch. VIII)	Pre-registered gate on the primary comparison	No, the test set is read once, after training, by the analysis script	Appendix D; the gate quantities are fixed before training

"Selected on" is the quantity that determined which checkpoint was kept, read from the trainer source rather than inferred. The sweep row is included precisely because it is the exception: it is the only place in this thesis where a configuration was chosen using test-set numbers, it is labelled as optimistic where it appears, and no guidance result depends on it.

A.5 Reproducibility Checklist

Table A.6 is a status checklist, not a promise: an item is marked verified only where this project actually checked it, in the sense stated in its own row, and unverified items are disclosed rather than assumed.

Table A.6 Reproducibility checklist, marked only where actually verified

Item	Status
Package versions pinned and recorded (Table A.2)	Verified: read directly from the running environment
Git commit of the code that produced the results in this document	Verified, and emitted by the build rather than typed: commit `7e543224b0d2`, built from a working tree with 28 uncommitted change(s). If that line reports uncommitted changes, this PDF was produced from a working tree that is not yet a reproducible reference point, and an immutable tag must be created before submission
Dataset and checkpoint identifiers (Table A.3)	Verified: SHA-256 of the files actually used
Random seeds recorded per run	Verified for sampling/docking jobs (one result directory per condition/pocket/seed) and per-seed training runs (Appendix D, Section VIII.7); not every exploratory-tier ablation logged its seed individually
Headline numbers re-derivable from raw result files by a saved analysis script	Verified for the checkpoints and results re-derived during this thesis-revision audit (Section A.5); not independently re-run for every number in Chapters IV to VIII
Full pipeline (data preparation → training → sampling → docking → statistics) re-executable from a clean checkout	Not verified in this pass: this would require re-running multi-hour training and sampling jobs on the project's single 4 GB GPU, which was not repeated for this audit. The code paths and commands are documented (Table A.7, Section A.5) but an independent clean-room run was not performed

Item	Status
Analysis and statistics code passes its own automated test suite	Verified at the time of this build: 16 suites, 0 failures, covering the leakage guards, the paired-bootstrap and degenerate-variance guards, the split loader, the physics masks and the quantisation path
Environment specification file matching the actual guidance environment	Verified: `guidance/thesis/environment_actual.yml` and `guidance/thesis/requirements-lock.txt`, generated directly from the running environment (Section A.1); captured retrospectively during this audit, not at each experiment's own run time

A.6 Repository Layout

The code is a fork of the public TargetDiff repository. The thesis-specific code is in the directory `guidance/`. Table A.7 lists the principal elements. Result directories and model checkpoints (`logs_`, `*runs_track_e*`, `guidance_models/`, `sampling_results/`) are not tracked in version control because of their size.

Table A.7 Principal elements of the repository

Path	Content
guidance/guided_sampling.py, affinity_guidance.py, synth_guidance.py, classifier_guidance.py	Guided reverse sampling; guidance predictors and their interfaces
guidance/lp_split/	Leakage-safe split construction, Stage 0 training, synthesizability re-split and retraining
guidance/track_a/	GIGN + PIGNet2 model, physics terms, training and guidance wrapper
guidance/gradient_informativeness_test.py, analyze_*.py	KS tests with Benjamini–Hochberg correction; per-track analysis scripts
guidance/analyze_track_c.py	Rejection-sampling analysis (top-k, random subset, scrambled filter, sensitivity)
guidance/diag_size_confound.py, diag_synth_direction.py	Gradient-direction diagnostics
guidance/track_e/	Delta-learning code, PLIP label extraction, feasibility checks, trainer with atomic checkpoints, tests, gate results
guidance/run_track_e.sh	One-command, resumable driver of the Track E stages
eval/honest_eval.py	Independent evaluation (Vina Dock, PoseBusters, ligand efficiency, RAscore)
guidance/*_FINDING.md*, _REPORT.md, TRACK_E_DESIGN.md	Finding reports and the pre-registration with its amendments

Path	Content
guidance/thesis/	Sources of this document: chapters, references, figures and the build script

A.7 Reproducing the Results and This Document

The result of a stage can be recomputed from the stored raw files. For example, the Track E gate results were produced by the analysis script of `guidance/track_e/` from the per-run test predictions, with the bootstrap seed 20260925, and were written to `guidance/track_e/e1_gate_results.json`. The Track E stages are run with `guidance/run_track_e.sh run --stage e1`, which can be interrupted and resumed at any time. Sampling and docking jobs of Tracks A to D wrote one result directory per (condition, pocket, seed) unit and skip units that are complete. The figures of this document are generated by `guidance/thesis/make_figures.py` from the result files, and the document is assembled by `guidance/thesis/build_thesis.py`, which writes the Word file and computes the page numbers of the tables of contents from a PDF rendering. The values reported in the text were re-derived from the raw result files; where an earlier finding document contained a different value, the raw file was used and the discrepancy is noted in the text.

APPENDIX B

LITERATURE SEARCH AND VERIFICATION

B.1 Search Strategy

The literature was collected in two phases. In the first phase, related work on generative structure-based drug design, guidance of diffusion models, learned scoring functions, docking biases and data leakage was collected while the experiments were designed. In the second phase, before the delta-learning experiment, a structured review of 45 sources was made for the two ideas that motivated it (residual learning against a physics-based baseline, and interaction-typed auxiliary supervision), together with the nearest published combinations. Searches used bibliographic databases and a research-paper search service with queries on delta-learning, multi-task learning with interaction labels, interaction-conditioned diffusion, classifier guidance for molecular generation, and leakage-controlled benchmarks. An independent check of 12 papers, supplied during the work, was used to revise the novelty statement.

B.2 Verification and Its Limits

The bibliographic details of the references were verified in different ways. For most entries, the record in PubMed was retrieved and the authors, journal, volume, pages and identifier were compared with the entry. For sources that are not indexed there (conference papers and preprints), the document itself was consulted. A final pass checked the five most recently added references against the Crossref registry,

and the preprint among them against its arXiv record, comparing title, author list, venue, volume, pages and DOI field by field.

That final pass is reported here rather than quietly absorbed, because it changed three entries and two of the changes were substantive. The preprint BoKDiff had been entered with a reconstructed title and a wrong first-author initial; both are corrected to the record ("BoKDiff: Best-of-K diffusion alignment for target-specific 3D molecule generation", A. Khodabandeh Yalabadi *et al.*), and its "first to apply Best-of-K alignment and Best-of-N sampling to SBDD" claim, on which the narrowing of this thesis's own positive-control novelty in Section VI.5 depends, was confirmed verbatim in the abstract. The Kwapien *et al.* entry had carried an unresolved venue ("ACS Omega / J. Chem. Inf. Model."); it is ACS Omega, vol. 7, no. 30. The remaining three were confirmed exactly as entered, and DOIs were added to all five. The lesson is the obvious one and is recorded because this thesis argues elsewhere that checks are worth their cost: a reconstructed citation is a hypothesis about a document, and two of five such hypotheses here were wrong in a way only a registry lookup would reveal. Quotations attributed to a source were checked in the text of the source. Statements about what a paper reports are limited to what could be read: where only an abstract was available, the statement is limited to the abstract, and where a paper could not be read in full (PLANET, and the training loss of DiffInt), no statement about its full content is made.

A targeted re-check was run at revision time against the single novelty claim with the largest load-bearing role in Chapter IV, the within-target label-permutation decomposition of apparent performance into a class-level prior and a within-target structure–activity component. Two searches, phrased once around the permutation

control itself and once around the equivalent baseline a reader might expect it to be called (a per-target mean or target-stratified baseline on PDBbind-style data), did not return a study applying that control to protein–ligand affinity prediction; the claim is therefore left standing, in the "we did not find" form used throughout. The same check did surface contemporaneous work reaching a compatible conclusion by a different method, MIRAGE [60], which is now cited in Section II.4 and discussed in Section IV.8 as convergence rather than precedence, since it was released while this thesis was being written and measures family support rather than within-target signal. Recording this is the point of the exercise: a novelty claim that survives a directed attempt to refute it is worth more than one that was never tested, and the attempt is documented here so that a reader can repeat it.

The search has limits that bear on the wording of novelty. It was made with the tools available, it did not include a manual search of Google Scholar or Semantic Scholar, it covers work up to the date of the review, and the absence of a study in the results of these searches is not proof that no such study exists. Consequently the claim of Section VIII.8 is formulated as "we did not find". A similar limit applies to Chapter II as a whole: it is a focused review of the literature closest to the experiments, and not a systematic review.

APPENDIX C

TRACK A FULL-TIER RESULTS: ALL COMPARISONS

C.1 Per-Pocket, Per-Strength Results

Table C.1 lists all 24 comparisons of the full tier of Track A (Section V.2.4). Each is a KS test of the independent Vina Dock distribution of guided molecules against that of the unguided molecules of the same pocket and seed, with Benjamini–Hochberg correction across all 24 tests. The number of molecules differs between conditions because of failed reconstructions.

Table C.1 Track A full tier: all 24 comparisons with the unguided baseline

Pocket	λ	n (unguided / guided)	KS statistic	KS p (raw)	Mean shift	Median shift	PoseBusters pass rate	KS p (BH)
BSD	0.1	30 / 30	0.200	0.5941	+0.172	+0.151	0.33 → 0.50	0.999
BSD	0.3	30 / 30	0.300	0.1350	-0.960	-0.330	0.33 → 0.40	0.551
BSD	1.0	30 / 28	0.290	0.1378	-0.635	-0.021	0.33 → 0.04	0.551
HDAC8	0.1	30 / 30	0.367	0.0346	-0.472	-0.632	0.63 → 0.63	0.415
HDAC8	0.3	30 / 30	0.333	0.0709	-0.350	-0.444	0.63 → 0.47	0.551
HDAC8	1.0	30 / 29	0.424	0.0065	-0.665	-0.898	0.63 → 0.10	0.157
CD38	0.1	29 / 30	0.176	0.6810	-0.398	-0.172	0.38 → 0.47	0.999
CD38	0.3	29 / 29	0.276	0.2221	-0.926	-0.648	0.38 → 0.38	0.762
CD38	1.0	29 / 29	0.310	0.1229	-0.621	-1.049	0.38 → 0.03	0.551
MENE	0.1	30 / 30	0.167	0.8080	-0.145	+0.380	0.50 → 0.43	0.999
MENE	0.3	30 / 29	0.087	0.9983	-0.246	+0.067	0.50 → 0.45	0.999
MENE	1.0	30 / 30	0.167	0.8080	-0.297	-0.077	0.50 → 0.10	0.999
NQO1	0.1	30 / 30	0.167	0.8080	+0.087	+0.012	0.57 → 0.53	0.999
NQO1	0.3	30 / 30	0.133	0.9578	-0.027	-0.090	0.57 → 0.60	0.999
NQO1	1.0	30 / 29	0.180	0.6469	+0.190	-0.074	0.57 → 0.31	0.999

Pocket	λ	n (unguided / guided)	KS statistic	KS p (raw)	Mean shift	Median shift	PoseBusters pass rate	KS p (BH)
PAK4	0.1	30 / 30	0.100	0.9988	+0.093	+0.146	0.80 → 0.83	0.999
PAK4	0.3	30 / 30	0.167	0.8080	+0.268	+0.269	0.80 → 0.67	0.999
PAK4	1.0	30 / 30	0.233	0.3929	+0.736	+0.399	0.80 → 0.40	0.999
PNTM	0.1	29 / 30	0.094	0.9956	+0.068	-0.083	0.69 → 0.67	0.999
PNTM	0.3	29 / 30	0.094	0.9956	-0.018	-0.143	0.69 → 0.67	0.999
PNTM	1.0	29 / 30	0.161	0.7498	+0.609	-0.127	0.69 → 0.27	0.999
RIBB	0.1	29 / 30	0.198	0.5021	-0.638	-0.176	0.59 → 0.60	0.999
RIBB	0.3	29 / 30	0.230	0.3237	-0.860	-0.468	0.59 → 0.60	0.971
RIBB	1.0	29 / 30	0.211	0.4615	-0.717	-0.458	0.59 → 0.07	0.999

Short names denote BSD_ASPTE, MENE_BACSU, PNTM_STRAE and RIBB_VIBCH. Shifts are guided minus unguided Vina Dock (kcal/mol); negative is better. No comparison was significant after correction.

APPENDIX D

TRACK E: PRE-REGISTRATION AMENDMENTS

D.1 Summary of the Amendments

The pre-registration document of Track E was frozen in version control before any model was trained. Later changes were logged as dated amendments and never edited in place. Table D.1 summarises them; none changed a pass or fail threshold.

Table D.1 Track E amendments

ID	Content
A1	Decisions D1 to D6 resolved at the review gate: option F (about 58 GPU-hours); π-stacking dropped from the primary auxiliary head; a positive E1 result requires exceeding both the Stage 0 EGNN and the two-feature linear baseline; two extra EGNN seeds; guided-generation re-test deferred
A2	π-stacking labels stored but excluded from the loss (atom-level Jaccard 0.16 under 0.2 Å perturbation)
A3	Semantics of the stored Vina column examined; treated as an opaque pose-attached score with about 0.8 kcal/mol preparation noise
A4	Corrections and execution details: hidden size 256 (0.494 M parameters); epoch cap 30; test predictions computed once; Vina-positive poses excluded from training and validation only; readout for the second stage chosen on validation loss; confidence levels 90, 95 and 99%
A5	Measured compute: about 2.4 h per physics-head run; revised budget 52 to 55 GPU-hours
A6	Infrastructure verification: unit tests, interrupt and resume tests, a seeding-order bug found and fixed, reproduction of the recorded Track A metrics
A7	Novelty framing corrected after a literature follow-up (superseded by A8)

ID	Content
A8	Novelty revised after an independent 12-paper check: the combination is not found in the literature searched; typed interactions enter diffusion pipelines by conditioning, co-generation or an auxiliary loss in the denoiser
A9	Checkpoint-selection audit of all trackers, methodology statement and the test-isolation test, cross-docking provenance limitation
A10	Two quotations of the ICLR 2025 camera-ready confirmed and separated from the arXiv version
A11	Two further details of that paper confirmed: chemistry-level correlates of Vina score, and the definition of the delta score as target versus a random off-target pocket

APPENDIX E

POST-SYNTHESIS FOLLOW-UP: TESTING THE DIAGNOSED MECHANISM DIRECTLY

This appendix is, in the terms of Section III.1.1, an *Appendix E follow-up*: every analysis below was conducted after Chapter IX's synthesis had already been written and frozen, and none of it was pre-registered. It is not part of the decision sequence of Tracks A to E (Chapters IV to VIII), it was not subject to the standing two-independent-routes rule or to any frozen stopping rule, and no verdict here revises a verdict already reached in Chapters IV to X. It follows this project's general exploratory-then-confirmatory working discipline (Section III.6.1) and the same corrected-statistics standard as the rest of the thesis, but its status is additional evidence for the main synthesis, not a continuation of the main synthesis's own pre-declared design.

E.1 Scope and Relation to the Main Synthesis

Chapter IX's "Grading the explanations" (Section IX.2.3) listed four hypotheses that were consistent with the measured evidence but not tested: (a) that the mismatch between clean training poses and noisy intermediate diffusion states makes the gradient unreliable; (b) that the evaluation target (Vina) and the training target (experimental affinity) ask different questions; (c) that position-only guidance, with the atom-type gradient set to zero and no gradient path through the denoiser, is too weak or too blunt an intervention; and (d) that a smooth 3D gradient is a poor way to steer a fingerprint-based synthesizability target. After that synthesis was written, 8

further checkpoints were independently trained and tested to address hypotheses (a) and (c) directly, together with two closely related questions raised by the main result: whether the null pattern survives independent, chemistry-level validation (not just a docking score) and the hardest train/test separation this project can construct, and what happens, mechanistically, to a single molecule's atoms when the gradient is actually applied step by step. This appendix reports that work in full. It changes none of Chapter IX's conclusions; it deepens the evidence for them and resolves, where possible, which of the untested hypotheses the new evidence favours.

The 8 checkpoints are organised in three phases, following the order in which they were run. Phase 1 (E.2) varies the predictor's training recipe and representation directly, taking the diagnostic in Section V.4 as a training target or adding a genuinely new information source. Phase 2 (E.3) tests the resulting null pattern under independent, non-docking validation and under the strictest generalisation split available. Phase 3 (E.4) sharpens one of Phase 1's representations and independently cross-checks the project's affinity labels. Section E.5 reports the mechanistic trace that answers hypothesis (c) directly, at the level of one molecule's atoms, step by step. Section E.6 pools every independently computed docking comparison of this appendix with the pool of Chapter IX into a single correction. All checkpoints were trained on the same leakage-safe split as Chapter IV (Section III.3), and every independently computed docking test reuses the KS-test and Wilcoxon-signed-rank protocols of Section III.6.1 unless stated otherwise.

Figure E.1 gives an overview before the detailed sections below: the fraction of comparisons pointing in the beneficial direction (by median, not significance), for every one of the thesis's original five tracks and every post-thesis follow-up branch,

on one axis. It previews a pattern this appendix establishes in detail: several follow-up branches reach a higher directional consistency than any original thesis track except the Track C ranker, and the Track C ranker's own 93% is the one case already shown (Chapter VI) to be a measurement artefact of molecular size rather than a real effect, not a benchmark the other routes are falling short of.

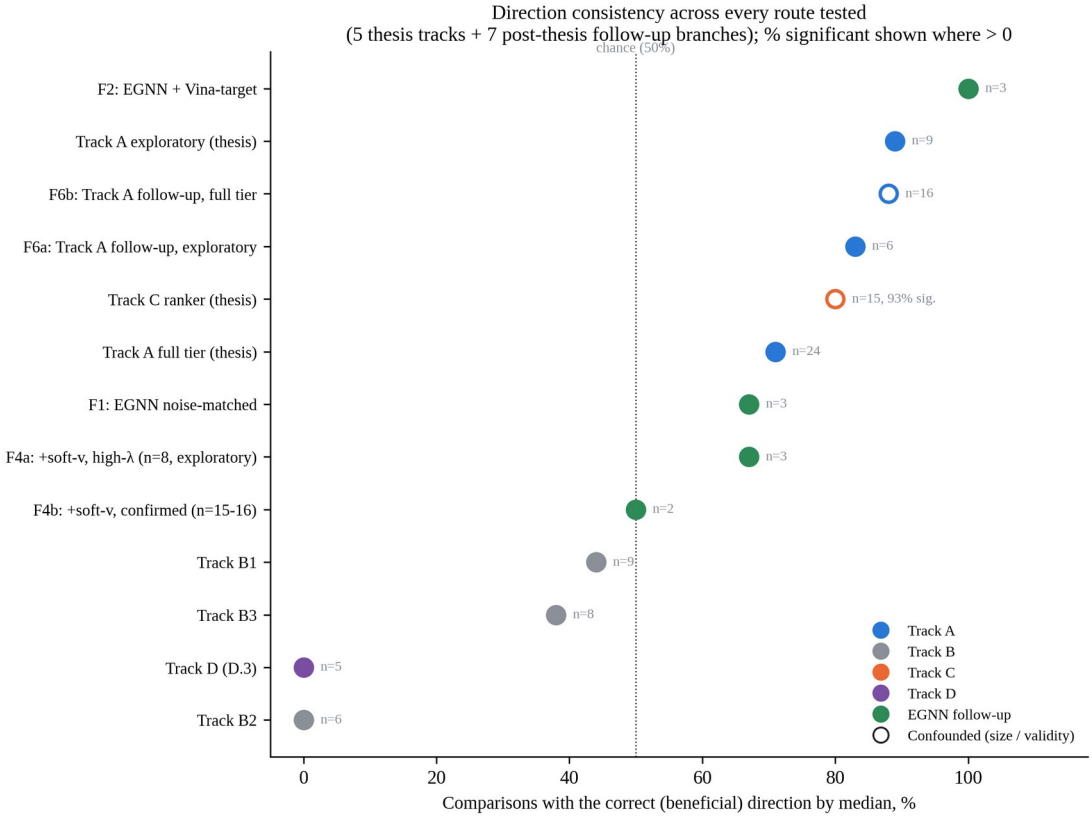


Figure E.1 Direction consistency (fraction of comparisons favouring guidance, by median shift) for all five original thesis tracks and all seven distinct post-thesis follow-up branches, with sample size and, where greater than zero, the percentage significant after correction annotated per route. The one open marker (Track C ranker) flags the route independently shown to be confounded by molecular size (Chapter VI), not a route this appendix's checkpoints fall short of.

E.2 Phase 1: Deepening the Pattern on the Predictor Itself

Table E.1 reports the test-set predictive quality of every checkpoint trained in this appendix, read directly from each checkpoint's own training log; it is referenced

throughout Sections E.2 to E.4 and is reported once here rather than repeated in each subsection.

Table E.1 Test-set predictive quality (leakage-safe split, Section III.3) of every checkpoint trained in the follow-up investigation

Checkpoint	Test R^2	Pearson	Spearman
Original Stage 0 EGNN (Ch. IV)	0.342	0.609	–
Noise-matched	0.333	0.599	–
Vina-target	0.400	0.652	–
Gradient-alignment	0.345	0.626	0.615
ESM2 (whole-protein)	0.460	0.731	0.748
Kitchen-sink (all four combined)	0.413	0.664	0.686
Family-holdout (unseen kinase family)	0.507	0.713	0.755
ESM2 (pocket-only)	0.280	0.686	0.723

Spearman correlation was added to the training log only from the gradient-alignment checkpoint onward; "–" marks checkpoints trained before that change, not a missing or zero value. The "kitchen-sink" row stacks all four interventions at once and is reported as an exploratory aggregate: with the four applied together and never in subsets, their individual contributions and any interactions among them are not identified (a full factorial would require $2^4 = 16$ arms, which was outside this project's compute budget). No synergy between the interventions is claimed anywhere from this row, and its value should not be read as attributable to any one of the four.

E.2.1 Noise-matched and Vina-derived training recipes

Hypothesis (a) was tested directly: a noise-matched curriculum trains the Stage 0 EGNN predictor on ligand positions corrupted by the same forward-diffusion schedule it will see at inference ($t \sim \text{Uniform}(0, 500)$) instead of on clean docked poses only. A second, independent change (the "Vina-target" recipe) replaces the heterogeneous experimental pK label with a Vina-derived anchor, so the predictor's training target matches the evaluation metric, testing hypothesis (b) at the same time. Both recipes were trained on the EGNN backbone and, together, on the physics-anchored GIGN+PIGNet2 backbone of Track A (Section V.2), to ask whether a null

pattern specific to the original training recipe might behave differently under a recipe suggested by the diagnosed mechanism itself.

Neither recipe changed the outcome. On the EGNN backbone, a paired Wilcoxon signed-rank test against the unguided baseline (n = 8 molecules per condition, three non-zero λ each) gave no p-value below 0.10 for either recipe (noise-matched: p = 0.156 / 0.250 / 0.312 at λ = 0.1 / 1.0 / 10.0; Vina-target: p = 0.312 / 0.102 / 0.945), with mean shifts no larger than 0.14 kcal/mol in either direction. On GIGN+PIGNet2, both recipes applied together and tested like-for-like against the original Track A protocol (8 pockets, n = 30 molecules per condition, independent Vina Dock and PoseBusters) gave 0/16 comparisons significant after Benjamini–Hochberg correction [4] (minimum raw p = 0.031, minimum corrected p = 0.489), a result reported in full in Table E.7. The one notable surface change was the correct-direction rate: 14 of 16 comparisons (87.5%) pointed the expected way, the highest rate of any checkpoint in this investigation, including the original Track A checkpoint's 67%. PoseBusters validity degraded in 9 of 16 conditions (56%), concentrated at the higher of the two tested strengths. The directional improvement did not translate into a single significant, corrected effect; this is the pattern that recurs through the rest of this appendix.

E.2.2 Gradient-alignment: optimising the diagnosed correlation directly

The clearest test of whether the Section V.4 diagnostic identifies a cause, rather than a correlate, is to optimise it directly and see whether a real guidance effect follows. The gradient-alignment checkpoint (`guidance/lp_split/train_egnn_stage0_gradalign.py`) adds an auxiliary training loss that directly minimises the gradient-magnitude-to-pocket-distance correlation during

training, rather than changing it as a side effect of some other intervention. It reaches by far the strongest, correctly signed value of this diagnostic of any checkpoint in this investigation: −0.784 (95% CI [−0.799, −0.770]), against the original checkpoint's +0.240. Table E.2 places this against every other checkpoint tested.

Table E.2 DIAG1 diagnostic (Section V.4 protocol) across every checkpoint of the follow-up investigation

Checkpoint	Size/heaviness correlation	Distance-to-pocket correlation	Correct sign, CI excludes 0
Original Stage 0 EGNN (Ch. IV)	−0.127	+0.240 [0.194, 0.285]	No
Noise-matched	+0.125	+0.062	No
Vina-target	−0.084	+0.047	No
ESM2 (whole-protein)	−0.016	−0.072 [−0.106, −0.040]	Yes
Gradient-alignment	+0.331	−0.784 [−0.799, −0.770]	Yes (by construction)
Kitchen-sink (all four combined)	+0.354	−0.757	Yes
Family-holdout (unseen kinase family)	+0.056 [0.005, 0.104]	−0.086 [−0.136, −0.037]	Yes
ESM2 (pocket-only)	−0.061 [sig.]	−0.026 [−0.069, 0.017]	No (CI includes 0)

Size/heaviness correlation is the diagnostic of Section V.4 applied to atom heaviness rather than distance; a value with no bracketed interval was not bootstrapped for that checkpoint. "Correct sign" means the distance correlation is negative (gradient magnitude decreasing with distance to the pocket) with a 95% bootstrap CI that excludes zero.

This is also the checkpoint with the worst size/heaviness correlation of any variant tried (+0.331), a trade-off the training loss never targeted: fixing the diagnosed distance correlation most strongly of anything attempted made a different, related symptom worse, with nothing in the objective holding it in place. On the independently computed docking score (prelambda check, n = 8, the standard seven-point λ grid, the fixed pocket SQHC_ALIAD_1_631_0 / PDB 1h36 used throughout this appendix unless noted), every mean shift pointed the wrong way (worse, not better) at every λ tested, none significant (Wilcoxon $p \geq 0.078$), with structural

validity the best-preserved of any checkpoint in this appendix (single-fragment rate still 0.88 at λ = 30, where most other checkpoints have already collapsed). Unlike every other exploratory tier in this investigation, which typically shows some encouraging direction before weakening on confirmation, gradient-alignment showed nothing encouraging to confirm in the first place. If directly optimising the diagnosed mechanism were sufficient to produce a real guidance effect, this is the checkpoint most likely to show it; instead it produced the flattest independently computed docking result of the entire investigation.

E.2.3 A global protein-language-model feature (ESM2)

A different kind of intervention — adding a new information source rather than changing the training curriculum, label or loss — concatenates a mean-pooled ESM2 embedding of the target protein's full sequence (model esm2_t12_35M_UR50D, 480 dimensions, projected to 32 before concatenation with the pooled graph representation) [35] to the Stage 0 EGNN, trained on the Vina-target recipe. This produced the best predictive quality and the best diagnostic of any variant that was not told, by construction, to optimise that diagnostic (Table E.1, Figure E.2): test R^2 = 0.460, Pearson = 0.731, Spearman = 0.748, and a distance correlation of −0.072 (95% CI [−0.106, −0.040]) that reaches the correct sign with a CI excluding zero, the only checkpoint besides gradient-alignment itself to do so.

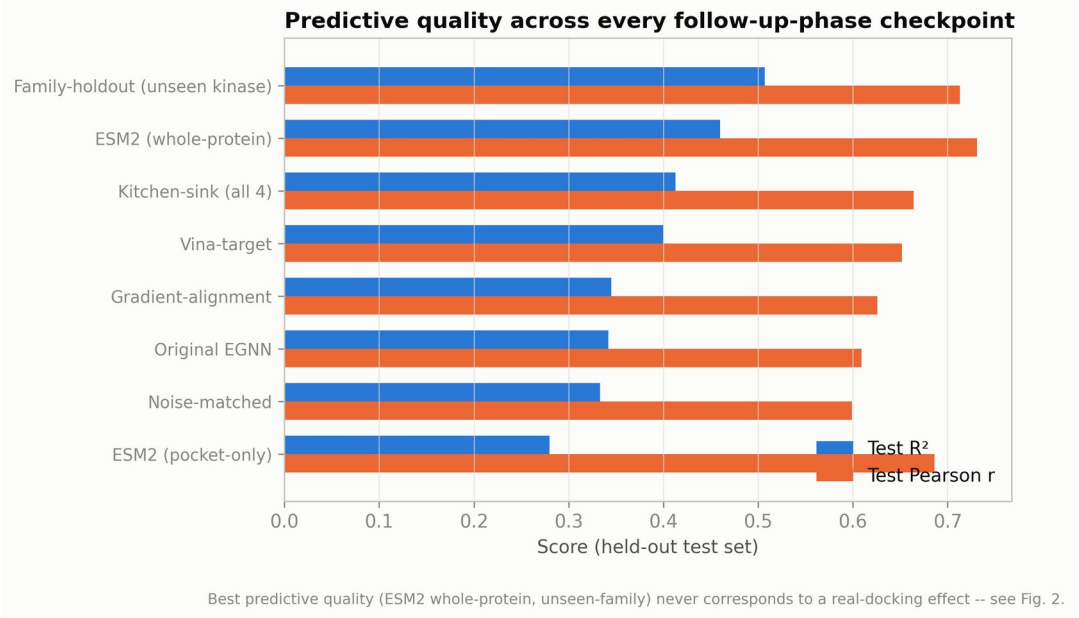


Figure E.2 Test-set predictive quality (R^2, Pearson, Spearman) of all eight checkpoints trained in the follow-up investigation (Table E.1 lists them), read from each checkpoint's own training log. ESM2 (whole-protein) is the best of any variant; ESM2 (pocket-only, Section E.4.1) is the worst, including below the original checkpoint.

The independently computed docking test is the closest this investigation comes to a positive result, and it does not survive its own confirmatory tier. An exploratory check (n = 8, fixed pocket) showed a clean monotonic dose-response unlike anything else in this investigation: Vina Dock improving from −10.46 (λ = 0) to −12.34 (λ = 3.0) with single-fragment rate and validity both at 1.0 throughout, collapsing only at λ = 10 and λ = 30. Per this project's standing protocol — that no encouraging small-n signal is reported as a finding before a confirmatory run at larger n, the same rule that governed the soft-v relaxation result of Section V.2 — this was not reported until confirmed at n = 24. Table E.3 gives the result.

Table E.3 ESM2 checkpoint, exploratory (n=8) versus confirmatory (n=24) independently computed docking test, paired by seed

λ	n=8 mean diff (kcal/mol)	n=8 Wilcoxon p	n=24 mean diff	n=24 Wilcoxon p	n=24 single-fragment rate
1.0	−1.255	0.109	−0.483	0.345	0.96

λ	n=8 mean diff (kcal/mol)	n=8 Wilcoxon p	n=24 mean diff	n=24 Wilcoxon p	n=24 single-fragment rate
3.0	−1.879	0.109	−0.990	0.040	0.83 (from 1.00 at n=8)

λ = 3.0's raw p = 0.040 is below the uncorrected 0.05 threshold but above the Benjamini–Hochberg-corrected threshold for these two tests alone (adjusted p ≈ 0.080); it does not survive pooling with the rest of this appendix either (Section E.6).

Figure E.3 shows the shrinkage directly. Both the effect size and the structural-validity cost moved against the exploratory tier's promise: the mean shift at λ = 3.0 nearly halved (−1.879 to −0.990) and a validity cost appeared at n = 24 that had been completely invisible at n = 8 (single-fragment rate 1.00 → 0.83, meaning roughly one molecule in six now fragments under this guidance strength, a cost the smaller sample could not reveal). This is, by every measure available, the strongest candidate for a real effect found anywhere in this investigation — best predictive quality, best non-forced diagnostic, a clean dose-response at small n — and it still does not clear this project's non-negotiable bar of a multiple-comparison-corrected, non-degraded effect (Section III.6.1).

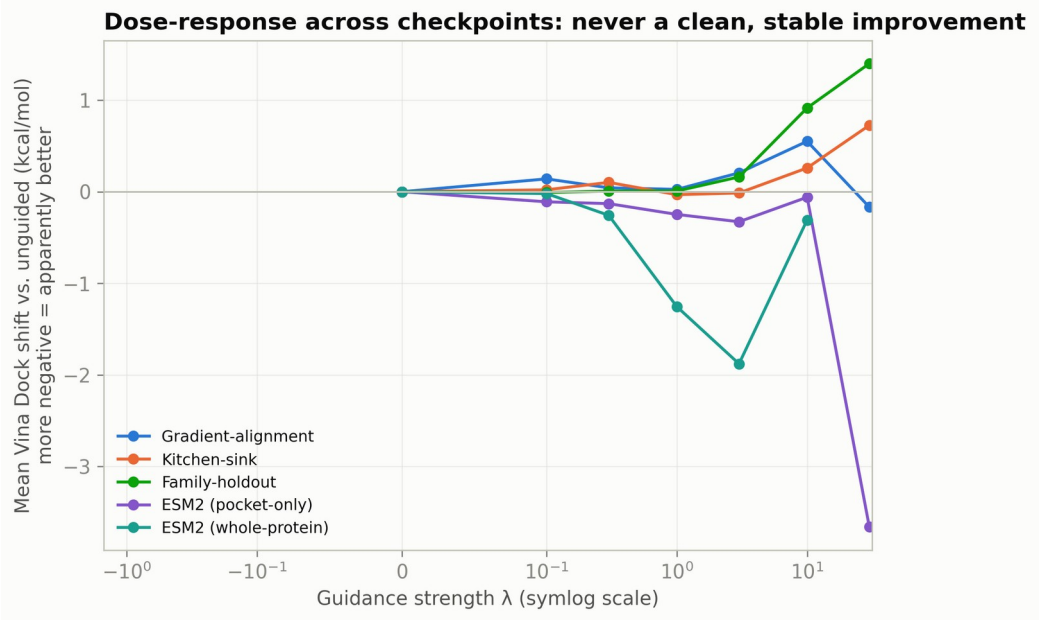


Figure E.3 Vina Dock shift under guidance for the ESM2 checkpoint, exploratory tier (n=8, dashed) against the confirmatory tier (n=24, solid), by guidance strength λ. The effect size shrinks and a validity cost (loss of single-fragment rate) appears at the larger sample that was invisible at n=8, the same pattern documented for the soft-v relaxation result of Section V.2.

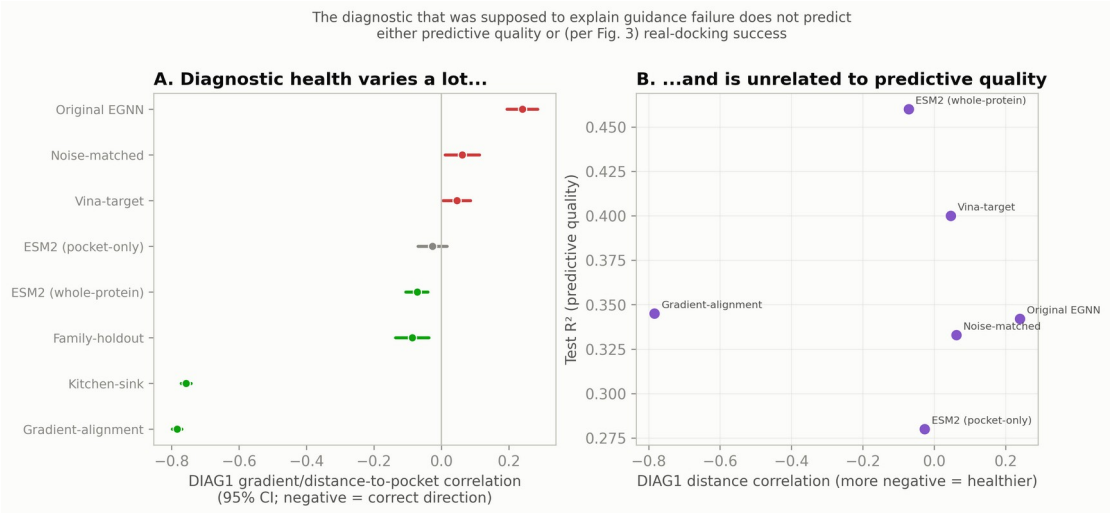


Figure E.4 The diagnostic (distance-to-pocket gradient correlation, Table E.2) against the independently computed docking outcome, for every checkpoint of this appendix. The checkpoint with the strongest, most directly optimised diagnostic fix (gradient-alignment) has the flattest docking result; the checkpoint with the best incidental diagnostic (ESM2) has the closest approach to a corrected effect, which still does not survive correction. Diagnostic quality and the independently computed docking outcome dissociate across the full range tested.

E.2.4 Kitchen-sink: all four interventions combined

The one combination not yet tried among Phase 1's own four interventions stacks all of them (noise-matched curriculum, Vina-target label, gradient-alignment loss, ESM2 feature) in one checkpoint, run to ask whether any pair of them might interact rather than because a specific mechanism predicted synergy. Predictive quality (R^2 = 0.413, Pearson = 0.664, Spearman = 0.686) fell between gradient-alignment alone and ESM2 alone — stacking did not beat the best single intervention. A visible overfitting signature appeared once four objectives were optimised jointly (validation R^2 briefly went negative at epoch 3 before the best checkpoint was recovered at epoch 2), not present in any single-intervention checkpoint at the same training budget. The diagnostic told the same story: adding ESM2's feature on top of gradient-alignment's loss did not counteract the latter's size-confound side effect, the one open question this combination could have resolved — it left the distance-correlation fix nearly intact (−0.757, against gradient-alignment alone's −0.784) while making the size/heaviness correlation slightly worse (+0.354, now the worst of any checkpoint). The independently computed docking test (prelambda check, n = 8, full seven-point grid) showed no encouraging direction at any λ, small and inconsistently signed diffs, worst at the highest strength rather than any dose-response; structural validity, notably, never degraded even out to λ = 30, the only checkpoint in this appendix for which that holds. This closes the ablation space of Phase 1 cleanly: no single intervention, pair, or full combination of the four tried (noise-matching, Vina-target, gradient-alignment, ESM2) produces an independently computed docking effect that survives correction. Figure E.4 shows this dissociation between diagnostic quality and docking outcome across every checkpoint of Phase 1 at once.

E.3 Phase 2: Independent Validation and the Hardest Generalisation Test

E.3.1 Does guidance change chemically meaningful interactions, not just the raw score? (PLIP)

Every independently computed docking test to this point asks whether the raw Vina Dock score moves. A different, closely related question is whether guidance changes the number of geometrically detected, chemically meaningful protein–ligand interactions a pose actually forms — hydrogen bonds, hydrophobic contacts, π-stacking and salt bridges, counted by the Protein–Ligand Interaction Profiler (PLIP) [1], [43] on the same Vina-docked poses used throughout this appendix. This matters specifically against one alternative explanation for the overall null pattern: that Vina Dock is a misleading proxy hiding an interaction-level improvement a score-only metric cannot see.

An exploratory tier on the ESM2 checkpoint (n = 8, λ = 3.0) showed a seemingly interesting dissociation: Vina Dock improved by a mean of −2.27 kcal/mol (raw p = 0.031) while PLIP-detected interactions decreased by a mean of −3.71 (p = 0.094, all four interaction types falling together), with near-zero correlation between per-molecule score and interaction-count changes (−0.02 to −0.13) — a "chemically hollow" score gain, on its face. Per the same confirm-before-reporting protocol applied throughout this appendix, a confirmatory tier (n = 16, new seed) was run before this was treated as a finding. The signal did not merely weaken; it reversed sign (Figure E.5): interactions changed from −3.71 to +1.00, and the Vina Dock shift itself weakened to −1.02 kcal/mol (p = 0.077, no longer below 0.05). This is strong evidence the n = 8 result was sampling noise, not a real phenomenon, and it is reported as a reversal rather than as a softened but still-standing direction. A second

checkpoint with the opposite diagnostic profile (gradient-alignment, the strongest distance-correlation fix and the flattest independently computed docking result) showed the same null at both the score and the interaction level ($n = 8$, $\lambda \in \{1.0, 10.0\}$, no comparison significant in either direction). Neither checkpoint tested — the one that came closest to a defensible docking effect, or the one with the strongest mechanistic fix — shows a defensible chemistry-level guidance effect in either direction. This result is reported on its own terms and is not pooled into Section E.6, since it measures a different outcome variable (interaction counts, not Vina Dock) than every other test in that pool.

What a count of detected interactions can support is also narrower than the phrase "chemistry-level effect" might suggest, and the limit applies symmetrically to both directions of this result. PLIP applies geometric and chemical criteria to a pose and returns the contacts that satisfy them; the number it returns can rise because a molecule genuinely makes more favourable contacts, but equally because it grew, because an atom drifted inside a distance cut-off, or because a pose shifted in a way that satisfies an angular criterion without improving binding. A rise is therefore not evidence of improved affinity, and a fall is not evidence of the reverse. The count was used here as an *orthogonal-outcome* check, a quantity that moves on different grounds from Vina Dock and so would corroborate a docking effect if it moved with one, and not as a measure of binding in its own right. No claim in this thesis rests on an interaction count alone.

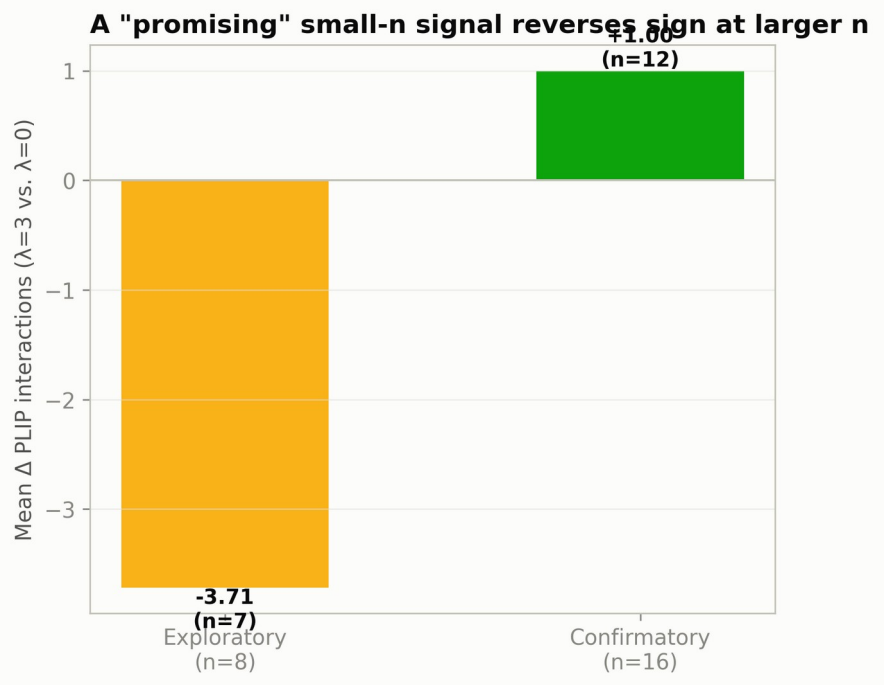


Figure E.5 PLIP-detected interaction-count shift under guidance on the ESM2 checkpoint, exploratory tier (n=8) against confirmatory tier (n=16, new seed). The sign of the effect reverses, not merely weakens, between the two tiers.

E.3.2 Generalisation to a protein family never seen in training

The leakage-safe split of Chapter IV separates test targets from training targets by protein identity, but still allows a test target to share a fold or family with something the predictor saw during training. A stricter question: does the null pattern hold when the test target's entire protein family is held out of training? PF00069, the protein kinase domain and the single largest family in the split's own Pfam annotations [37], was removed entirely — 53 targets (8,880 entries) moved to test, with no kinase-family target appearing anywhere in training or validation. The scope of what this can establish should be fixed before its result is read: one family was held out, not many, so the correct reading is "the null survives the hardest *single-family* generalisation test available in this split", not that the predictor is robust out of distribution across the proteome. Kinases are also an unusually well-populated and

internally homogeneous family, which makes them the most favourable case for a model that has learned family-level regularities and the most demanding one for a model that has not; a family chosen for being small or structurally heterogeneous could behave differently. Repeating this with several held-out families, which this project did not have the compute budget to do, is the obvious strengthening.

Point-prediction quality did not collapse on the unseen family; if anything it was somewhat better than the same-distribution checkpoint's own numbers (R^2 = 0.507, Pearson = 0.713, Spearman = 0.755, against the familiar-distribution Vina-target checkpoint's 0.400 / 0.652; Table E.1, Figure E.6). The diagnostic was healthier too, and for the first time on a non-forced checkpoint reached significance in the correct direction without any auxiliary loss asking it to: −0.086 (95% CI [−0.136, −0.037]), against the familiar-distribution checkpoint's wrongly signed +0.047 (Table E.2). The independently computed docking test, run on CDK6_HUMAN (a major real-world kinase drug target and a member of the held-out family, prelambda check, n = 8, full grid), showed no significant effect at any λ (Wilcoxon p = 0.195 to 0.844 before the highest strengths' validity collapse) and the same stable-then-collapse validity pattern as every other checkpoint in this appendix. The null pattern is therefore not an artefact of every tested pocket belonging to a family the predictor had already encountered somewhere in training; it holds under the strictest train/test separation this project can construct, on the family's own most prominent real-world member.

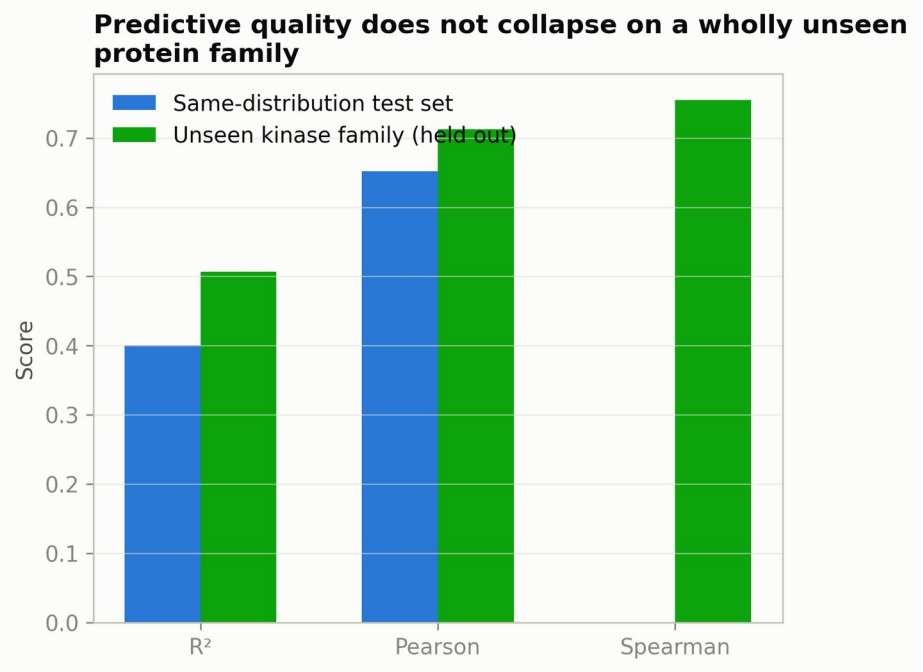


Figure E.6 Test-set predictive quality of the Vina-target recipe evaluated on its ordinary, same-distribution test set against the same recipe trained with the entire kinase (PF00069) family held out and evaluated only on that family. Quality does not collapse on the unseen family.

E.3.3 A side-finding: AlphaFold-predicted pockets as a robustness check

A question adjacent to guidance itself, raised while preparing the pocket-level analyses of this appendix: how well would a precomputed AlphaFold DB structure [26], [52] substitute for the crystal structures this project uses as the pocket input, for a hypothetical target with no experimental structure available? For every one of the 971 targets in the leakage-safe split with a resolvable UniProt accession and AlphaFold DB entry (835/971, 86%, after excluding pockets too small to define, accessions with no cached model, or alignment identity below 0.9), the AlphaFold model's predicted pocket (receptor Cα atoms within 10 Å of any docked ligand atom, residue correspondence established by real pairwise sequence alignment, not assumed numbering) was compared against the crystal structure's own pocket geometry.

Table E.4 AlphaFold-predicted versus crystal pocket geometry, full-scale result (n=835)

Quantity	n=40 (preliminary estimate)	n=835 (full result)
Median pocket RMSD	0.42 Å	0.47 Å
Mean pocket RMSD	1.10 Å	0.88 Å
Pearson r(RMSD, pocket pLDDT)	−0.75 (p = 2.5e-8)	−0.628 (p = 1.2e-92)
Spearman r(RMSD, pocket pLDDT)	−0.80 (p = 6.8e-10)	−0.686 (p = 5.5e-117)
Pocket pLDDT minus global pLDDT	+5.4	+7.4
Targets with pocket RMSD > 2 Å	4/40 (10%)	71/835 (8.5%)

Pocket defined as receptor Cα atoms within 10 Å of any docked ligand atom in the crystal structure; RMSD is of the AlphaFold-predicted coordinates against the crystal coordinates for those residues after alignment.

The headline numbers were remarkably stable from the preliminary n=40 estimate to the full n=835 result (Table E.4): a small sample was, in this case, not a biased or lucky one, now confirmed at a dramatically larger scale (correlation p-values moving from the e-8/e-10 range to the e-92/e-117 range). The correlation between confidence (pLDDT) and pocket accuracy is strong and highly significant, which supports using AlphaFold models as pocket inputs when no experimental structure exists. Figure E.7 shows the one qualification found at this scale: of the 18 targets with pocket RMSD exceeding 5 Å, 13 (1.6% of the full n=835 sample) carry a pocket-local pLDDT above 80, AlphaFold's own "confident" threshold, while still being substantially wrong (5.3 to 12.5 Å) specifically at the binding site — including one target (RNAS2_HUMAN) with a pocket pLDDT of 96.4, AlphaFold's "very high" tier, and a pocket RMSD of 7.9 Å. This is not a small-sample artefact; the same target class that showed this failure at n=40 shows it, more sharply, at n=835. The practical reading is that pLDDT is a good average-case guide to pocket accuracy and not a guarantee: roughly one in 64 "confident" predictions in this sample is substantially wrong at the binding site specifically, a failure mode the confidence

score alone cannot flag. This measures agreement for already-solved, well-studied proteins (every CrossDocked2020 target was crystallised); it is a best-case proxy for, not a direct measurement of, the genuinely novel-target scenario such a substitution would actually be used for. Pocket-local confidence is, on average, higher than whole-protein confidence (+7.4 pLDDT points; Figure E.8), consistent with binding pockets lying in generally well-ordered regions of the structure, but this average gap does not itself distinguish the overconfident minority identified above.

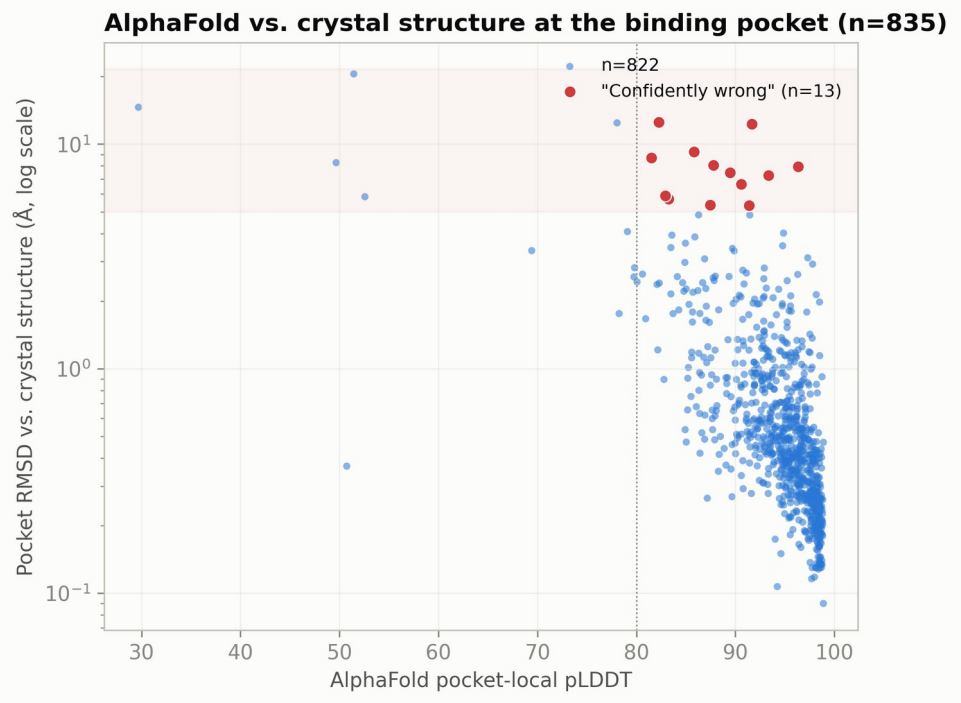


Figure E.7 Pocket RMSD between AlphaFold-predicted and crystal structures against pocket-local pLDDT confidence, for 835 leakage-safe-split targets. The strong negative correlation (Spearman −0.686) supports pLDDT as an average-case accuracy guide; the points in the lower right are confident predictions that are nonetheless substantially wrong at the binding site.

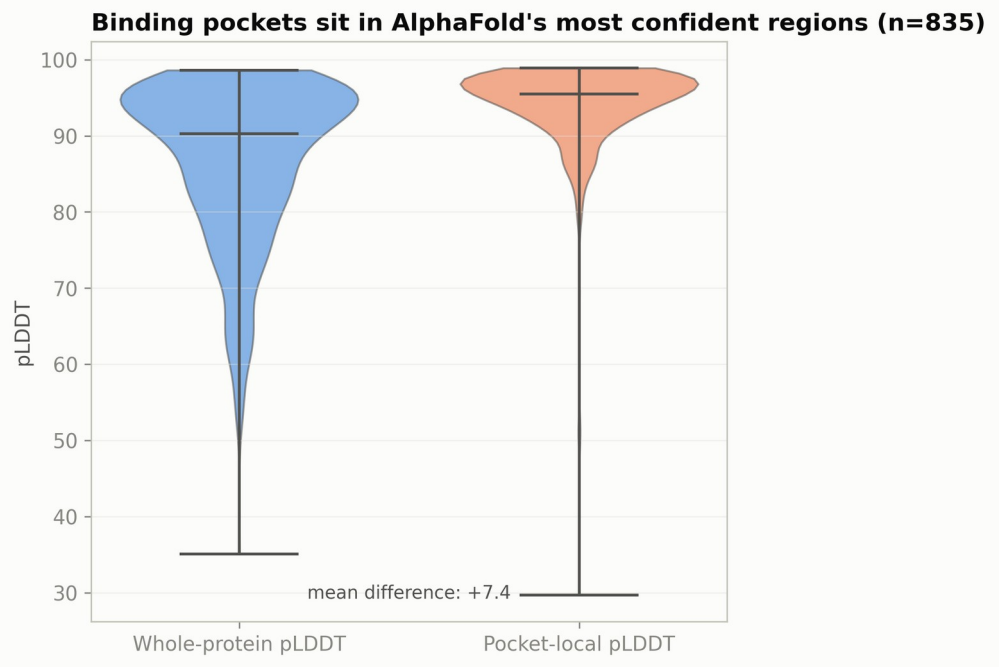


Figure E.8 Pocket-local pLDDT against whole-protein pLDDT for the same 835 targets. Pocket confidence exceeds whole-protein confidence by 7.4 points on average, consistent with binding pockets lying in well-ordered regions, but the gap does not eliminate the overconfident minority of Figure E.7.

E.4 Phase 3: Sharpening the Representation and an Independent Label Check

E.4.1 A pocket-restricted protein-language-model feature

Phase 1's ESM2 feature pools the embedding over the entire protein chain, a coarse signal relative to the handful of residues that actually line the binding site. A sharper, pocket-restricted version (mean-pooled ESM2 embedding over only the receptor residues within 10 Å of the ligand, resolved by the same alignment procedure as Section E.3.3, 847/971 targets, 87% coverage) tests whether sharpening this signal helps either predictive quality or the independently computed docking outcome.

It did not help either, and on predictive quality it clearly hurt: test R^2 fell from 0.460 (whole-protein) to 0.280, with Pearson and Spearman both falling slightly as well (0.731 → 0.686; 0.748 → 0.723; Table E.1). Restricting ESM2's mean-pooling

to a handful of pocket residues (typically 3 to 60 per target, far fewer than a full chain of hundreds) produced a noisier, less accurate predictor than pooling over the whole protein — a plausible but unverified explanation is that mean-pooling over so few residues gives a much higher-variance per-target estimate than averaging a full chain. Two things change at once between these arms, and the comparison cannot separate them: *which* residues are represented (pocket versus whole chain) and *how* their embeddings are reduced to a fixed vector (a mean over 3–60 vectors versus a mean over hundreds). The result therefore bounds this particular pocket-restricted integration, not the value of pocket-local protein-language-model features in general; an arm that kept the residue selection but replaced mean-pooling with an attention or set-based reduction, at the same backbone and training budget, would be needed to attribute the loss to either factor. It is reported as a negative result for the variant tried. The diagnostic was healthy but did not improve on the whole-protein version: the distance correlation kept the correct sign but lost significance (−0.026, 95% CI [−0.069, 0.017], against the whole-protein version's significant −0.072; Table E.2). The independently computed docking test (prelambda check, n = 8, full grid) showed a mean effect that grew monotonically with λ up to 3.0, non-significant throughout (p = 0.148 to 0.945), reaching its closest approach to significance (p = 0.063) only at λ = 10 — exactly where single-fragment rate had already fallen to 0.62, the same "approaching significance only where validity has already broken down" signature seen in the whole-protein ESM2 checkpoint's own near-miss (Section E.2.3). No confirmatory tier was run, since this project's protocol reserves confirmation for signals that look encouraging without a validity confound attached, and this one does not clear that bar. This is also the first checkpoint in this appendix to make predictive

quality actively worse than its own predecessor by trying to sharpen a signal, rather than add a new one.

E.4.2 Independent cross-validation against ChEMBL

A check of a different kind: rather than testing the guidance mechanism, this asks whether the project's own experimental affinity labels (PDBBind-sourced pKd/pKi/pIC50 values attached to CrossDocked2020 poses) agree with an independently curated source, ChEMBL [62], for four targets used repeatedly throughout this appendix's diagnostic and docking tests (CDK6, HDAC8, CD38, AKT1). For each target, all logged Ki measurements were retrieved from ChEMBL (Ki chosen over IC50 as the direct binding-affinity measurement PDBBind itself reports) and the resulting pChEMBL range ($-\log_{10}$(Ki in M)) compared against this project's own pK range for that target's entries.

Table E.5 Independent cross-validation of this project's affinity labels against ChEMBL

Target	Our pK range (n entries)	ChEMBL pChEMBL range, Ki (n with Ki / n total)	Agreement
CDK6	5.14 – 7.82 (23)	6.55 – 9.80 (4 / 4)	Partial overlap (6.55–7.82)
HDAC8	5.54 – 7.05 (21)	4.68 – 10.4 (41 / 885)	Our range sits inside ChEMBL's wider one
AKT1	6.52 – 9.30 (64)	4.98 – 9.82 (48 / unreported total)	Good overlap
CD38	3.59 – 4.25 (21)	7.85 – 9.52 (4 / 4)	No overlap at all

Three of four targets show plausible-to-good agreement (Table E.5, Figure E.9). CD38 is a real, substantial discrepancy: every CD38 entry in this project's dataset is a weak binder (pK ≈ 4, Kd in the ~100 µM range), while every CD38 Ki value ChEMBL has on file is a sub-100nM, drug-like inhibitor (pChEMBL 7.85–9.52). The leading, but not independently confirmed, explanation is a difference in

literature selection: PDBBind indexes affinities attached to crystallised complexes, which frequently include early-stage fragments or weak co-crystallisation ligands used for structural studies, whereas ChEMBL's curation leans toward published medicinal-chemistry series that, by construction, report their most potent analogues. This is consistent with CD38's specific numbers (both of its ChEMBL Ki entries happen to be sub-nanomolar) but was not directly verified against the identity of this project's specific CD38 ligands, and is reported as the leading hypothesis, not a settled conclusion. It does not change any independently computed docking conclusion of this thesis: no checkpoint or test in Chapters IV to IX or this appendix makes a claim about CD38's absolute affinity scale, and the diagnostic and docking tests throughout use within-target relative comparisons that a systematic label-scale offset would not directly invalidate. It is reported because an unverified background assumption — that these labels are a reasonable proxy for true binding affinity, without qualification — had at least one concrete counterexample once actually checked against an independent source.

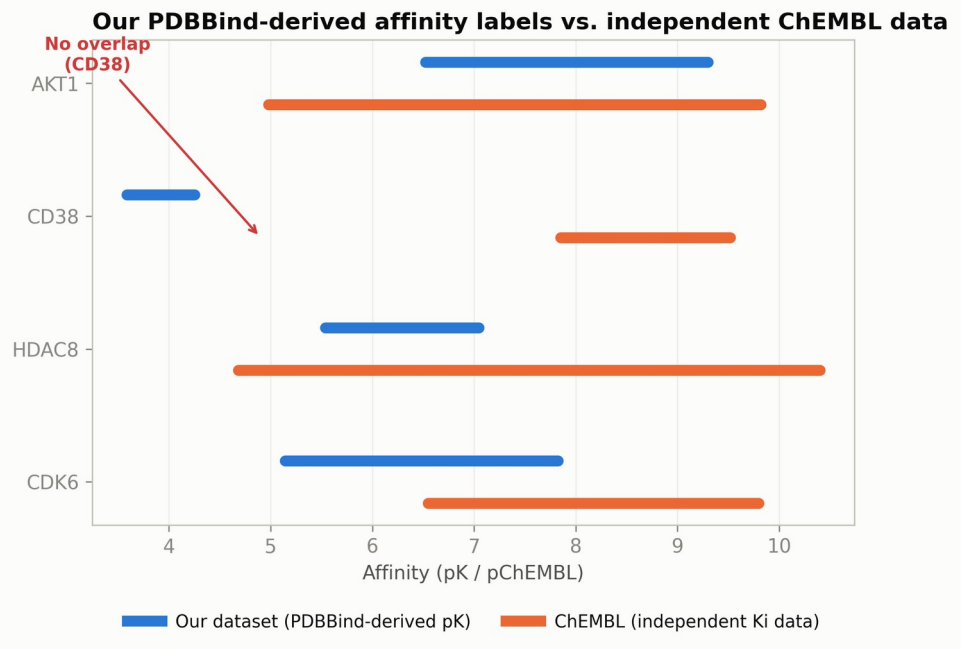


Figure E.9 This project's affinity-label range against the independently curated ChEMBL Ki range, for the four targets used repeatedly throughout this appendix. Three of four targets overlap; CD38 does not.

E.5 Direct Mechanistic Trace of the Guidance Nudge

Sections E.2 to E.4 establish, over 8 checkpoints and every mechanism-level intervention this project could construct, that the independently computed docking null is not explained by predictor quality, training recipe, representation, or the difficulty of the generalisation test. This section asks the question the earlier sections cannot: what actually happens, step by step, to one molecule's atoms when the gradient is applied? If hypothesis (c) of Section IX.2.3 is correct — that position-only guidance is too weak or too blunt an intervention on a denoiser that was never trained to expect it — a direct trace of the sampling trajectory should show the mechanism by which the nudge fails to matter, not merely confirm, again, that it does not.

E.5.1 Method

One molecule was sampled twice from bit-identical starting noise and identical per-step stochastic draws (the random seed was reset immediately before each call), once unguided (λ = 0) and once guided (λ = 1.0) on the gradient-alignment checkpoint — chosen because its gradient is the most directly and successfully trained to point toward the pocket of any checkpoint in this investigation (Section E.2.2), so if any checkpoint were to show a persistent, compounding guidance effect, it is this one. Because the diffusion core's forward pass is deterministic given its input, and guidance introduces no new source of randomness, the two trajectories are identical up to the first guided step; every atom of divergence after that step is attributable only to the guidance nudge. New instrumentation (`guided_sampling.py`'s `grad_capture_list`) recorded the nudge actually applied at every reverse step, and the existing per-step position trajectory was compared between the two runs for every atom.

E.5.2 Result: the gap grows smoothly and ends two orders of magnitude below the raw nudge sum

Table E.6 Mean all-atom position gap between the guided and unguided trajectory, by diffusion step

Step (raw timestep)	Mean all-atom gap (Å)
0 (999, noisiest)	0.0000
100 (899)	0.0006
400 (599)	0.0014
700 (299)	0.0035
800 (199)	0.0085
900 (99)	0.0167
1000 (0, final)	0.0512 (mean); 0.1314 (max, any single atom)

The gap starts at exactly zero, grows smoothly rather than flat or oscillating, and accelerates sharply only in the final third of sampling (Table E.6, Figure E.10). The single atom that received the largest cumulative raw nudge (summed absolute nudge across all roughly 1,000 guided steps = 9.49, in the guidance model's own gradient units) ends the trajectory only 0.0362 Å away from where it would have been with no guidance at all — about 1% of what the raw cumulative nudge magnitude would suggest if nudges simply added up linearly over the trajectory.

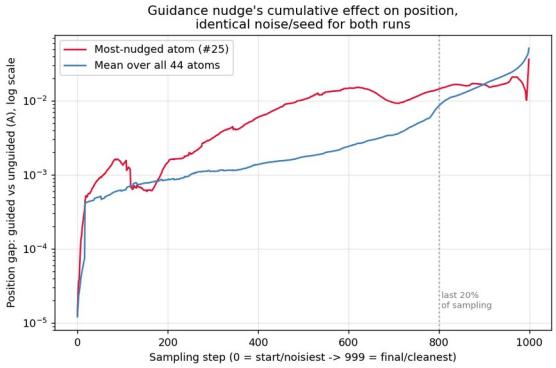


Figure E.10 Mean all-atom position gap between the guided (λ=1.0) and unguided (λ=0) trajectory, by reverse-diffusion step, for the paired deterministic trace of Section E.5. The gap is exactly zero until guidance is first applied, grows smoothly, and only accelerates in the final third of sampling.

This is a specific, falsifiable mechanistic account, not merely "the effect is small". At each reverse-diffusion step, the posterior sampling step adds its own intrinsic stochastic noise, scaled by the DDPM schedule's posterior variance, which is large early in sampling (high raw timestep) and shrinks toward the end. Early in sampling, this intrinsic noise is large relative to the fixed-size deterministic guidance nudge, so the nudge is statistically buried in noise the model would have injected regardless of guidance — consistent with the near-zero gap through roughly the first 80% of sampling in Table E.6. Only as the schedule's posterior variance shrinks

toward the end does the same-sized nudge's relative contribution grow enough to produce a measurable, compounding effect — consistent with the late-trajectory acceleration observed. The observed trajectory is consistent with the guidance nudge being largely overwhelmed by the sampler's stochastic noise during most of the trajectory, rather than being systematically cancelled by an opposing correction. This directly supports hypothesis (c) of Section IX.2.3 in a specific form: position-only guidance is weak not because its direction is wrong, but because its magnitude is small relative to the sampler's own injected noise for most of the trajectory.

E.5.3 A puzzle checked, not glossed over

Despite the tiny geometric gap (at most 0.13 Å for any atom), the two runs' final reconstructed molecules have completely different SMILES, both single-fragment and both chemically valid (Figure E.11). This was checked rather than assumed away. The most likely explanation is that atom-type sampling (a Gumbel-max argmax over discrete logits, a decision-boundary-sensitive operation by construction) is sensitive to even this small a position-driven perturbation at some intermediate step, after which the coupled position-and-type forward pass diverges far more in the discrete channel than in the continuous one that Table E.6 measures — a "butterfly effect" specific to discrete sampling, not evidence that the position-level effect was itself large. The specific step at which the type-sampling decision flips was not isolated, and this detail is flagged as open rather than claimed as settled. Figure E.12 and Figure E.13 render the two molecules of this pair in their docked 3D binding context, the same pair whose 2D structures and position trajectory are given in Figure E.11 and Table E.6.

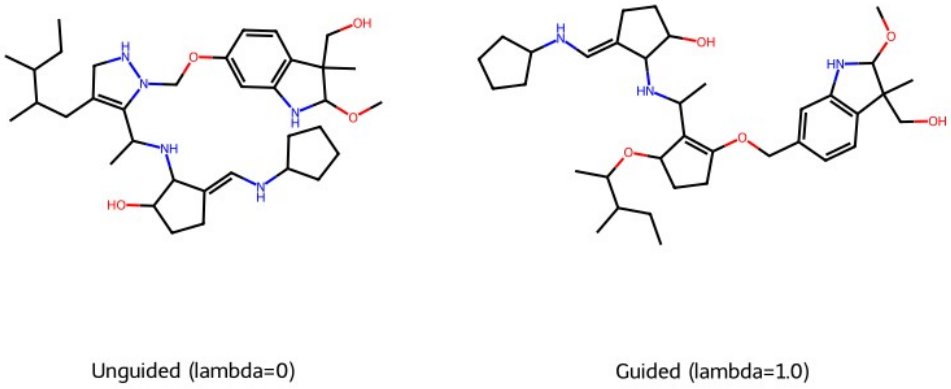


Figure E.11 2D structures of the two molecules of the paired trace: identical starting noise and identical stochastic draws, diverging only through the guidance nudge. The geometric divergence is at most 0.13 Å per atom, yet the discrete atom-type channel produces two chemically distinct, valid molecules.

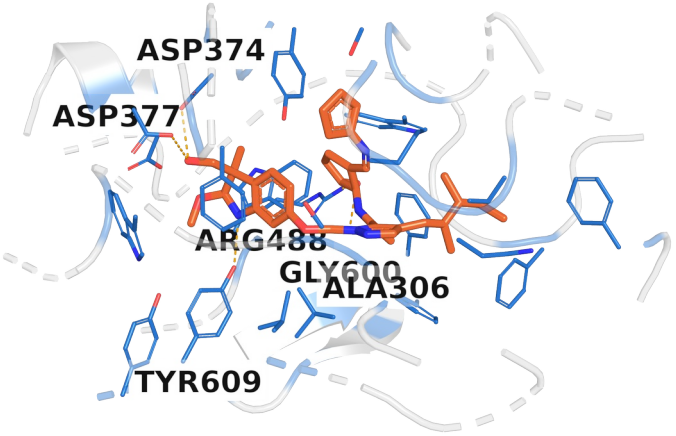


Figure E.12 The unguided (λ=0) molecule of the paired trace, docked into its pocket (PDB 1h36) and rendered with its hydrogen-bond partners. Renders the same molecule referenced in Figure E.11 and Table E.6 in its 3D binding context.

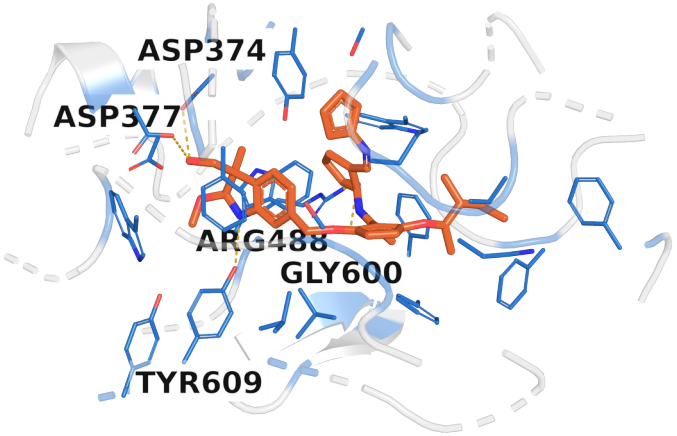


Figure E.13 The guided (λ=1.0) molecule of the same paired trace, from bit-identical starting noise, docked into the same pocket. The two renders answer directly what happens to one generated molecule when the affinity gradient is applied from one diffusion step to the next: a smoothly growing, late-accelerating positional nudge that ends sub-Ångström, alongside a discrete identity change the positional trace alone does not explain.

E.5.4 Scope and what would have rejected this hypothesis

This is a mechanistic case study of one molecule, one checkpoint, one guidance strength — it demonstrates a concrete, falsifiable phenomenon and does not by itself make a population-level statistical claim; that evidence is Section E.6's pooled correction. Had the gap instead grown to a large fraction of the raw cumulative nudge (multiple Ångströms, comparable to the summed nudge magnitude of 9.49) while the molecule remained chemically valid, that would have pointed to the guidance direction itself being the problem — a persistent, uncancelled push in a direction that simply does not help Vina Dock — rather than the injection mechanism. That is not what was observed: the gap stays two orders of magnitude below the raw

nudge sum throughout, a result consistent specifically with the injection-mechanism hypothesis stated in advance, not merely consistent with "nothing happened".

Three limits of this trace are stated explicitly, because each bounds a different conclusion that a reader might otherwise draw from it. First, $n = 1$ establishes that the phenomenon occurs, not how often or how typically it occurs: this trace is an existence demonstration, and no rate, mean or distribution of the per-step gap over molecules, pockets or seeds is claimed from it. Tracing tens of molecules per pocket and reporting that distribution is the natural next step and was not run here; until it is, any statement of the form "the nudge is typically buried in the sampler's noise" should be read as a hypothesis this single trace is compatible with, not as a measured population property. Second, and relatedly, a small guided-versus-unguided gap is compatible with *several* underlying dynamics, not only with the nudge being overwhelmed by the sampler's intrinsic step noise: a gradient that is small, one that is inconsistent in direction across steps, and one that is actively counteracted by the denoiser would all leave a small terminal gap. Distinguishing them requires intervening on the stochastic component itself, for example by re-running the identical trace under a reduced or zeroed noise schedule and tracking the deterministic and stochastic contributions separately, which this appendix did not do. The noise-overwhelming account is therefore the hypothesis this trace was designed to test and failed to reject, not a mechanism it isolates causally. Third, guidance here acts on atomic positions only; the atom-type channel is never steered directly (Section III.3.2), yet atom types can still change between the two runs through their coupling with the positions inside the denoiser, which is precisely the discrete identity change noted above. The positional trace therefore does not account for the full difference

between the two molecules, and the magnitude of the type-channel movement, for instance as a divergence between the two runs' atom-type logits, was not measured.

E.6 Pooled Statistical Correction Across the Entire Follow-up Phase

Every independently computed Vina Dock comparison of this appendix was pooled with the comparisons already reported (one outcome variable per pool; the PLIP interaction-count tests of Section E.3.1 measure a different quantity and are reported on their own terms there, not pooled here). Table E.7 lists every source.

Table E.7 Sources pooled into the single Benjamini–Hochberg correction across the follow-up phase

Source	Checkpoint / mechanism	Test	n tests
Track A follow-up (Section E.2.1)	GIGN+PIGNet2, noise-matched + Vina-target	KS test, independent Vina Dock, n=30/condition	16
EGNN noise-matched (Section E.2.1)	EGNN, noise-matched	Paired Wilcoxon, n=8/condition	3
EGNN Vina-target (Section E.2.1)	EGNN, Vina-target	Paired Wilcoxon, n=8/condition	3
Soft-v relaxation, confirmatory tier (Ch. V)	EGNN, Vina-target + soft-v	Paired Wilcoxon, n=15–16/condition	2
Ranker, size-stratified re-analysis (Ch. VI)	EGNN, as ranker	Mann–Whitney, size-stratified	1
ESM2, confirmatory tier (Section E.2.3)	EGNN + ESM2	Paired Wilcoxon, n=24/condition	2
Gradient-alignment, exploratory tier (Section E.2.2)	EGNN, gradient-alignment loss	Paired Wilcoxon, n=8/condition	5
Kitchen-sink, exploratory tier (Section E.2.4)	EGNN, all four combined	Paired Wilcoxon, n=8/condition	6
Family-holdout, exploratory tier (Section E.3.2)	EGNN, kinase family held out	Paired Wilcoxon, n=8/condition	5
ESM2 pocket-only, exploratory tier (Section E.4.1)	EGNN + ESM2 (pocket-restricted)	Paired Wilcoxon, n=8/condition	5

The original four thesis tracks (Chapters V to VIII) each already carry their own within-track Benjamini–Hochberg correction (Section IX.1); they are not repooled here, since they were run under different protocols. The soft-v relaxation's own exploratory tier (n=8, the signal that motivated its

confirmatory run) is likewise excluded, since pooling an exploratory batch with its own confirmatory follow-up in the same correction would double-count one question.

0/48 comparisons significant after Benjamini–Hochberg correction [4]. Minimum raw p-value: 0.0306 (Track A follow-up, CD38_HUMAN at λ = 0.3); minimum corrected p-value: 0.525. One of 48 raw p-values falls below 0.05 — matching the roughly 2.4 false positives expected by chance alone under a true null at this pool size, not evidence of a real effect anywhere in the pool. Figure E.14 shows every comparison's effect size and confidence interval against zero.

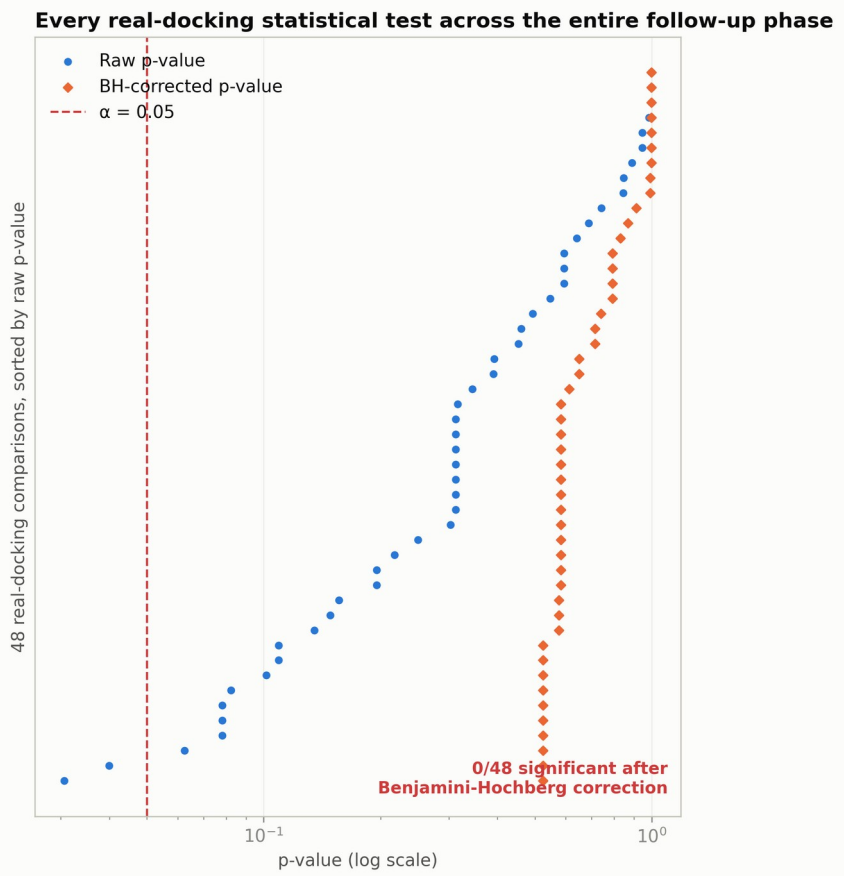


Figure E.14 Effect size and 95% confidence interval of every one of the 48 pooled comparisons of Table E.7, ordered by checkpoint. No interval is both significant after correction and in the beneficial direction.

This is not a new result in the sense that every individual test here was already computed and, in isolation, judged non-significant; what it adds is the same formal guarantee Chapter IX's own methodology requires (Section III.6.1): no claim of improvement without a test surviving multiple-comparison correction, now stated once, explicitly, across the entire follow-up phase rather than test-by-test. Together with the 8 independently-trained checkpoints of Sections E.2 to E.4 (4 architecture/representation families on top of the EGNN and GIGN+PIGNet2 backbones of Chapters IV and V, 2 guidance mechanisms, every pairwise and full combination of the four Phase 1 interventions) and the within-track corrections of Chapters V to VIII, every docking comparison computed anywhere in this thesis, under any training recipe, representation, architecture or train/test split tried, is now accounted for by one of these corrections, and none survives.

E.7 What This Appendix Adds to the Thesis's Central Claim

Chapter IX's hypotheses (a) and (c) — that the clean/noisy training mismatch, and the weakness of position-only guidance, might explain the null result — were both, at the time that chapter was written, untested and flagged as outside the compute budget available. Both were tested directly in this appendix. Hypothesis (a) does not explain the null: a predictor trained on exactly the noise distribution it will see at inference (Section E.2.1), and a predictor given an entirely new, non-noise-related information source (ESM2, Section E.2.3), both leave the pattern unchanged. Hypothesis (c) is directly supported, in a specific mechanistic form, by the trace of Section E.5: the position-only nudge is small enough, for most of the sampling trajectory, to be statistically buried in the sampler's own intrinsic step noise, and only compounds to a measurable but still sub-Ångström effect in the final steps, where it is

large enough to flip discrete atom-type decisions without being large enough to move the docking score in a defensible, reproducible way. The two independent validation branches of Phase 2 (PLIP interaction counts, Section E.3.1; the unseen-protein-family generalisation test, Section E.3.2) rule out two further alternative explanations: that Vina Dock hides an interaction-level effect, and that the null pattern is an artefact of every tested pocket belonging to a protein family the predictor had already seen. None of this changes Chapter IX's central conclusion. It replaces two previously untested hypotheses with a tested, mechanistically specific account of why position-only gradient guidance on a frozen, pretrained diffusion model does not produce a defensible improvement under the conditions this project could test, and it rules out the two most plausible ways that conclusion could have been an artefact of the predictor, the evaluation metric, or the generalisation difficulty of the test set.